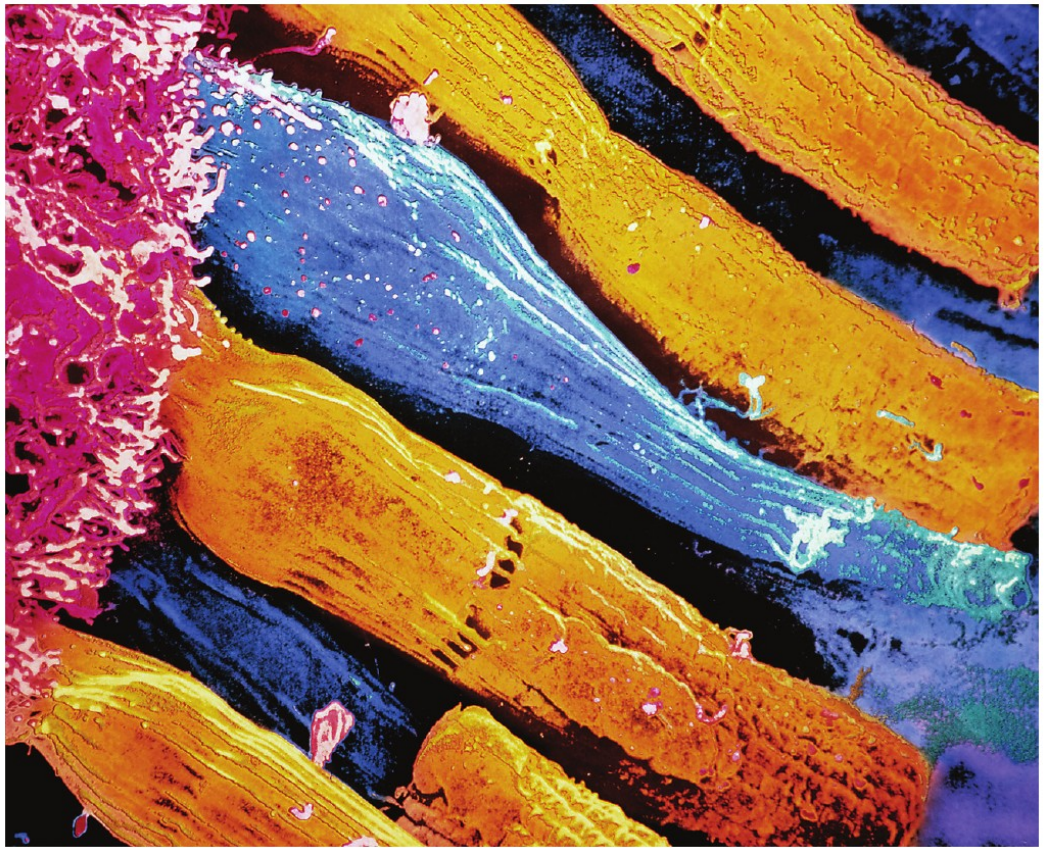


Carlson and Birkett (2022), chapter opener, p. 153. Photo: Bill Longcore / Science Source.

BEHAVIORAL NEUROSCIENCE 392

Vision, Illusions, and Neural Plasticity

From Light to Sight: How the Brain Builds the Visual World

Fall 2026

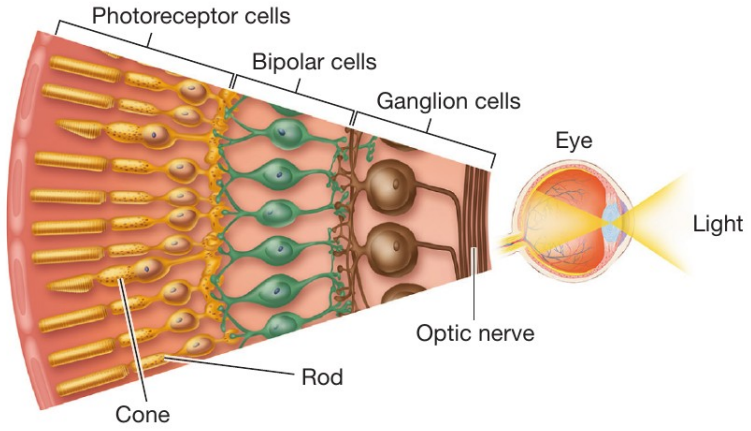
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Vision begins with changes in retinal activity

Sensation begins when cells detect physical energy. In vision, that energy is light entering the eye. Photoreceptors in the retina, the thin neural tissue lining the back of the eye, change their electrical activity when they absorb light. Other retinal neurons transform those changes into signals sent toward the brain.

Perception is the experience of seeing an object, its color, or its movement. That experience requires the brain to combine signals from many retinal locations. A patch of light may belong to a prey animal, a person's face, or a reflection, depending on its arrangement and the surrounding scene.

Visual signals also guide actions before an observer describes what is present. A hunting fish turns toward a small moving target and adjusts its eyes before capturing it. The position and movement of that target must be converted into commands for the eyes, head, and body while the target continues to move.



Carlson and Birkett (2022), Fig. 6.5, p. 158.



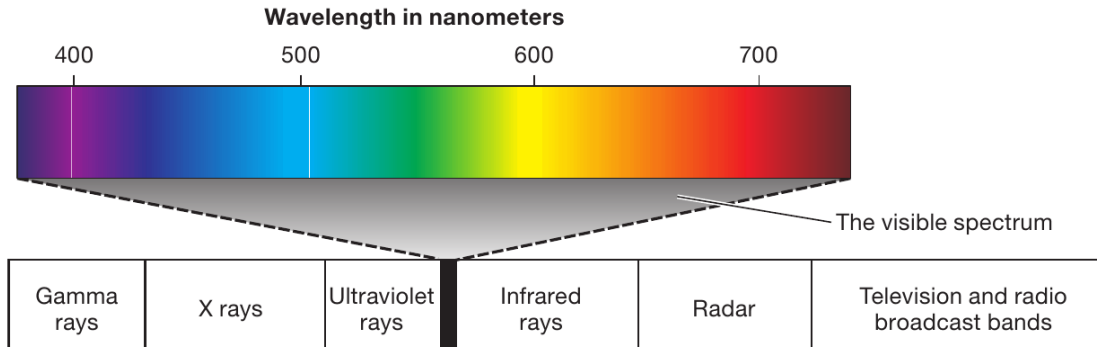
Mearns et al. (2025), 98347 Fig. 1 Video 2.

Wavelength and intensity describe different inputs

Wavelength is the distance between successive peaks of a light wave. Human photoreceptors detect only part of the electromagnetic spectrum. Within that range, changing the mixture of wavelengths can change perceived hue. Ultraviolet light has shorter wavelengths than humans normally see, although some other animals have photopigments that detect it.

Intensity describes how much light reaches a surface or receptor. Increasing intensity usually makes a stimulus appear brighter, but perceived brightness also depends on its surroundings and the eye's adaptation. A dim object against a dark background can be easier to detect than a brighter object against an even brighter background.

Saturation describes how strongly a color differs from a neutral gray or white appearance. Mixing wavelengths can reduce saturation without simply making a light dimmer. Hue, brightness, and saturation describe perceptual dimensions, whereas wavelength and intensity describe physical inputs that the visual system must measure and compare.



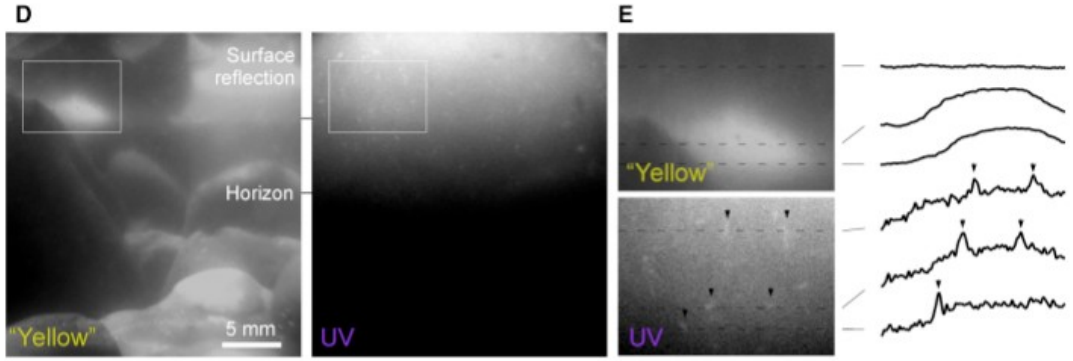
Carlson and Birkett (2022), Fig. 6.2, p. 156.

Ultraviolet light makes tiny prey stand out

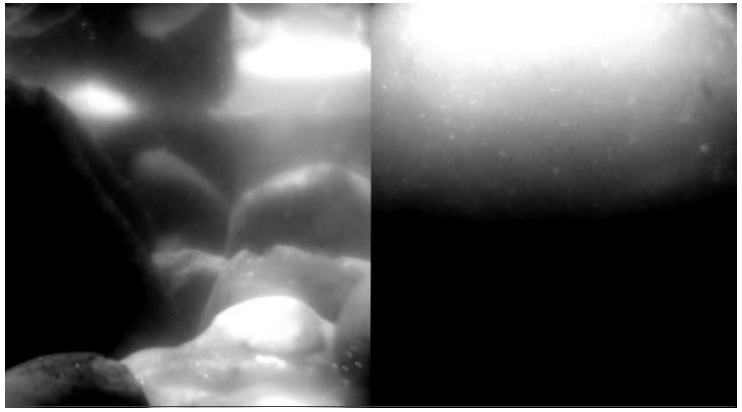
Larval zebrafish eat tiny swimming organisms called **paramecia**. Yoshimatsu and colleagues filmed these organisms through different wavelength filters in a naturalistic tank. In the longer-wavelength view, background detail competed with the prey. In the ultraviolet view, illuminated paramecia appeared as bright points against a less cluttered background.

The researchers then presented real prey under ultraviolet or yellow illumination. Larvae produced prey-capture responses more readily under ultraviolet light. Those responses included inward eye rotation and tail movements. Removing ultraviolet-sensitive cones eliminated the usual advantage of ultraviolet illumination, identifying these photoreceptors as a source of the behaviorally useful signal.

The useful contrast depends on the animal's receptors and its environment. Human observers do not normally see this ultraviolet channel, so a scene that looks difficult to us can contain conspicuous prey signals for a fish. Underwater illumination, background reflections, and the prey's optical properties determine which receptor population receives the clearest signal.



Yoshimatsu et al. (2020), Fig. 1, selected panels.



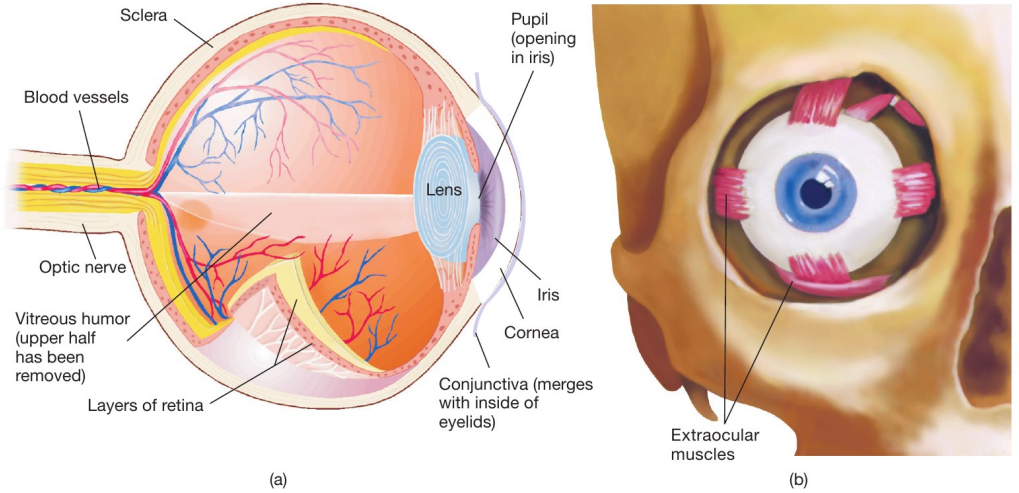
Yoshimatsu et al. (2020), Video S1.

The cornea and lens focus light on the retina

The **cornea**, the transparent front surface of the eye, bends incoming light. The lens provides additional focusing so that rays from an object meet at the retina. Because these optical structures redirect the rays, the retinal image is inverted vertically and reversed horizontally relative to the arrangement of objects outside the eye.

The **pupil** is the opening through the iris, the colored ring of muscle surrounding it. Changing pupil size regulates the amount of light entering the eye. The lens changes shape during **accommodation**, allowing near and distant objects to be brought into focus on the same retinal surface.

The six **extraocular muscles** rotate the eyeball to bring selected objects onto different retinal locations. Moving the eye changes which photoreceptors sample an object without moving the object itself. The brain must therefore combine the retinal signal with information about eye position to locate the object in the surrounding world.



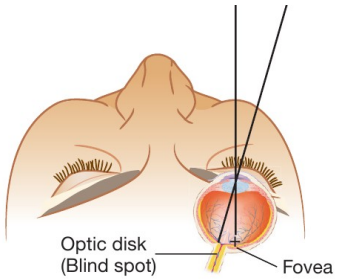
Carlson and Birkett (2022), Fig. 6.4, p. 157.

A blind spot interrupts the retinal image

The **optic disc** is the location where ganglion-cell axons leave the retina to form the optic nerve. It contains no rods or cones. Light falling there cannot activate photoreceptors, producing a small blind region within each eye's visual field even when the surrounding retina is functioning normally.

In the original demonstration, keeping one eye closed and fixating the cross can make the separate colored circle disappear at a particular viewing distance. Changing distance moves the circle's retinal image onto or away from the optic disc. The disappearance occurs while the cross remains visible because the two images occupy different retinal locations.

The missing region usually escapes awareness during ordinary viewing. The other eye can supply overlapping information, and the surrounding visual pattern influences what is perceived at the blind location. The continuous appearance of a scene therefore does not mean that every point in it was sampled by a photoreceptor in each eye.



+



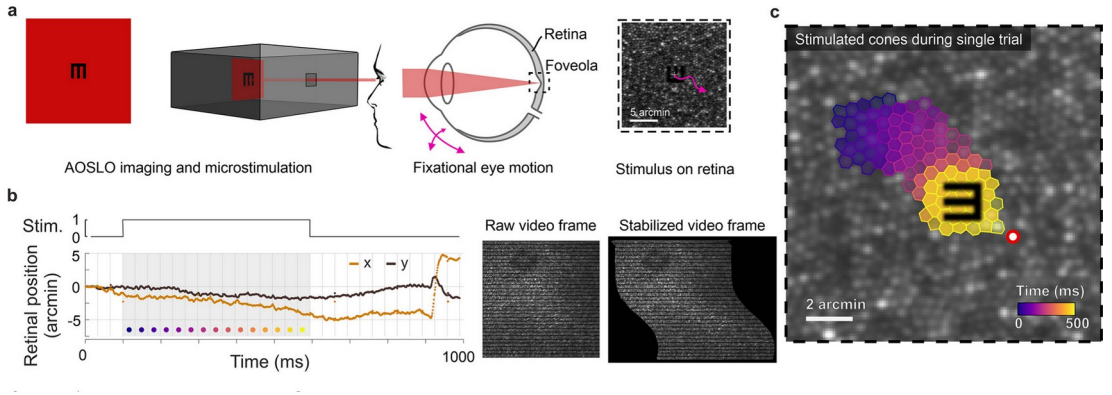
Carlson and Birkett (2022), Fig. 6.6, p. 158.

Living human cones can be imaged during seeing

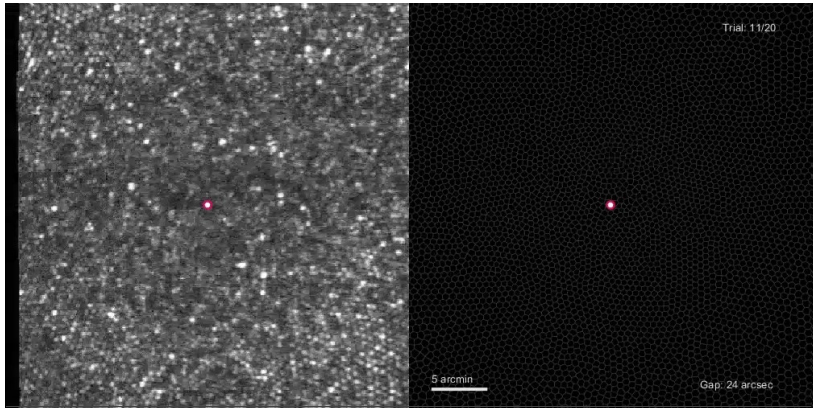
The **foveola** is the small central region that supports the finest human spatial vision. Witten and colleagues used adaptive optics, a method that corrects optical distortions, to image individual cones there. Their instrument also delivered a tiny letter-shaped target while recording where that target fell on the living retinal mosaic.

Participants identified the orientation of an E-shaped target whose size changed across trials. The researchers measured both the spacing of the cones and the target's path across them. This made it possible to compare visual performance with the physical receptor arrangement that sampled the stimulus in each eye.

Even during attempted steady fixation, the target moved across neighboring cones because the eye made small movements. These movements changed the receptor signals available during the brief presentation. The experiment therefore measured spatial vision as an active sampling process, with receptor spacing and eye motion recorded together rather than inferred from a static photograph.



Witten et al. (2024), Fig. 1, selected panels.



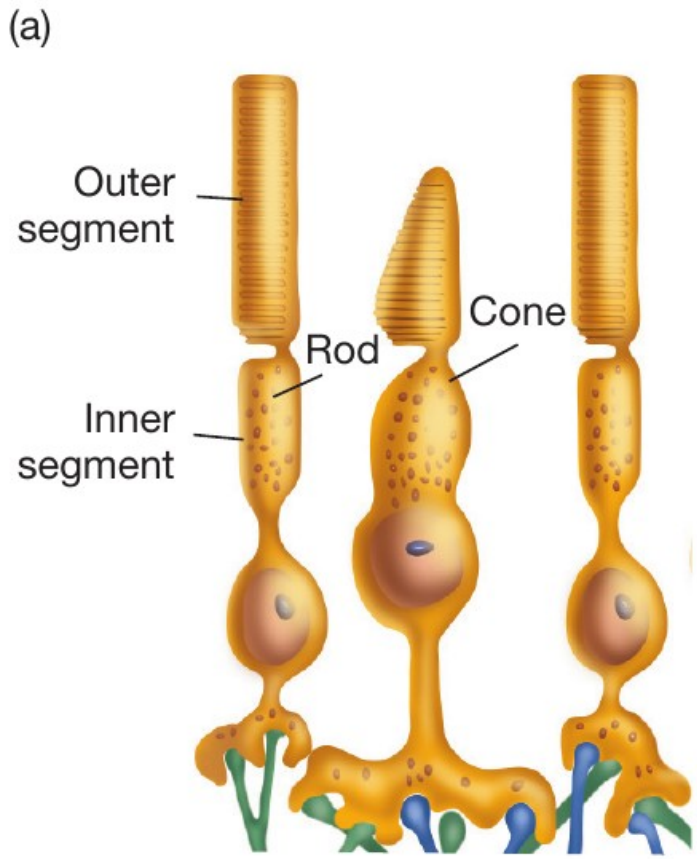
Witten et al. (2024), 98648 Video 1.

Rods and cones trade sensitivity for detail

Rods and **cones** are photoreceptors named for the shapes of their outer segments, the membrane-rich structures containing light-sensitive pigment. Rod pathways work effectively in dim conditions. Cone pathways support daylight color vision and fine detail. Their different sensitivity and wiring make the same scene appear different as illumination changes.

The central human fovea contains densely packed cones and no rods. Small retinal signals remain relatively separate there, preserving spatial detail. Away from the fovea, many receptors can contribute to a shared downstream neuron. Pooling helps detect weak light, while making the exact source within that pooled region harder to localize.

A faint star can become easier to detect when its image falls slightly away from the center of gaze. That places the light on rod-rich retina rather than the rod-free foveal center. Under these dim conditions, detecting the light and identifying its color or fine structure depend on different amounts and kinds of available information.



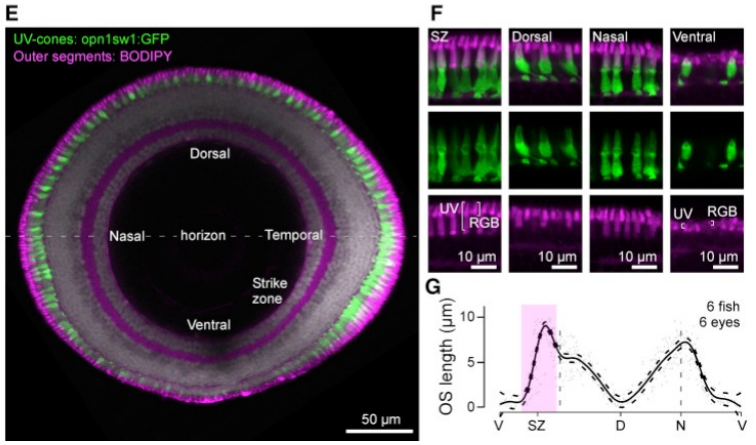
Carlson and Birkett (2022), Fig. 6.3, p. 157.

An acute retinal zone improves prey detection

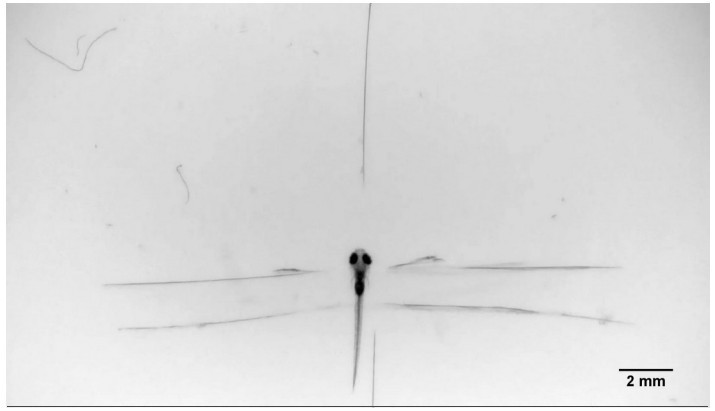
The larval zebrafish retina contains an **acute zone**, a region specialized for a behaviorally important part of visual space. Ultraviolet cones in this region view the upper frontal field where prey are captured. Their outer segments are unusually long, increasing the amount of pigment-containing membrane available to absorb incoming ultraviolet photons.

A tiny paramecium viewed from a distance can cover only one ultraviolet cone at a time. Increasing that cone's sensitivity therefore improves the chance that the prey produces a usable signal before it approaches closely. Regional specialization begins at the individual photoreceptor rather than waiting for many weak inputs to be combined in the brain.

During a hunting response, the eyes rotate inward and change where the prey falls on the two retinas. The same retinal specialization therefore samples different parts of external space as eye position changes. Receptor placement and eye movement work together to bring the target into the region with enhanced ultraviolet sensitivity.



Yoshimatsu et al. (2020), Fig. 2, selected panels.



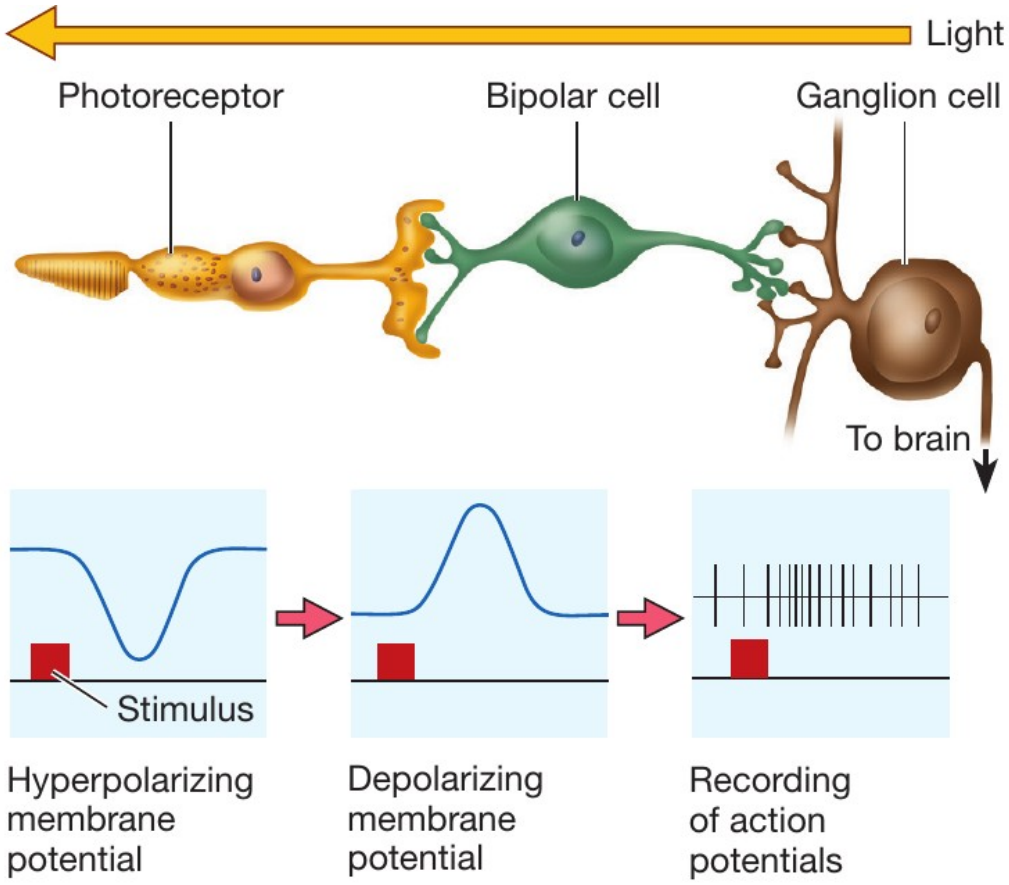
Yoshimatsu et al. (2020), Video S2.

Light reduces photoreceptor glutamate release

A **photopigment** contains an opsin protein bound to retinal, a light-sensitive molecule derived from vitamin A. Absorbing a photon changes retinal's shape and activates the opsin. In vertebrate rods and cones, the resulting chemical reactions reduce the concentration of cyclic GMP, a molecule that normally keeps certain membrane channels open.

As those channels close, inward positive current decreases and the photoreceptor **hyperpolarizes**, meaning its interior becomes more negative. This differs from a neuron becoming excited by opening an inward-current channel. The photoreceptor responds to increased light by reducing its electrical drive toward transmitter release at its synaptic ending.

In darkness, photoreceptors are relatively depolarized and release glutamate continuously. Light-induced hyperpolarization reduces calcium entry into the terminal and lowers glutamate release. Bipolar cells detect that decrease. Different bipolar-cell receptors then convert the same reduction in glutamate into either an increase or a decrease in their own electrical response.



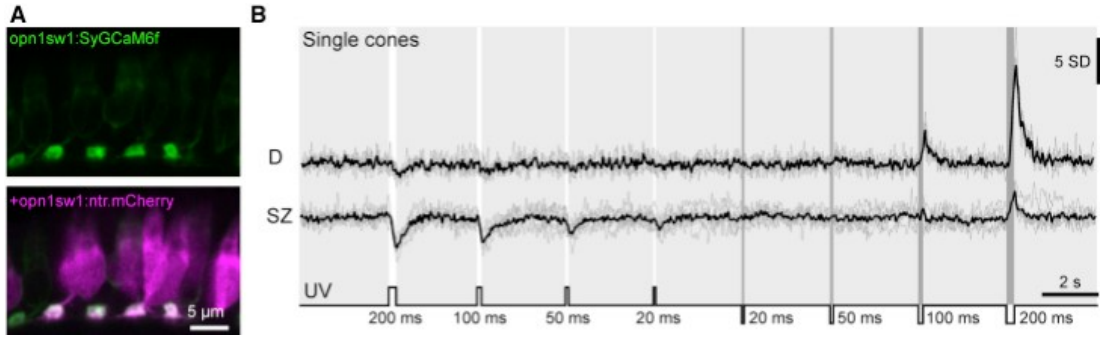
Carlson and Birkett (2022), Fig. 6.8, p. 160.

Synaptic calcium changes during a flash

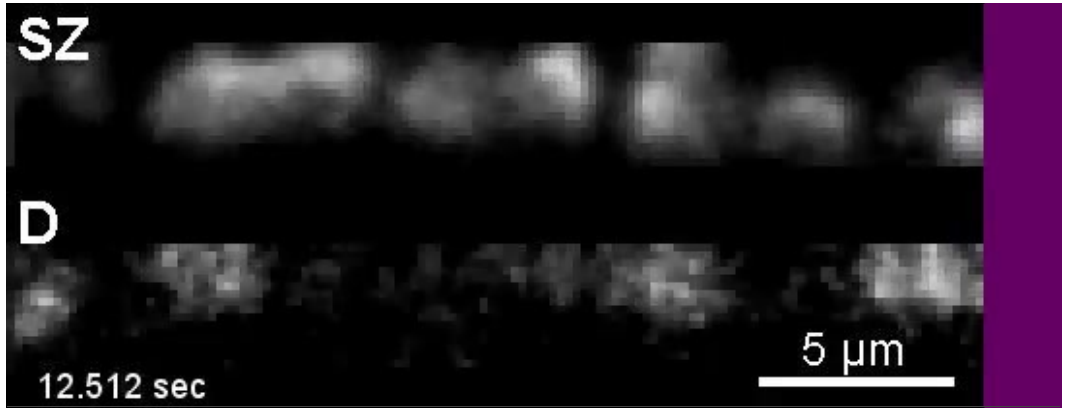
Calcium entering a photoreceptor terminal helps trigger the release of glutamate-filled vesicles. Yoshimatsu and colleagues placed a fluorescent calcium indicator in ultraviolet-cone terminals in living zebrafish. The indicator changed brightness as calcium changed, allowing the researchers to follow individual cones during brief increases and decreases in ultraviolet illumination.

Bright flashes reduced terminal calcium, while dark flashes increased it. The response depended on retinal location. Cones in the prey-capture acute zone responded strongly to brief bright signals and recovered differently from cones elsewhere. A shared ultraviolet-sensitive opsin therefore operated within cells whose electrical and synaptic behavior had been regionally adjusted.

The fluorescent measurement follows calcium near the release machinery; it does not directly count released glutamate molecules. The researchers also measured glutamate with a separate sensor. Comparing the two signals allowed them to examine how an electrical response reaching the terminal was transformed into the chemical message received by neighboring retinal cells.



Yoshimatsu et al. (2020), Fig. 3, selected panels.



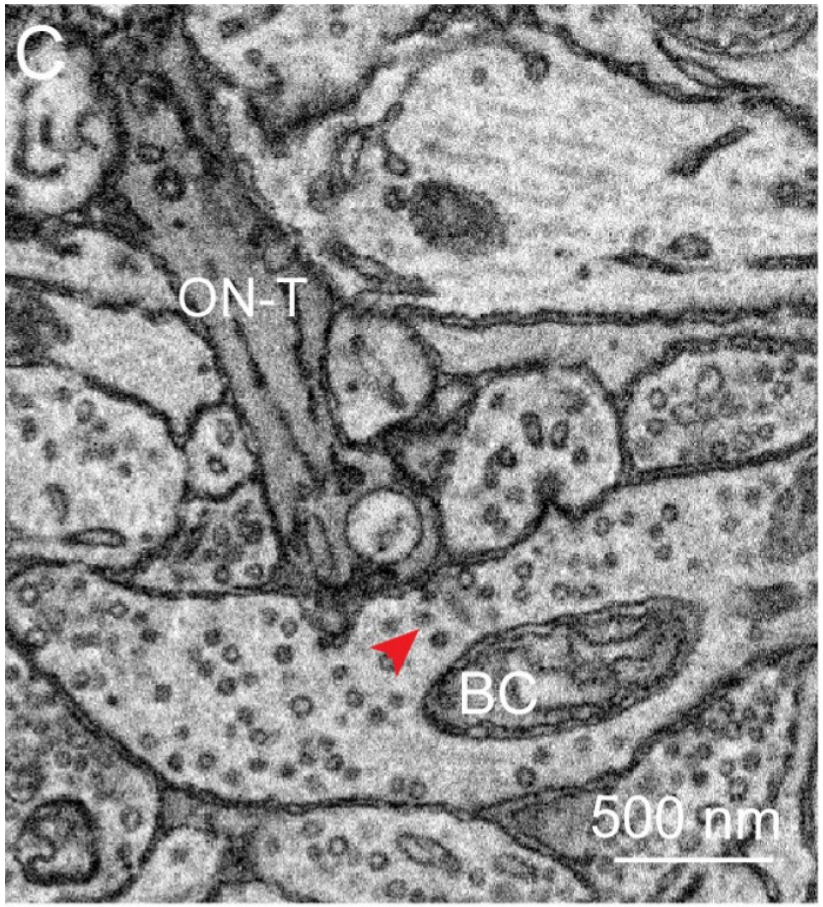
Yoshimatsu et al. (2020), Video S3.

Ribbon synapses support ongoing retinal signaling

Photoreceptors and bipolar cells release glutamate at specialized **ribbon synapses**. A ribbon is a protein structure that holds vesicles near the release site, helping organize transmitter delivery during continuing changes in membrane potential. The synapse can therefore convey both a sudden change and information that persists while illumination remains elevated.

In the mouse retina, transient and sustained ON ganglion cells receive input from different combinations of bipolar cells. The investigators reconstructed these connections with electron microscopy, which resolves the membranes and contact sites of individual cells. A visible ribbon at a bipolar terminal helped identify an excitatory contact onto a ganglion-cell dendrite.

The number and arrangement of vesicles near the ribbon can influence how long release continues after a stimulus begins. Differences in these vesicle pools were associated with the distinct output pathways. The study located an important transformation at the bipolar-cell terminal, where similar incoming voltage responses could produce differently timed glutamate signals.



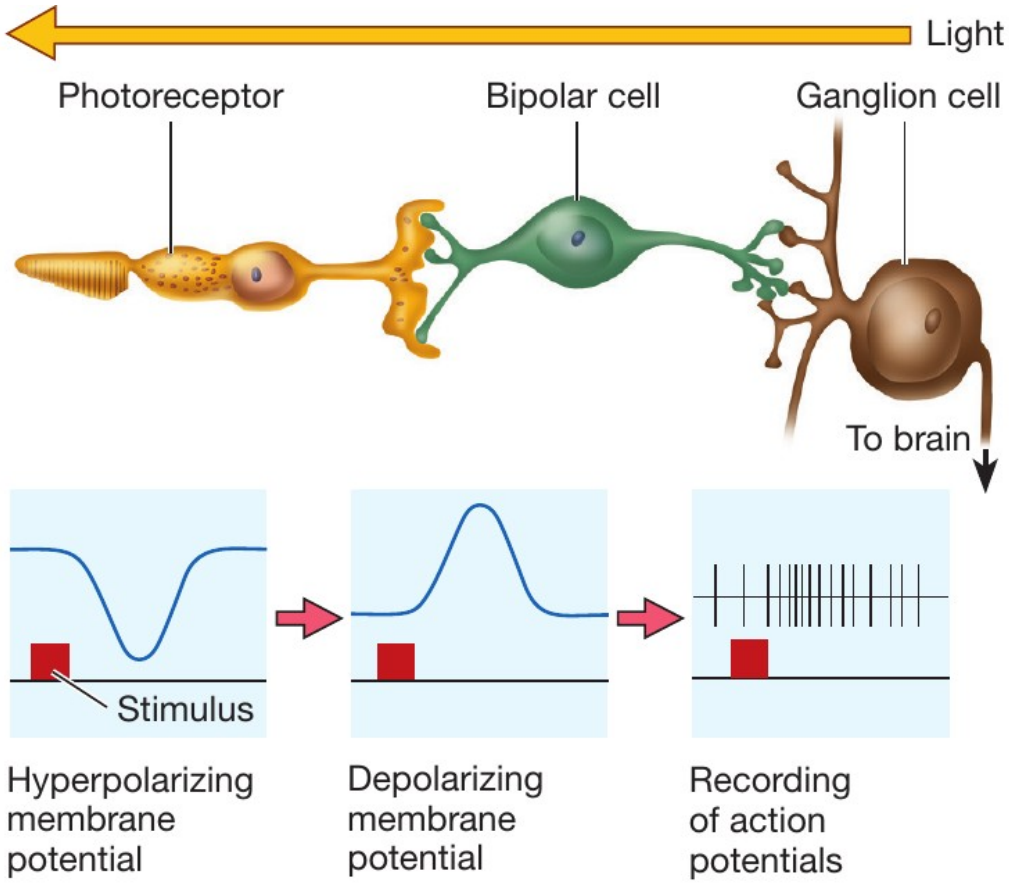
Kuo et al. (2025), Fig. 3, selected panels.

ON and OFF pathways separate light changes

An **ON bipolar cell** becomes more active when light increases in its receptive-field center. In darkness, glutamate from the photoreceptor activates a receptor pathway that keeps a depolarizing channel closed. When light reduces glutamate release, that suppression is relieved and the bipolar cell depolarizes, increasing its output to downstream neurons.

An **OFF bipolar cell** uses a different glutamate receptor. Glutamate directly opens an excitatory channel, so reduced glutamate release during light makes the cell less active. A decrease in light increases photoreceptor release and excites the OFF pathway. The same photoreceptor signal can therefore be separated into complementary messages about brightening and darkening.

Ganglion cells convert these graded retinal signals into action potentials that travel along the optic nerve. An ON ganglion cell can signal a bright edge entering its center, while an OFF cell can signal a dark edge. Both pathways can use increased firing to announce their preferred kind of change.



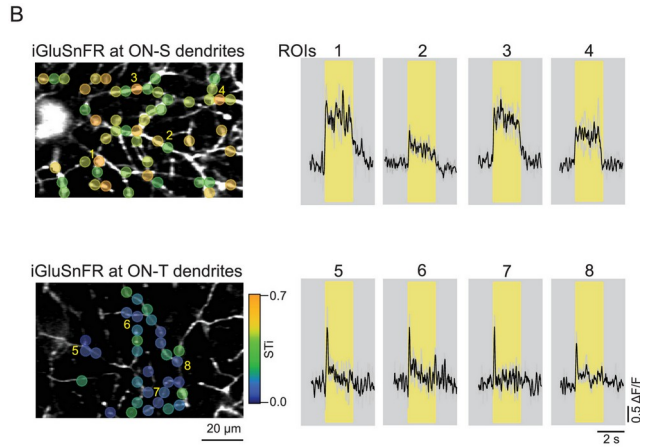
Carlson and Birkett (2022), Fig. 6.8, p. 160.

Glutamate release shapes response duration

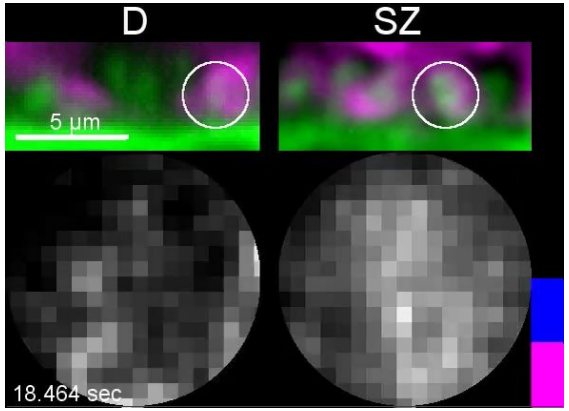
A **transient** response is strongest near a change in illumination and then declines. A **sustained** response remains elevated while the stimulus continues. In the mouse retina, different ON ganglion cells display these two patterns even when they receive the same light stimulus, providing separate signals about change and continuing illumination.

The investigators measured light-driven bipolar-cell voltage and glutamate near ganglion-cell dendrites. Several presynaptic bipolar types had similar voltage responses, but the glutamate arriving at transient and sustained ganglion cells had different time courses. The difference therefore emerged during transmission through the bipolar output synapses rather than simply copying two different voltage inputs.

Repeated flashes also affected the pathways differently. Excitatory currents reaching transient ganglion cells were more strongly reduced by a closely following flash. This short-term change in synaptic transmission helps explain why one pathway emphasizes a new event while another continues carrying information about a stimulus that remains present.



Kuo et al. (2025), Fig. 6, selected panels.



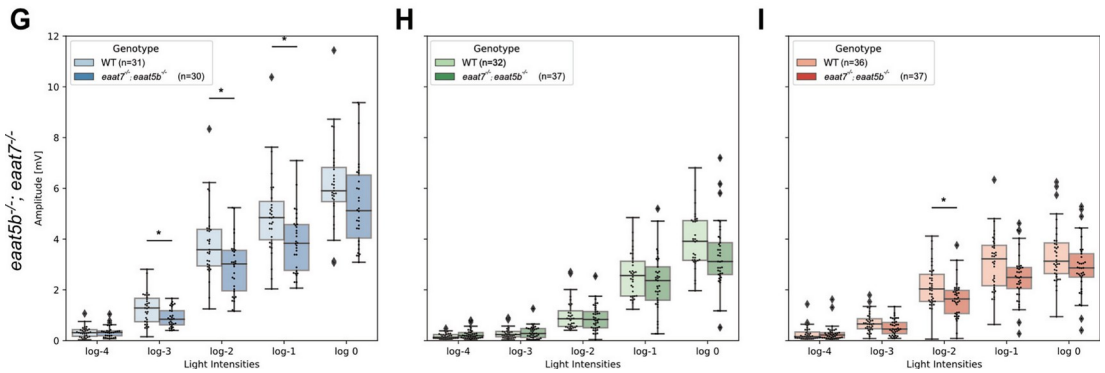
Yoshimatsu et al. (2020), Video S7.

Retinal transporters alter wavelength-specific signals

An **excitatory amino acid transporter**, or EAAT, is a membrane protein that handles glutamate. Some retinal EAAT proteins also conduct chloride ions and participate in the electrical response to glutamate. In zebrafish ON pathways, these transporter-associated currents operate alongside metabotropic glutamate receptors at the first retinal synapse.

The researchers removed the genes for EAAT5b, EAAT7, or both and recorded an **electroretinogram**, an electrical response summed across retinal tissue. Removing both transporters reduced responses to ultraviolet-blue and red illumination while leaving the tested green-light response relatively preserved. The disruption therefore depended on which wavelength channel was stimulated.

The proteins also had different distributions across the retina, including the region specialized for prey capture. A moving ultraviolet target evoked altered hunting responses in transporter mutants. Transporter-associated chloride currents help transmit cone signals into ON bipolar cells, so disrupting these proteins changes the retinal message available for detecting a hunting target.



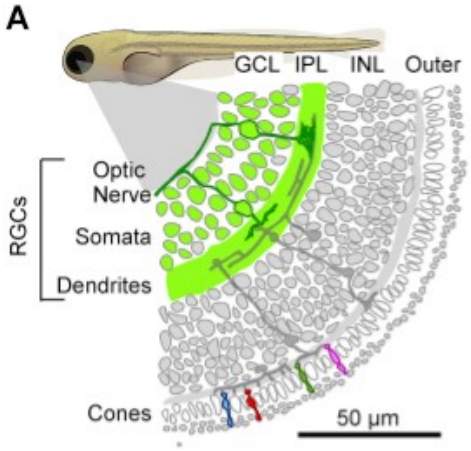
Garbelli et al. (2025), Fig. 1, selected panels.

Ganglion-cell activity can be followed in the live eye

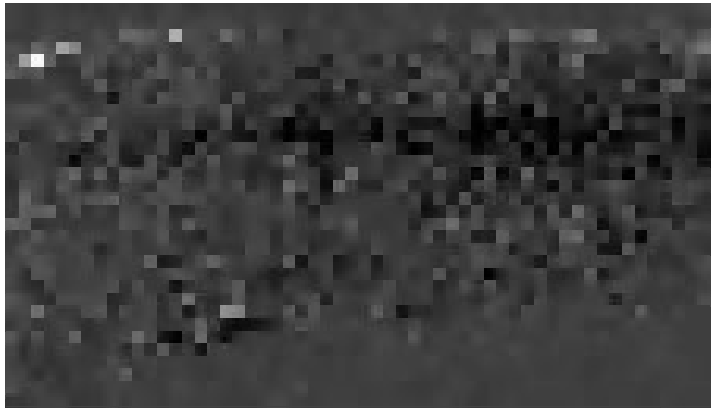
Retinal ganglion cells receive processed signals from bipolar and amacrine cells and send their axons toward the brain. Zhou, Bear, and colleagues expressed a calcium indicator in zebrafish ganglion cells. Their microscope followed activity in both the cells' branching dendrites and the cell bodies during controlled visual stimulation.

The retina curves around the eye, so a straight scan would pass through different tissue layers. The researchers instead scanned along the curved ganglion-cell and dendritic layers. This kept the measured structures in view and allowed responses from different parts of the retina to be compared within a common anatomical arrangement.

Red, green, blue, and ultraviolet stimulus components changed independently. A ganglion cell could increase its calcium signal after one component brightened and after another darkened. Measuring these combinations revealed which spectral changes each output cell emphasized before its axon carried the processed message into the brain.



Zhou et al. (2020), Fig. 1, selected panels.



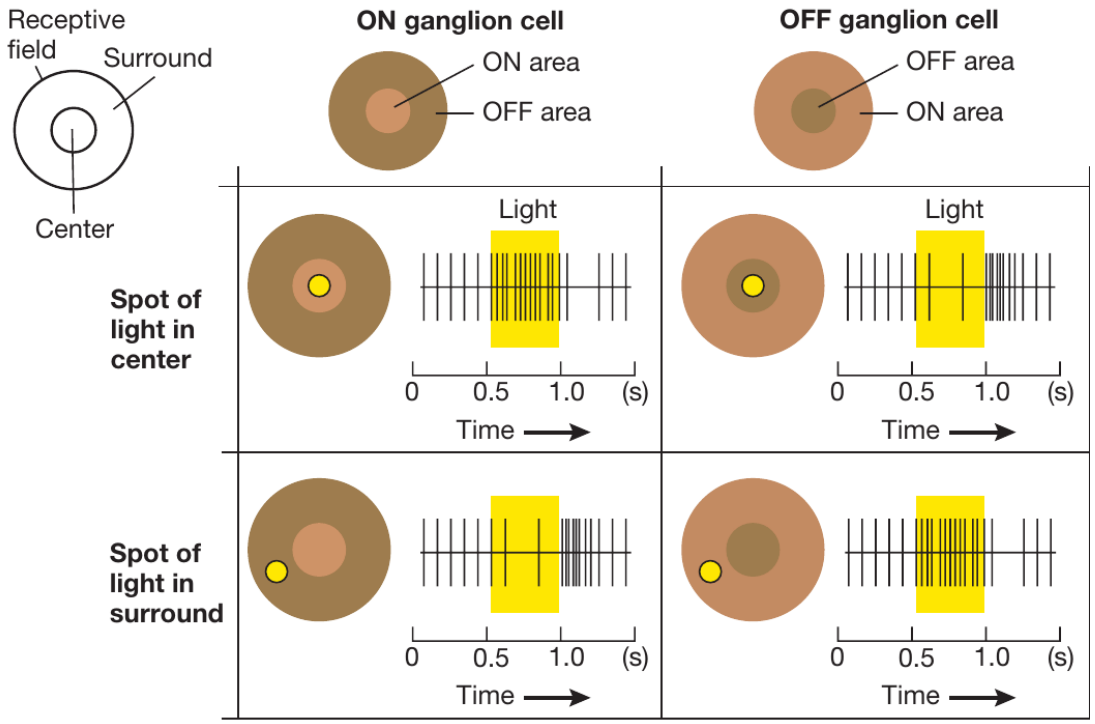
Zhou et al. (2020), Video S1.

Center and surround emphasize local contrast

A neuron's **receptive field** is the region of visual space in which a stimulus changes its activity. Many retinal ganglion cells have a central region and an antagonistic surround. Light that excites an ON center can reduce the same cell's firing when it falls in the surrounding ring.

Horizontal cells combine signals across neighboring photoreceptors, while other retinal connections further shape center and surround responses. Because the two regions oppose one another, illuminating both can produce less change than illuminating the center alone. The output depends on a local brightness difference rather than simply the total number of photons reaching the eye.

An edge places different light levels across neighboring receptive fields. Cells whose centers lie on one side and surrounds extend across the boundary can respond strongly. Uniform illumination produces more cancellation between center and surround. This arrangement emphasizes boundaries that help separate objects from backgrounds and makes surrounding brightness influence the appearance of a patch.



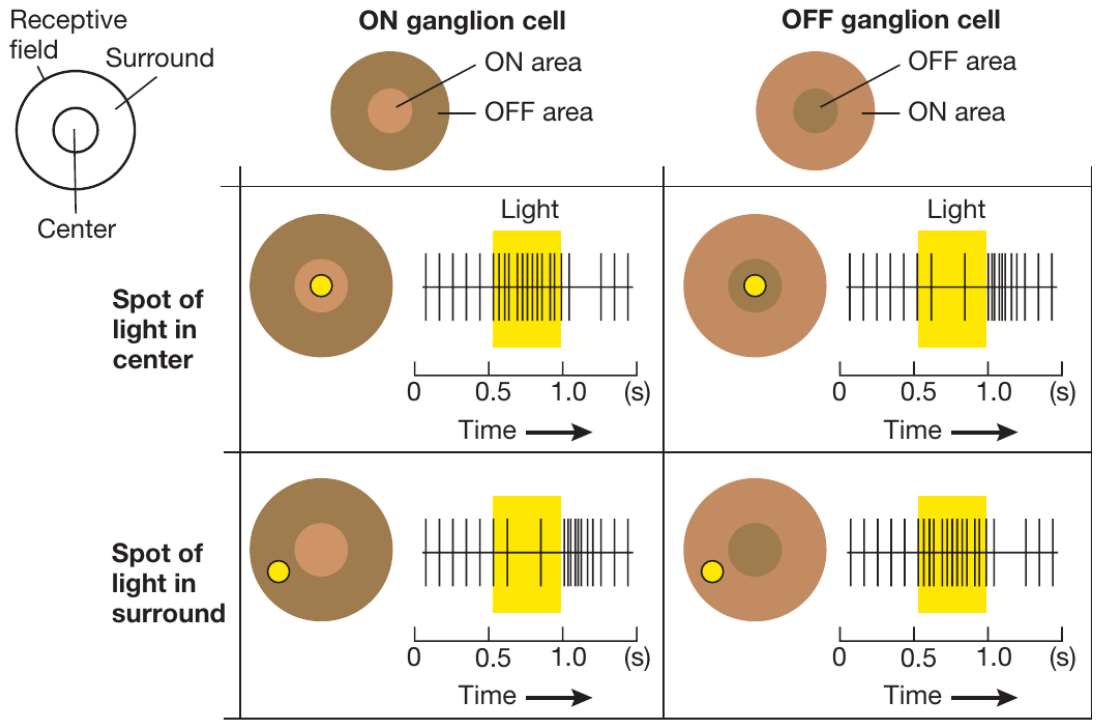
Carlson and Birkett (2022), Fig. 6.15, p. 168.

Surrounding light changes apparent brightness

Two patches reflecting the same amount of light can look different when their surroundings differ. A patch surrounded by darkness often appears brighter than the same patch surrounded by a bright region. **Brightness contrast** depends on comparisons between neighboring image regions rather than only the amount of light reaching one small retinal location.

Center-surround receptive fields provide one source of such comparisons. A bright surround can oppose the activity driven by an ON center, while a darker surround produces less opposition. Different surrounding stimulation can therefore change a cell's output even when the physical light falling in its center remains constant.

Perceived brightness also depends on larger-scale cortical interpretations of surfaces and illumination. A visible boundary may represent a change in paint, a shadow, or a change in surface orientation. Local retinal comparisons provide contrast information, while additional processing helps determine how that contrast belongs to the objects and lighting of the scene.



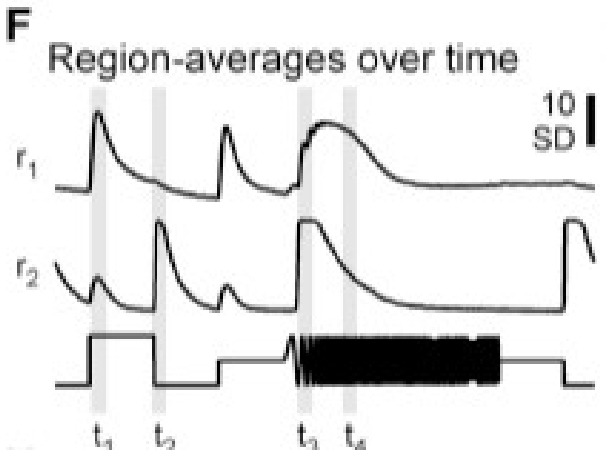
Carlson and Birkett (2022), Fig. 6.15, p. 168.

Retinal output separates onset from continuing light

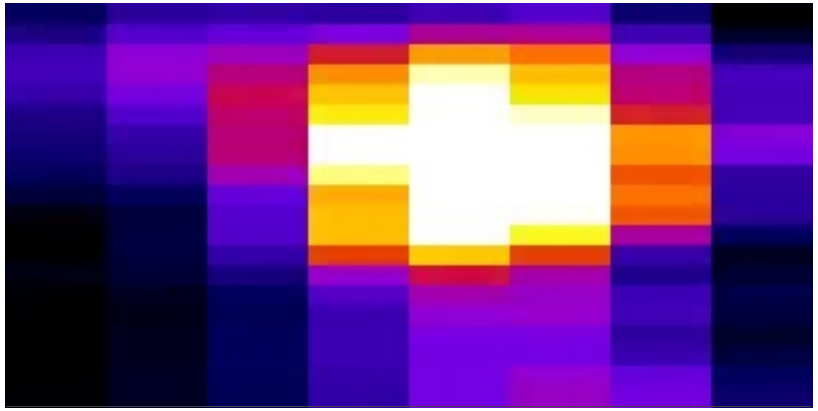
Ganglion-cell dendrites occupy different depths within the **inner plexiform layer**, the retinal layer where bipolar cells contact ganglion cells. Some dendritic regions respond strongly when light begins, while others respond when it ends. Mapping activity by retinal position and layer depth reveals how these complementary pathways are arranged within the living eye.

The researchers presented a step of light followed by changing flicker. Different ganglion-cell populations emphasized the onset, offset, or rapidly varying part of the stimulus. A light remaining present and a sequence of light changes could therefore generate different output patterns even when they stimulated the same broad retinal region.

These response patterns were not distributed uniformly around the eye. Retinal regions viewing different parts of the world contained different mixtures of temporal and spectral signals. The optic nerve consequently carries multiple specialized descriptions of a scene, including where a change happened, which wavelengths changed, and how quickly the change developed.



Zhou et al. (2020), Fig. 2, selected panels.



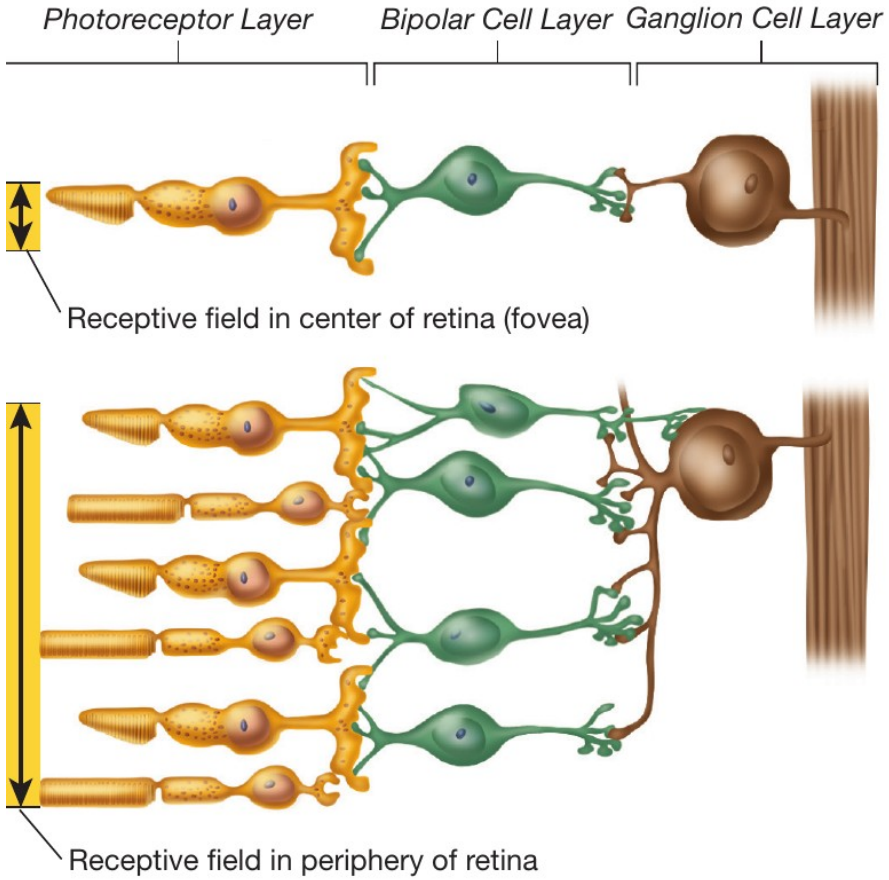
Zhou et al. (2020), Video S2.

Pooling changes sensitivity and spatial resolution

Convergence occurs when several neurons provide input to a shared downstream cell. In peripheral retina, many photoreceptors can contribute to one ganglion-cell output. A weak stimulus activating several receptors can then produce a combined response large enough to be transmitted, improving sensitivity when individual receptor signals are small.

The same pooling reduces spatial precision. If two nearby points activate receptors feeding the same output cell, that output alone cannot identify which receptor received each point. Separate signals from densely packed central cones preserve smaller differences in position and support the fine distinctions needed for reading or examining an object's surface.

Eye movements allow these different retinal regions to work together. Peripheral vision can detect an event that attracts attention, and a rapid gaze shift can bring it onto central retina for detailed inspection. The first detection and the later identification can therefore rely on different receptive-field sizes and different patterns of retinal convergence.



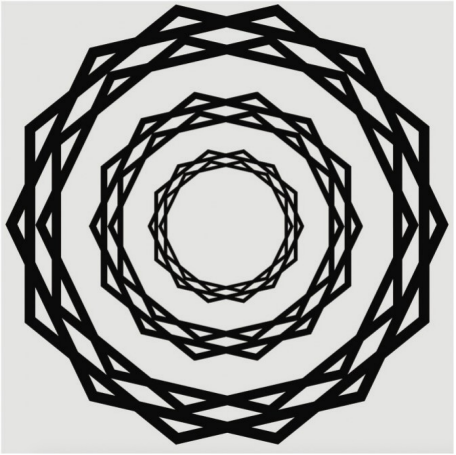
Carlson and Birkett (2022), Fig. 6.9, p. 160.

A static starburst can appear to emit bright rays

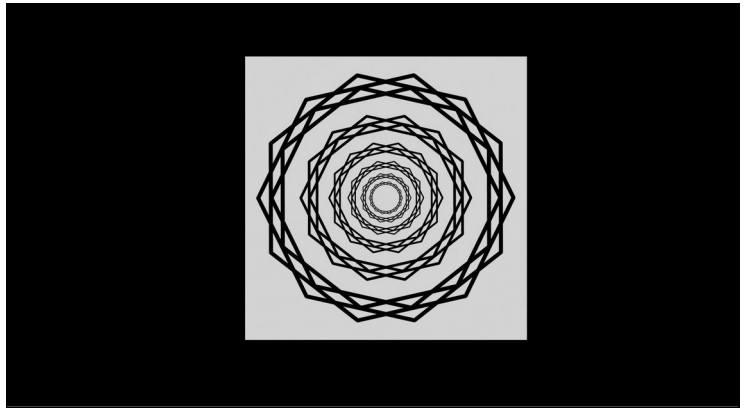
The **Scintillating Starburst** is an original visual stimulus built from overlapping concentric star polygons. Many observers perceive fleeting bright rays extending outward from its center. Those extended rays are not drawn as continuous bright lines in the physical image; their appearance changes as the observer inspects the pattern.

Karlovich and Wallisch tested how the arrangement of the polygons affected the reported rays. Contrast, the number of repeated rings, and the pattern's intersections changed the perceptual effect. The experiment therefore related a specific visual experience to controlled changes in an existing image rather than merely presenting an unusual decorative pattern.

The authors proposed an interaction between central and peripheral visual processing. Intersections provide local signals that can be interpreted differently when sampled away from fixation, contributing to apparent extended brightness. This explanation concerns how the pattern may be processed; the directly measured result was the observers' reported strength of the illusory rays.



Karlovich et al. (2021), Fig. 1.



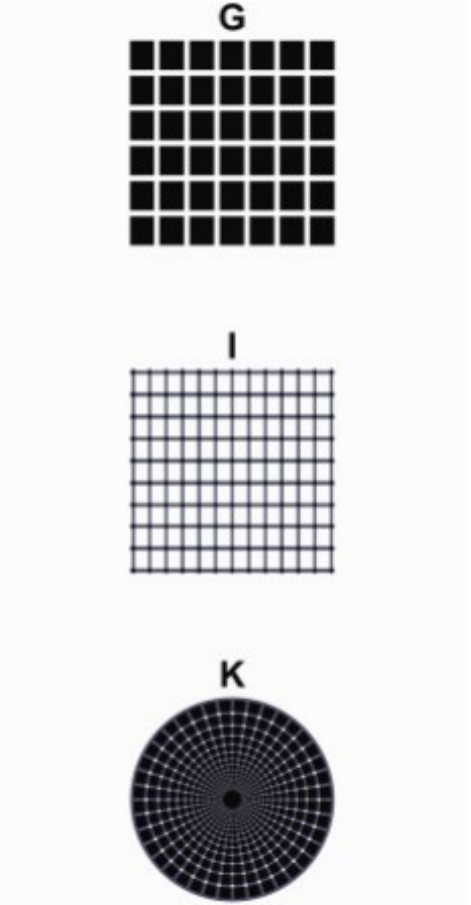
Karlovich et al. (2021), Scintillating Starburst, original Movie 2.

Grid patterns expose interactions across visual space

In grid illusions, brightness can appear at locations where no corresponding spot or line was printed. The original starburst study compared its stimulus with several grid patterns, including the **pincushion grid**. Their arrangements differ, but each places repeated intersections and neighboring contours across central and peripheral visual space.

Moving fixation changes which intersections fall on central retina and which fall on more broadly pooled peripheral regions. Apparent features can strengthen, weaken, or disappear during inspection. The physical pattern remains the same while the retinal sampling and the balance of surrounding signals change with each new gaze position.

The authors compared the starburst and grid patterns to distinguish their properties rather than treating all of them as one illusion. Similar-looking percepts need not have identical causes. Testing how contrast and geometry change each effect provides more specific evidence than assuming that any apparent brightness at an intersection must follow one simple retinal explanation.



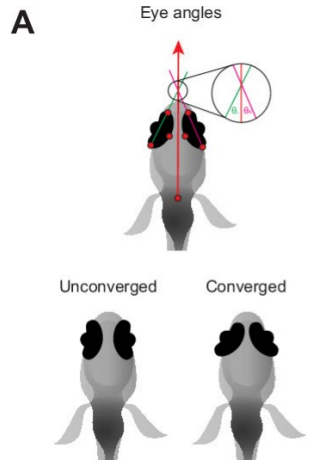
Karlovich et al. (2021), Fig. 3, selected panels.

Eye convergence moves prey into shared visual space

Binocular overlap is the part of the environment visible to both eyes. Mearns and colleagues compared hunting in several species of larval fish using high-speed recordings of eye position, tail shape, and prey location. In the cichlid species, the eyes rotated inward during hunting, expanding the shared view in front of the head.

The researchers tracked each eye relative to the animal's body axis rather than measuring only the animal's direction of travel. This separated eye convergence from a body turn toward the prey. During capture, cichlids tended to position prey centrally in the frontal region while coordinating eye and body movements.

Different species used different combinations of approach bouts and capture strikes. Their eye positions therefore had to be interpreted together with the prey's changing location. An inward eye rotation changes the two retinal images available for targeting, while the body movement changes the distance that the final strike must cover.



Mearns et al. (2025), Fig. 2, selected panels.



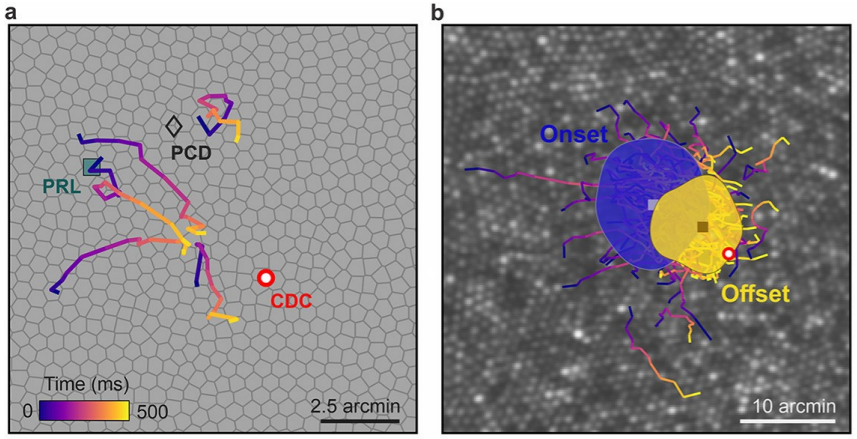
Mearns et al. (2025), 98347 Fig. 1 Video 1.

Small retinal drifts improve visual sampling

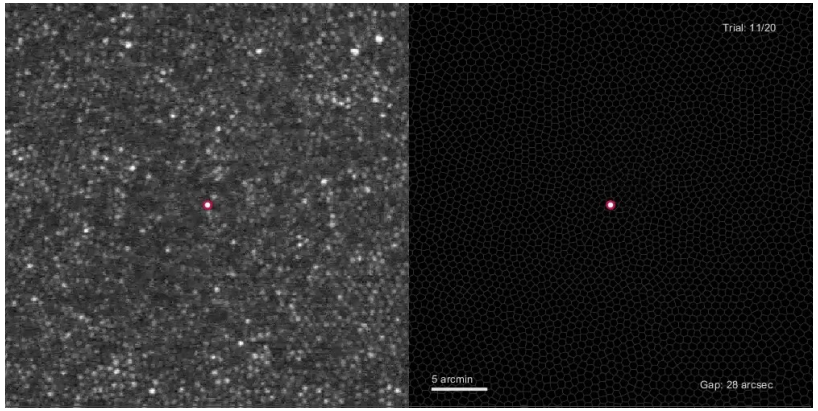
During fixation, the human eye makes slow movements called **ocular drift** in addition to occasional rapid microsaccades. In the cone-resolved experiment, drift moved the tiny target across the receptor mosaic during its presentation. The researchers excluded trials with larger interruptions so they could examine the contribution of these small continuous movements.

The target's path tended to carry it toward retinal regions with denser cone sampling. Resolution was better than predicted from a model that treated the receptor array as completely stationary. Successive receptor measurements supplied changing information about the target's edges during the time available for the observer's orientation judgment.

The direction and extent of drift differed among eyes, and their contributions depended on the local cone arrangement. A target crossing a densely sampled region could generate more useful spatial comparisons than one remaining over a less favorable location. Fine human vision therefore depends partly on how the eye moves across its own receptor mosaic.



Witten et al. (2024), Fig. 4, selected panels.



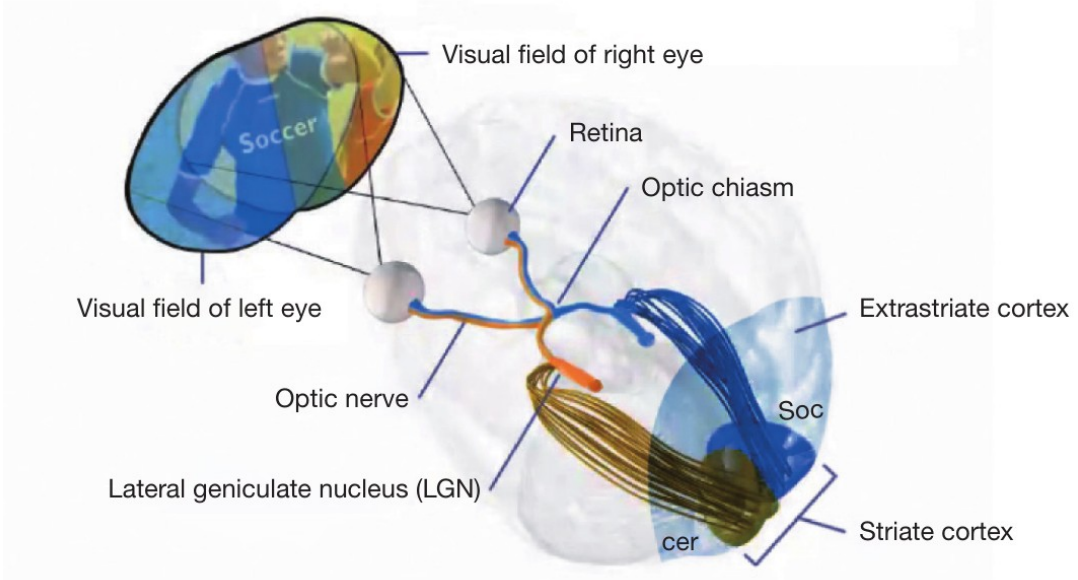
Witten et al. (2024), 98648 Video 2.

Each hemisphere receives the opposite visual field

The two optic nerves meet at the **optic chiasm**. Axons from the nasal half of each retina cross to the opposite side, while axons from the temporal half remain on the same side. Because the optics reverse the image, this partial crossing brings information from the same half of external space into one hemisphere.

The left visual field falls on the nasal retina of the left eye and the temporal retina of the right eye. Signals from both reach the right hemisphere. The right visual field reaches the left hemisphere through the complementary arrangement. Each hemisphere therefore receives information from both eyes rather than receiving an entire opposite eye.

Damage at different points produces different patterns of visual-field loss. An optic-nerve lesion removes input from one eye, while a lesion beyond the chiasm can remove corresponding portions of both eyes' visual fields. The arrangement of surviving vision can help identify which part of the pathway has been interrupted.



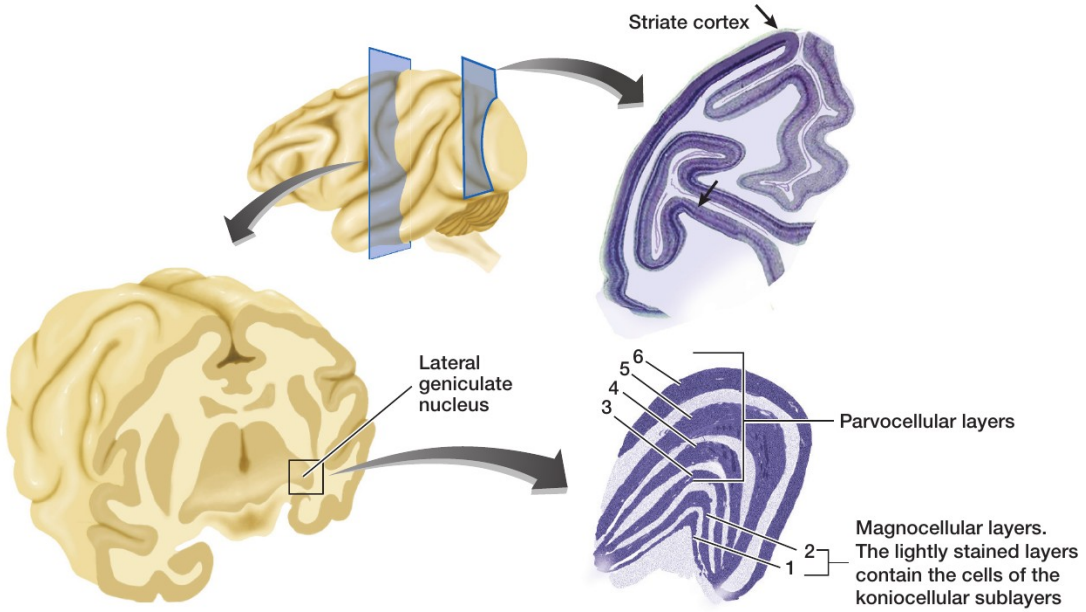
Carlson and Birkett (2022), Fig. 6.10, p. 161.

The LGN preserves eye and signal differences

The **lateral geniculate nucleus**, or LGN, is a thalamic relay between the retina and primary visual cortex. In primates, its main layers receive input from one eye at a time. Neighboring relay cells preserve an orderly relationship between locations in visual space, carrying a spatially organized signal toward the occipital cortex.

The **magnocellular** layers emphasize rapid changes and light-dark contrast, while **parvocellular** pathways preserve finer spatial detail and important cone comparisons. **Koniocellular** populations contribute additional signals, including short-wavelength cone information. These names describe distinct cell populations and routes, rather than three completely isolated versions of the visual world.

LGN neurons project to different layers and compartments of primary visual cortex. Cortical cells can then combine inputs that began with separate eyes or different retinal response properties. The pathway preserves useful distinctions during transmission while providing the connections needed for binocular and more complex visual responses to emerge.



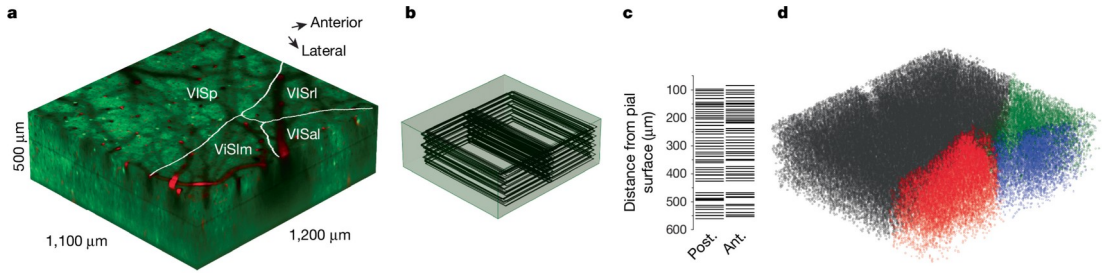
Carlson and Birkett (2022), Fig. 6.11, p. 164.

Visual responses can be matched to individual synapses

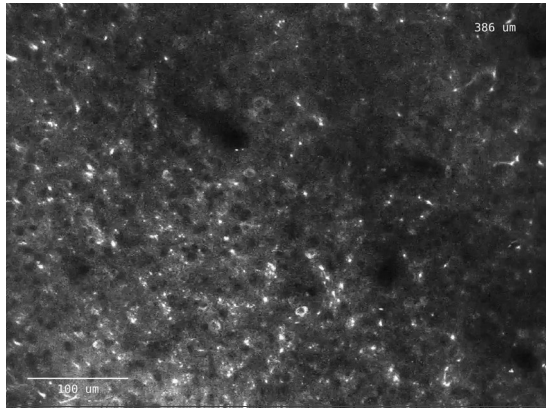
The **MICrONS** project combined recordings of mouse visual-cortex activity with a detailed reconstruction of the same tissue. The living animal viewed visual stimuli while calcium imaging measured responses from many neurons. Later, electron microscopy resolved the cells' branches and the synapses through which they contacted one another.

Matching the living and fixed-tissue images required recognizable anatomical landmarks and careful alignment. Once a recorded cell was identified in the reconstruction, its axon and dendrites could be traced through the volume. This linked a measured visual response to the physical input and output connections of that particular neuron.

The dataset includes primary visual cortex and neighboring higher visual areas. Researchers can therefore ask whether cells responding to similar visual features actually contact each other within an area or across area boundaries. The accompanying optical recording depicts visual-cortex imaging; the study's matched reconstruction supplies the evidence about individual synaptic contacts.



Bae et al. (2025), Fig. 3, selected panels.



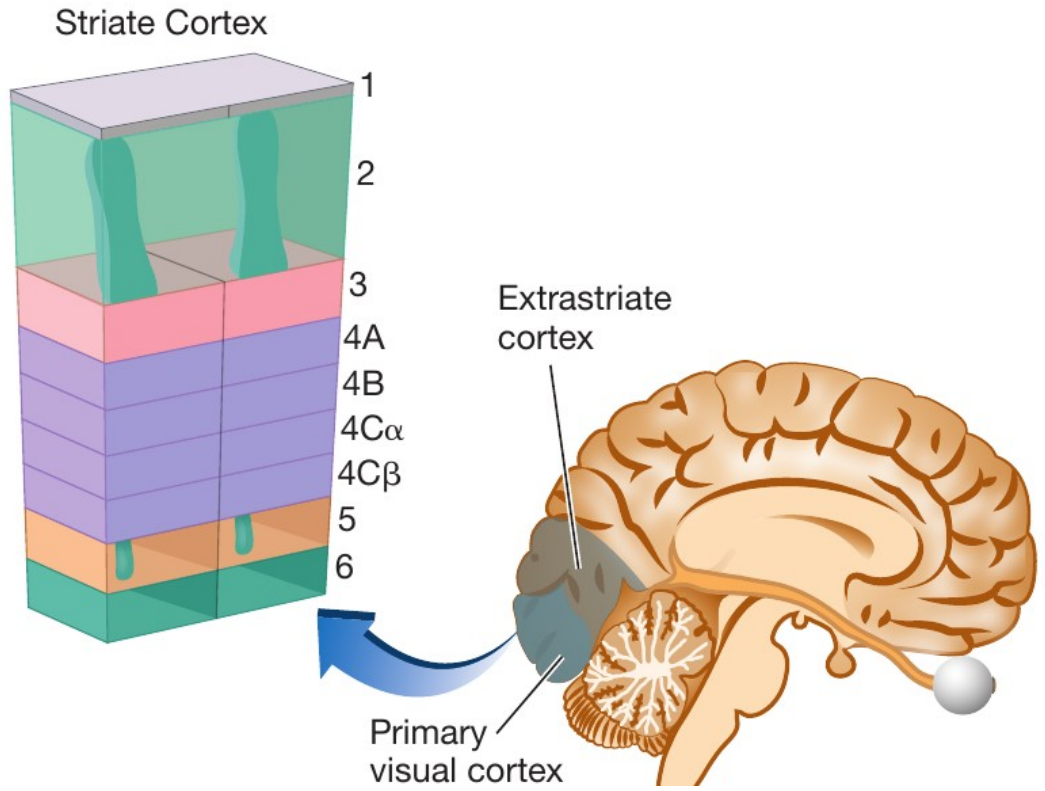
Sofroniew et al. (2016), normal mouse visual cortex, Movie 1.

V1 maps visual position and combines inputs

Primary visual cortex, or V1, lies in the occipital lobe and receives a major projection from the LGN. Its **retinotopic map** preserves neighborhood relationships in visual space. Nearby cortical locations generally process nearby parts of the visual field, although the amount of cortical tissue devoted to each region is unequal.

Central vision occupies a disproportionately large cortical representation because it supplies densely sampled spatial information. A small region near fixation can therefore recruit more cortical tissue than an equally sized peripheral region. This **cortical magnification** reflects the high-resolution input provided by central retina and the processing devoted to it.

Inputs arriving in cortical layers are combined through local connections and passed to other layers and visual areas. Many cortical neurons respond to stimulation through either eye, unlike the eye-specific inputs in the principal primate LGN layers. Combining those signals makes comparisons between the two retinal images possible within the cortex.



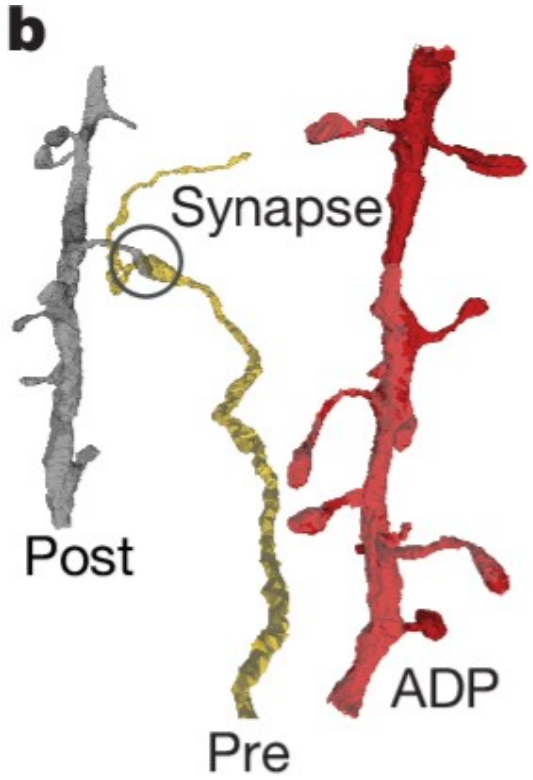
Carlson and Birkett (2022), Fig. 6.12, p. 164.

Similar visual tuning predicts cortical connections

Two neurons have similar **visual tuning** when the same kinds of visual changes tend to activate them. Using the MICrONS dataset, the investigators compared that functional similarity with actual reconstructed synapses. Neurons with more similar responses were more likely to be connected, both within visual areas and across area boundaries.

Nearby axons and dendrites have more opportunities to meet, so physical proximity had to be separated from selective contact formation. The analysis compared connected neurons with nearby alternatives whose branches also passed close together. Similar feature preferences still predicted finer details of synaptic connectivity beyond the opportunity created by neighboring branches.

A computational model separated which visual features activated a neuron from where its receptive field lay. Feature similarity predicted synaptic organization more strongly than receptive-field location alone. A visual circuit therefore contains selective connections among cells with related responses, rather than relying only on the fact that their branches occupy the same tissue.



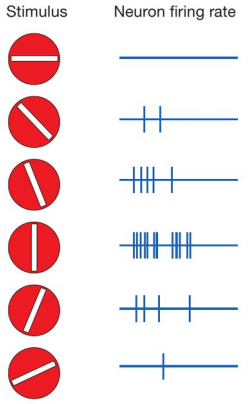
Ding et al. (2025), Fig. 2, selected panels.

Oriented edges activate selected cortical neurons

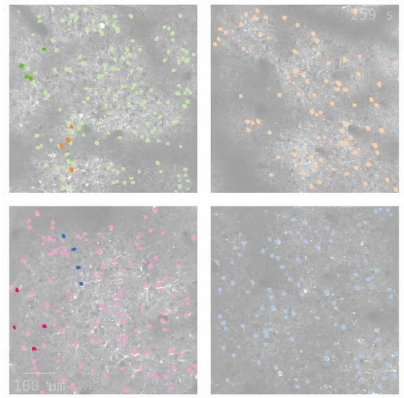
An **orientation-selective neuron** fires more strongly for an edge or bar at one angle than for the same feature at other angles. This can arise when several inputs with appropriately aligned receptive fields converge. An elongated light pattern can then activate the combined input more effectively than a small isolated spot.

A **simple cell** has spatially separated regions where light increases or decreases its firing. Both the edge's orientation and its precise position within the receptive field affect the response. A **complex cell** can preserve an orientation preference across a wider range of positions and often responds well as the preferred feature moves.

Different cells prefer different angles, so a contour activates a pattern across the population. The brain can use that population response to describe the edge's orientation while other neurons signal its location, color, and movement. Recognizing a complete object requires combining these local measurements across a much larger region of visual space.



Carlson and Birkett (2022), Fig. 6.36, p. 185.



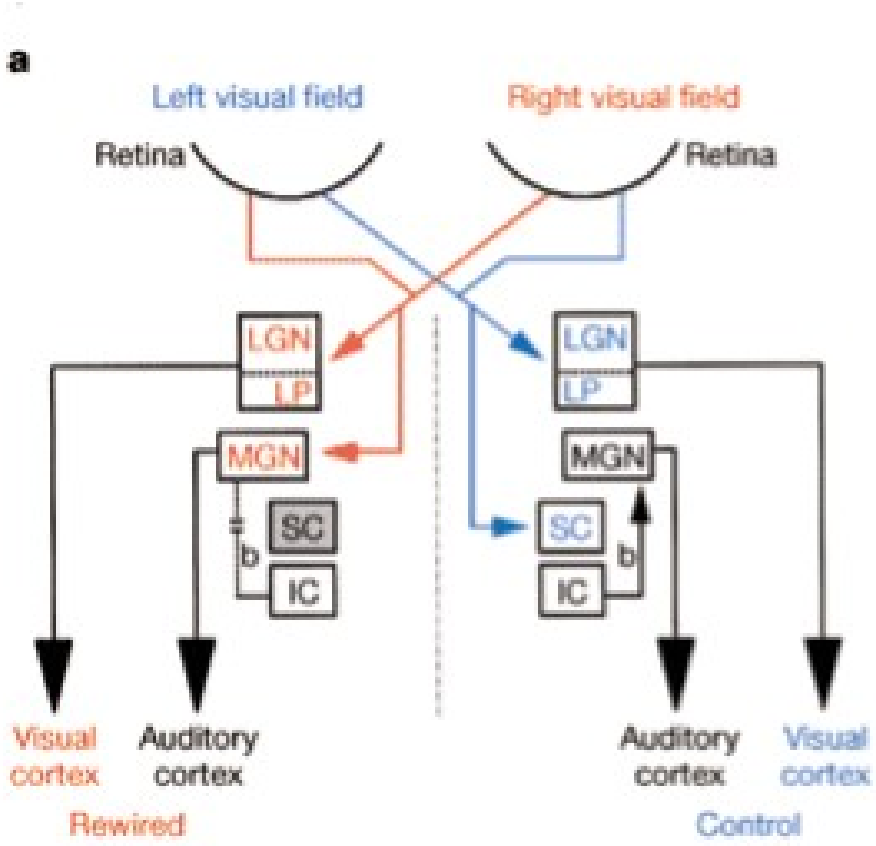
Sofroniew et al. (2016), normal mouse visual cortex, Movie 3.

Early retinal input can enter an auditory pathway

Normally, retinal axons project to visual targets such as the LGN, while ascending auditory input reaches the **medial geniculate nucleus**, or MGN. The MGN sends signals to primary auditory cortex, called A1. In the rewired-ferret preparation, this normally auditory thalamus-to-cortex connection remained available while its incoming input was experimentally altered.

Neonatal manipulations reduced the normal retinal targets and removed major auditory input to the MGN. Developing retinal axons then innervated the deprived auditory thalamus. The resulting route carried signals from the retina through the MGN to A1, allowing a cortical area normally associated with hearing to receive patterned visual input.

The intervention occurred while the pathways were developing, not after an adult cortex had been fully formed. Researchers later recorded visual responses in the altered pathway. The comparison therefore examined how early incoming activity helped shape a cortical circuit during development, rather than whether an adult ferret could learn to hear light through its ears.



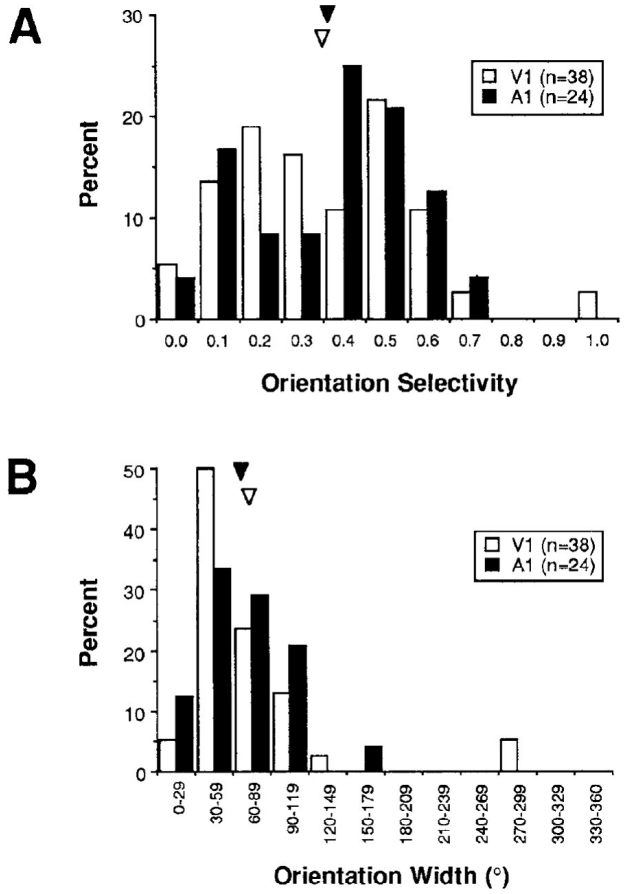
von Melchner et al. (2000), Fig. 1, selected panels.

Rewired auditory neurons acquired visual selectivity

Roe and colleagues recorded individual neurons in auditory cortex after retinal input had been redirected through auditory thalamus. Moving light bars within particular parts of the visual field changed these cells' firing. The neurons therefore had **visual receptive fields**, despite being located in a cortical region that normally processed auditory input.

Some rewired A1 neurons preferred particular bar orientations or movement directions. Simple and complex receptive-field organizations were also present. These properties resembled features measured in normal visual cortex, indicating that organized visual selectivity could develop within cortical tissue receiving a different kind of early sensory input.

The responses were not identical to those in normal V1. Rewired A1 cells tended to have larger visual receptive fields, weaker visual responses, and longer electrically evoked input latencies. Early input changed important response properties while the altered pathway retained measurable differences in the precision and timing of its visual processing.



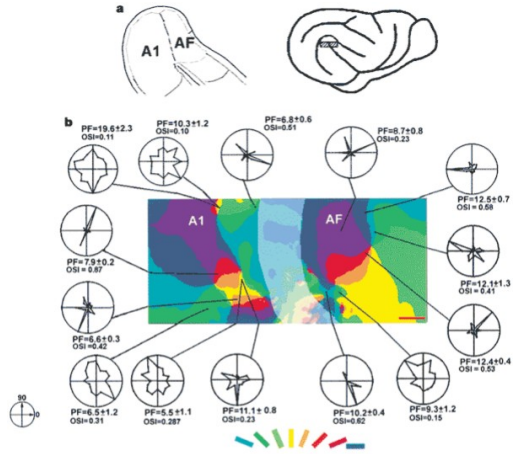
Roe et al. (1992), Fig. 8.

Visual orientation modules formed in auditory cortex

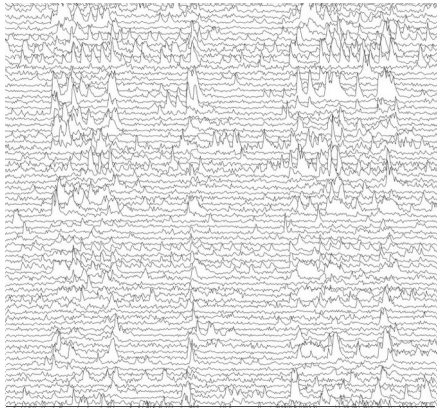
An **orientation module** contains neighboring cortical neurons with similar preferred edge angles. Sharma and colleagues measured the spatial organization of visual responses in rewired ferret A1. Regions preferring different orientations formed an organized map, extending the comparison beyond the response properties of one isolated neuron at a time.

Individual neurons within these regions had orientation tuning comparable to that of cells in normal visual cortex, but the overall map was less orderly. Long-range horizontal connections in rewired A1 also became more patchy and periodic than those in normal auditory cortex, although less so than the connections measured in V1.

Incoming visual activity therefore affected both local response selectivity and the organization of connections across cortex. The auditory cortical location did not prevent visual modules from developing. At the same time, the less orderly map and intermediate connection pattern preserved evidence that altered input had acted on tissue with its own developmental and anatomical constraints.



Sharma et al. (2000), Fig. 2, selected panels.



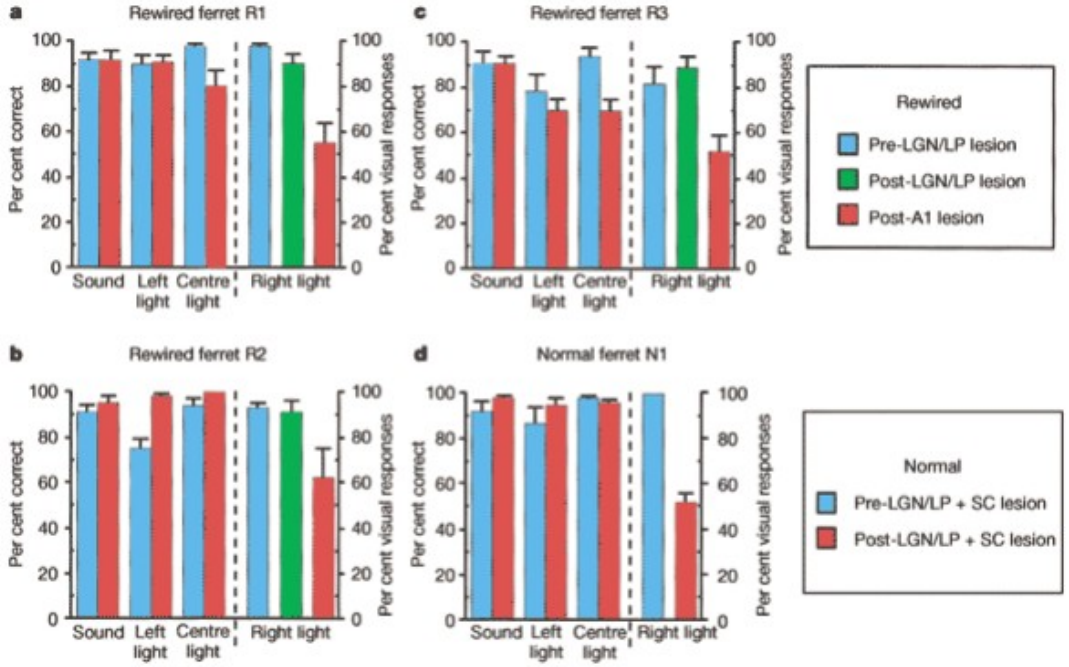
Sofroniew et al. (2016), normal mouse visual cortex, Movie 2.

The rewired pathway could guide visual behavior

A visually responsive neuron does not by itself establish that an animal can use its activity to make a visual decision. Von Melchner, Pallas, and Sur tested behavior in ferrets with retinal projections redirected into the auditory pathway. They presented stimuli within a region of visual space served by that altered projection.

The animals responded to light in a way consistent with their learned visual response rather than their auditory response. The result distinguished visual behavior mediated through the changed pathway from the mere presence of an electrically active retinal connection. Input arriving through auditory thalamus could support a behavior that treated the event as visual.

The animals could also discriminate gratings differing in spatial frequency through the rewired route, although acuity was lower than through the normal visual pathway. Early rerouting therefore supplied usable visual information while preserving limits on the detail that could be resolved. Coarse pattern differences could guide a choice even when finer differences exceeded the resolving ability of the altered pathway.



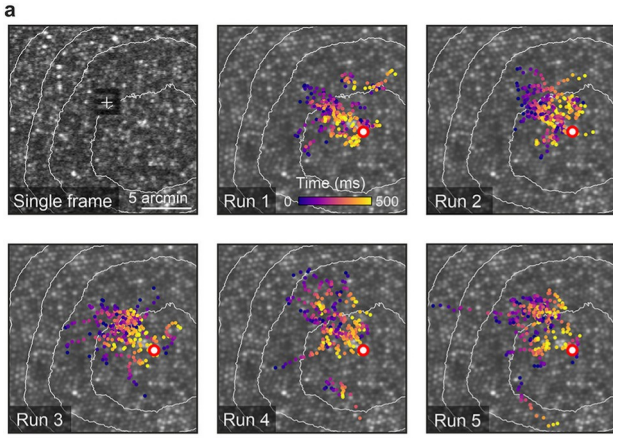
von Melchner et al. (2000), Fig. 2.

Cone spacing and eye drift act together

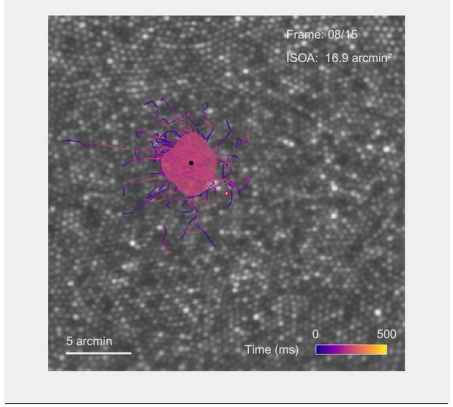
Cone density varies across the foveola and differs among human eyes. Closely packed receptors sample smaller spatial differences, but the retinal image does not remain fixed over them. Witten and colleagues measured the distance and direction of drift while participants judged tiny targets, relating these movements to the cone mosaic in each eye.

Drift length alone did not provide a complete account of performance. The target's starting position and the direction of its movement determined whether it approached a more densely sampled region. A movement of a given size could therefore change the available receptor information differently at different starting locations within the same small retinal area.

The recorded paths also varied across trials rather than following one identical trajectory. The researchers compared their distributions with the positions of dense cone regions and with measured resolution. These comparisons distinguish the physical receptor array from the changing sequence of receptor samples that the visual system receives during an actual visual judgment.



Witten et al. (2024), Fig. 3, selected panels.



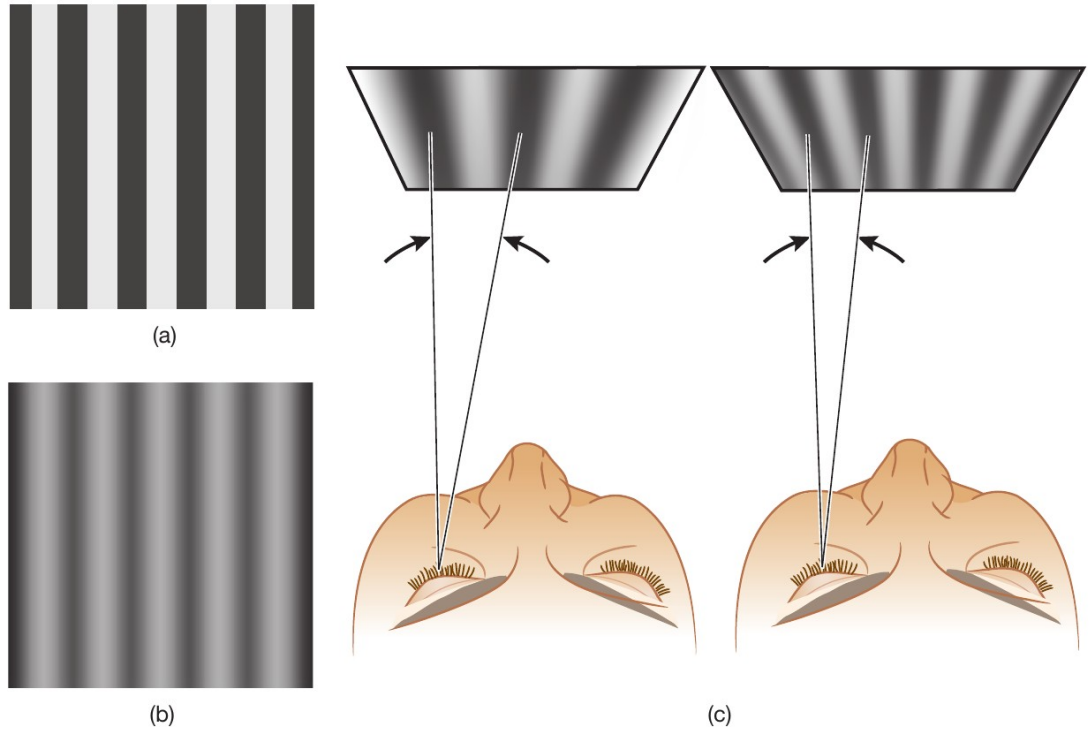
Witten et al. (2024), 98648 Video 3.

Spatial frequency describes coarse and fine structure

Spatial frequency describes how often light and dark alternate across visual space. Widely spaced stripes have low spatial frequency; tightly spaced stripes have high spatial frequency. The same object contains many spatial frequencies because broad light-dark regions, thin contours, and small surface details change brightness over different distances.

Visual frequency is often expressed in cycles per degree of visual angle. A cycle contains one complete light-dark repetition. Using visual angle describes the pattern reaching the eye, which changes with viewing distance. The same physical stripes occupy smaller retinal distances when the observer moves farther away from them.

Neurons respond preferentially to different ranges of spatial frequency. Low-frequency signals describe broad arrangements and large-scale shape, while high-frequency signals preserve fine edges and texture. An object's identity can depend on combining these ranges rather than treating any single spatial scale as a complete description of its appearance.



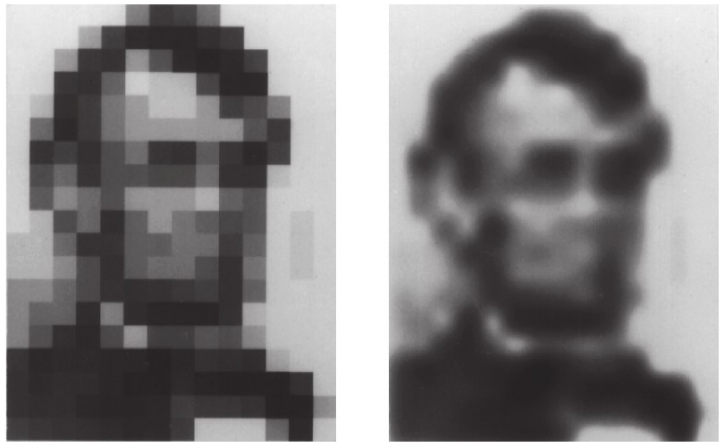
Carlson and Birkett (2022), Fig. 6.23, p. 175.

Removing sharp boundaries can improve recognition

A block-averaged picture can preserve an object's broad brightness pattern while adding conspicuous edges around each block. Those artificial boundaries introduce high spatial frequencies that were not part of the original object's structure. They can compete with the lower-frequency information carrying the broad arrangement needed to recognize the image.

In the original Lincoln demonstration, filtering out the high-frequency block edges makes the face easier to recognize even though fine detail has been removed. The filtered version preserves the broad light-dark arrangement while reducing an irrelevant set of strong boundaries. More visible edges do not always provide more useful information about identity.

Viewing from farther away produces a related change because the eye can no longer resolve the smallest boundaries. The two versions then look more alike. Spatial-frequency sensitivity therefore affects which image structures remain available at different viewing distances and which of those structures dominate the percept of an object.



Carlson and Birkett (2022), Fig. 6.24, p. 176.



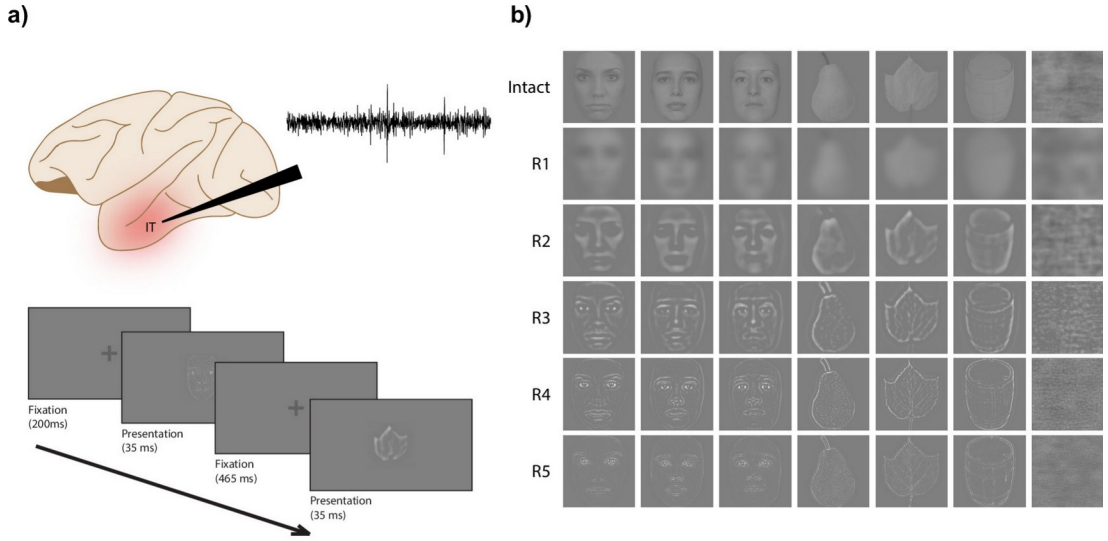
Yan et al. (2024), character-category sequence.

Inferior temporal neurons retain spatial-scale signals

The **inferior temporal cortex**, or IT, is part of the primate ventral visual pathway involved in object recognition. Toosi and colleagues recorded individual macaque IT neurons while presenting faces and other images filtered into different spatial-frequency bands. The filtering changed which broad shapes or fine details remained available in each image.

The researchers also scrambled image structure to separate sensitivity to spatial scale from sensitivity to recognizable content. Neuronal responses still contained information about spatial frequency when the original objects were no longer recognizable. Spatial scale therefore remained measurable within a cortical region that also carries more complex information about visual categories.

Different cells preferred different filtered versions, and a population readout could distinguish the presented frequency bands. These responses preserve information about how an image is constructed across spatial scales. Recognizing an object does not require all early visual measurements to disappear as the signal reaches higher visual cortex.



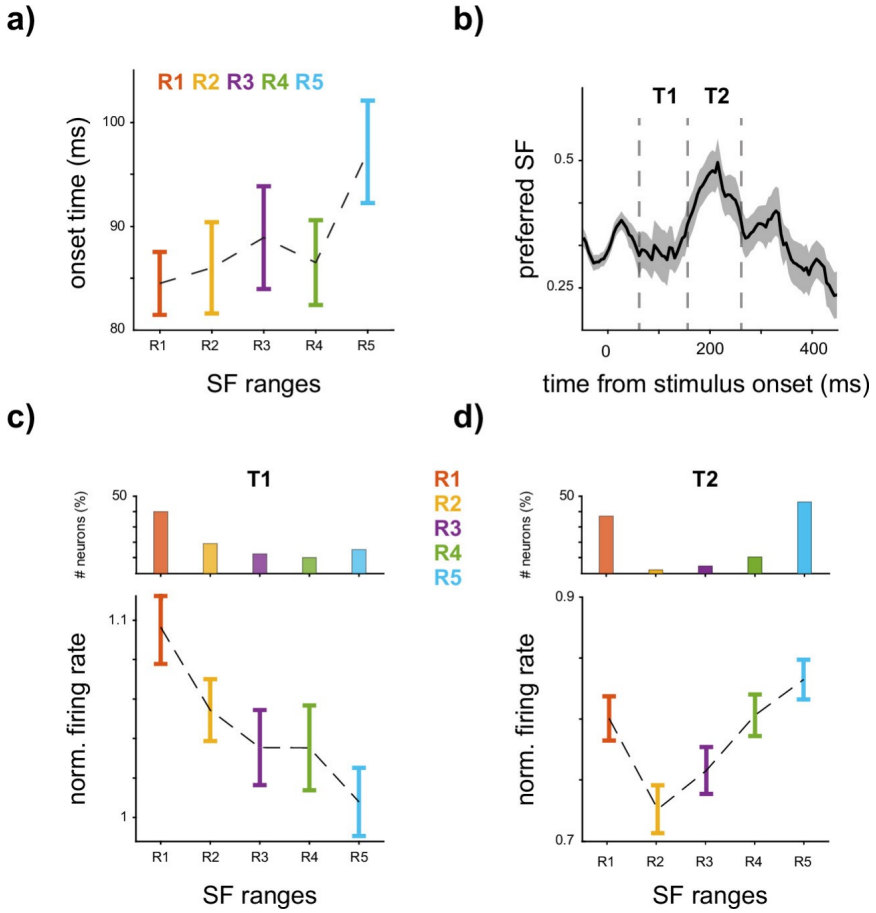
Toosi et al. (2025), Fig. 1, selected panels.

Coarse visual information emerges before fine detail

A neuron's response changes over the milliseconds following image onset. In the macaque IT recordings, information about low spatial frequencies could be recovered earlier than information about high spatial frequencies. The initial population response therefore emphasized coarse image structure before the later response placed greater weight on finer spatial details.

The average spatial-frequency preference shifted toward higher frequencies as the response developed. This shift was measured from changing firing patterns rather than by assuming that an object was recognized in one instantaneous step. A visual representation can evolve while the original image remains unchanged on the display.

Broad image structure can constrain possible object interpretations before fine features become available. Later detail can distinguish objects sharing the same general outline or arrangement. The temporal sequence recorded in IT supplies separate measurements of these stages, including an early preference for coarse structure and a later increase in fine-scale responses.



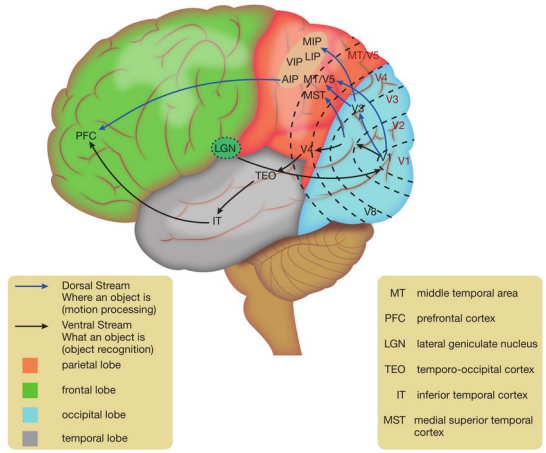
Toosi et al. (2025), Fig. 2.

Ventral and dorsal pathways support different tasks

The **ventral visual stream** extends toward the temporal lobe and contributes to identifying objects, faces, and scenes. The **dorsal visual stream** extends toward posterior parietal cortex and contributes to spatial judgments and visually guided actions. Both receive processed visual information and exchange signals with other cortical regions.

Identifying a cup requires combining its contours, color, and familiar appearance. Picking it up also requires estimating its current position, orientation, and width relative to the hand. These are related computations, but successful recognition and successful control of a reach can be disrupted separately after different kinds of cortical damage.

A hunting animal faces the same distinction at another scale. Detecting a potential prey item must be followed by continuously updated estimates of where it lies and how to move toward it. Visual pathways can preserve information useful for identifying the target while other circuits transform its changing position into an appropriate movement.



Carlson and Birkett (2022), Fig. 6.14, p. 166.



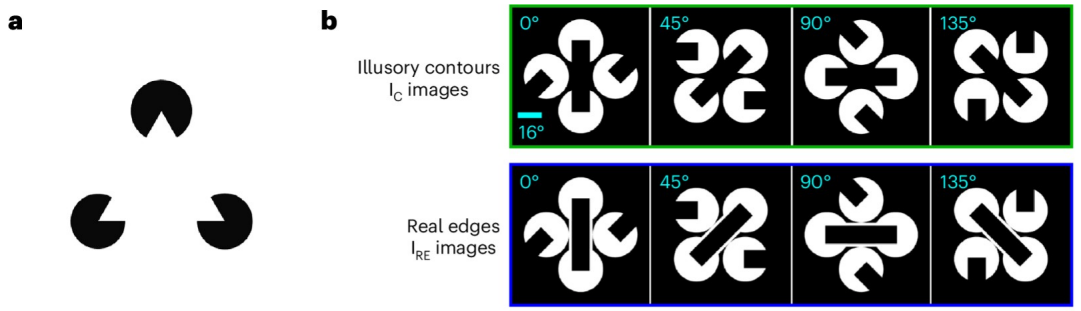
Mearns et al. (2025), 98347 Fig. 1 Video 5.

Visual neurons respond to contours that are absent

In a **Kanizsa illusion**, separated contour fragments can produce the appearance of an edge or surface spanning the empty region between them. The original study presented illusory bars and corresponding real bars to mice while recording visual-cortical neurons. Some V1 cells responded even though the region of their receptive field contained no printed bar.

The investigators changed the arrangement of the component fragments to distinguish responses to a completed bar from responses to one local segment. A specialized population responded to the global illusory arrangement rather than merely to each contributing piece. These were measured neural responses, not verbal reports of what the mice consciously perceived.

Targeted optical activation of the identified cells recreated a related population activity pattern without presenting the visual image. This **pattern completion** recruits a larger stored response from activation of part of the circuit. Feedback and recurrent cortical connections can therefore contribute information about an inferred contour beyond the contrast physically present at one retinal location.



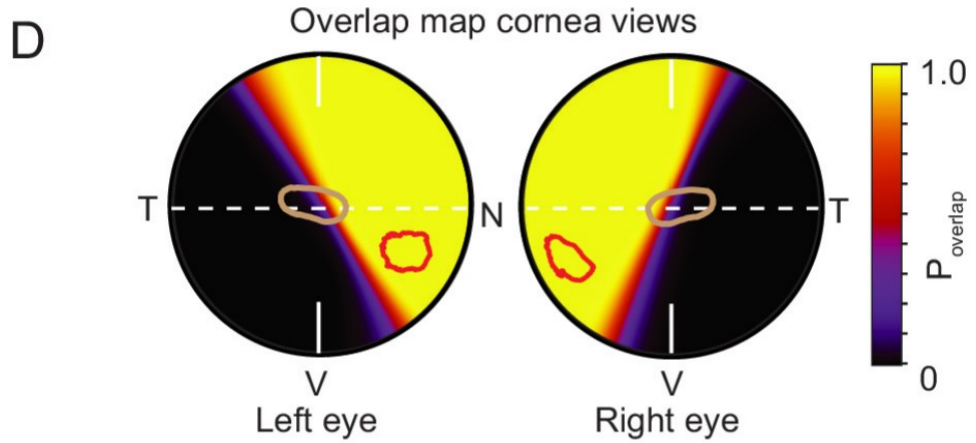
Shin et al. (2025), Fig. 1, selected panels.

Pursuit keeps prey near a stable retinal region

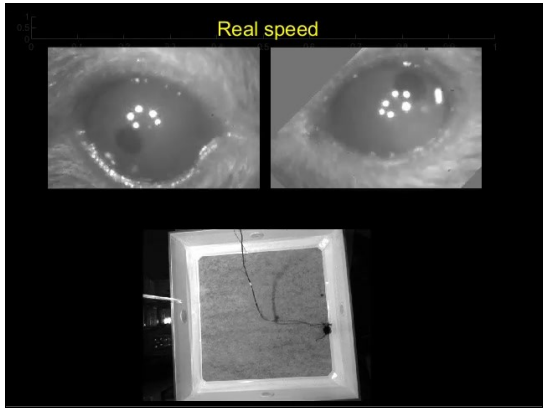
Holmgren and colleagues measured head and eye movements as mice pursued crickets. They also reconstructed the surrounding environment so they could estimate what fell within each eye's view. This allowed the prey's trajectory to be expressed relative to the animal's changing gaze rather than only relative to the laboratory camera.

During pursuit, the mouse tended to hold the cricket within a restricted part of its visual field. That region also experienced relatively little background motion generated by the animal's own movement. Keeping the prey there reduced the amount of surrounding image displacement competing with the signal needed to track the target.

The eyes did not simply rotate independently to follow every movement of the cricket. Head orientation and compensatory eye movements worked together to establish the changing view. A prey item moving across the arena could therefore remain within a comparatively stable retinal region while the mouse itself changed position and approached.



Holmgren et al. (2021), Fig. 4, selected panels.



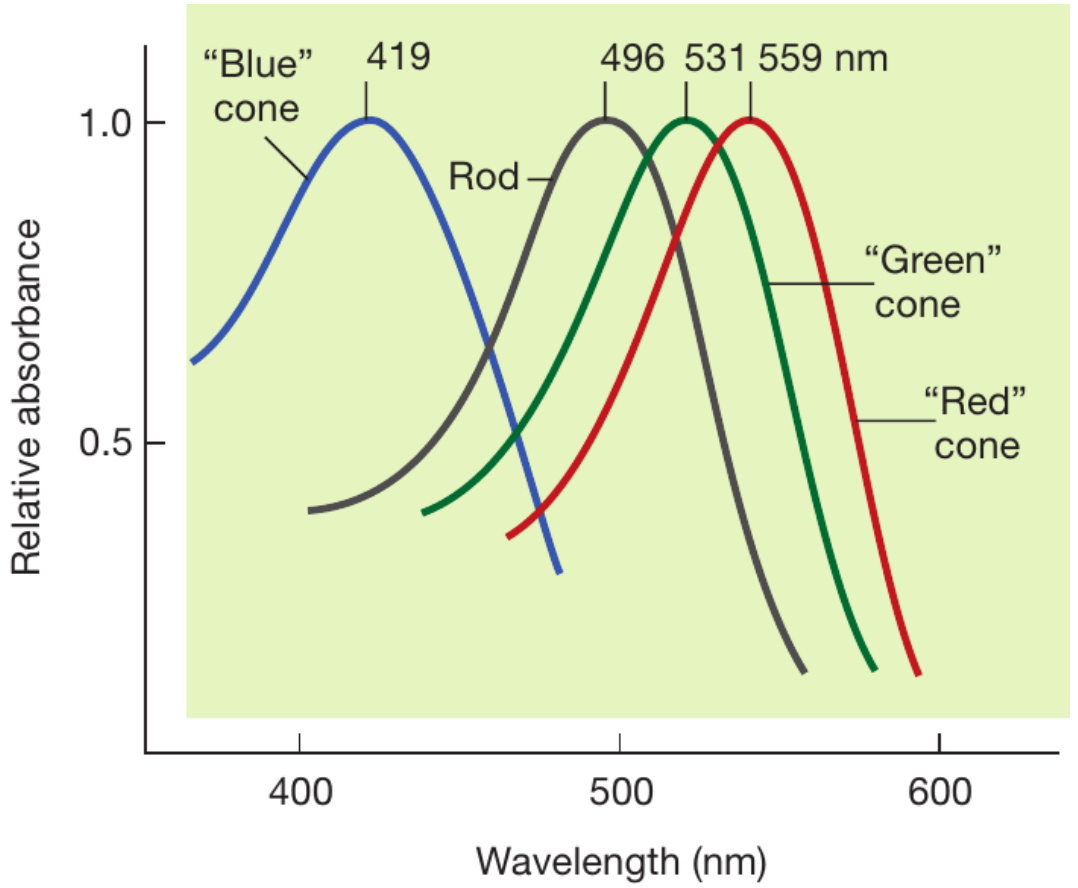
Holmgren et al. (2021), 70838 Video 4.

Cone comparisons support human color vision

Human color vision normally begins with three cone classes whose pigments have overlapping spectral sensitivities. They are called **S**, **M**, and **L** cones for short-, medium-, and long-wavelength sensitivity. Each class responds across a range of wavelengths, so none functions as a detector for only one named color.

A single cone cannot determine whether its activation came from a favored wavelength at low intensity or a less effective wavelength at higher intensity. Comparing activity across different cone classes reduces this ambiguity. Two lights producing different relative S, M, and L responses can be distinguished even if they appear similarly bright.

Different mixtures of wavelengths can also produce the same three cone responses. Such mixtures can look identical despite having different physical spectra. Color perception therefore depends on the information available in receptor comparisons, followed by additional retinal and cortical processing, rather than on a direct readout of every wavelength present.



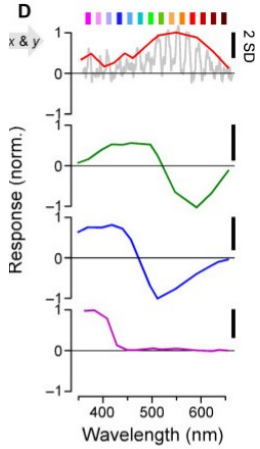
Carlson and Birkett (2022), Fig. 6.16, p. 169.

Four cone classes send differently tuned outputs

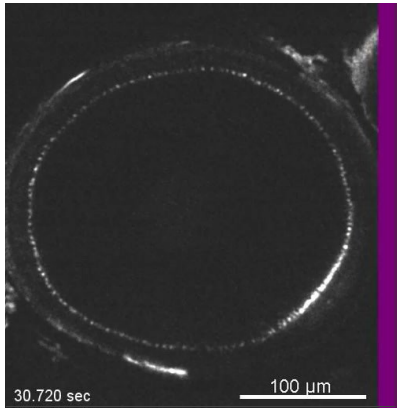
Zebrafish have cone classes sensitive to ultraviolet, blue, green, and red portions of the spectrum. Yoshimatsu and colleagues measured calcium signals at the output terminals of each class while presenting lights of different wavelengths. Measuring the terminal response captured both the cone's own light sensitivity and influences arriving through retinal connections.

The output of a cone did not always follow the absorption spectrum of its opsin alone. Some wavelengths drove calcium changes in opposite directions within the same cone class. This **spectral opponency** means that retinal interactions had already converted wavelength differences into opposing signals at the first stage of synaptic output.

Red-cone outputs largely emphasized changes in overall light, while other cone outputs carried different wavelength comparisons. Ultraviolet cones also supplied signals useful for detecting bright prey. Multiple receptor classes therefore provided parallel outputs suited to different visual demands rather than four interchangeable copies of the same brightness measurement.



Yoshimatsu et al. (2021), Fig. 1, selected panels.



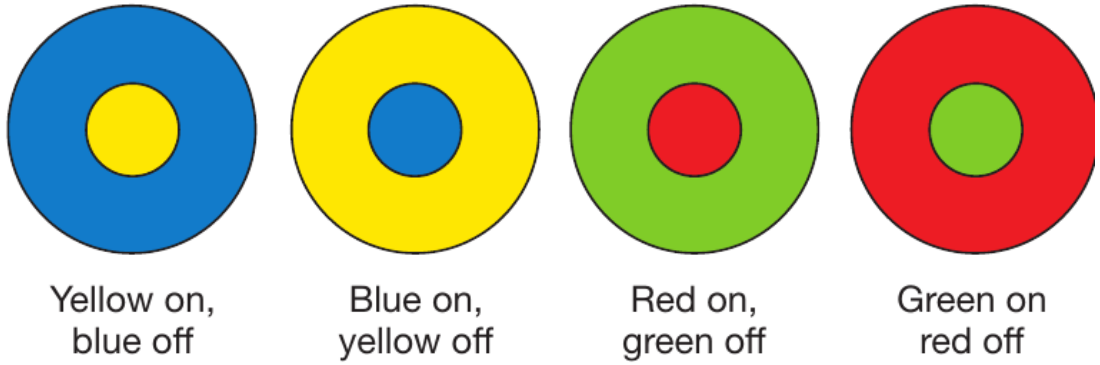
Yoshimatsu et al. (2020), Video S5.

Opponent circuits compare cone signals

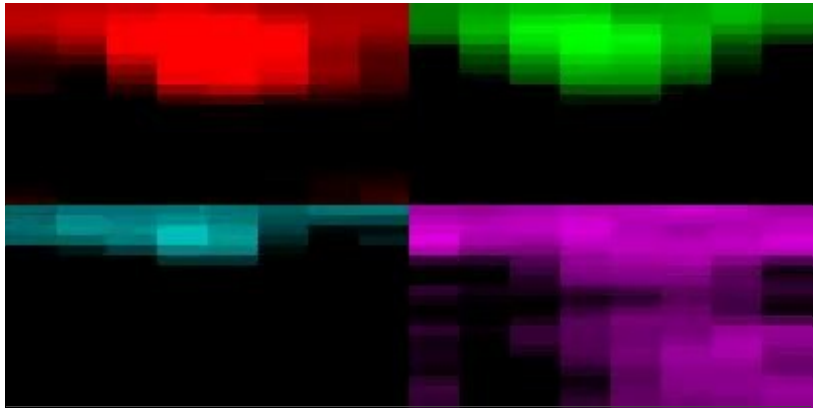
An **opponent channel** responds in opposite directions to different combinations of cone activity. In primates, important comparisons include L against M cone signals and S against a combination of L and M signals. These comparisons create signals related to color differences while other pathways emphasize changes shared across cone classes.

The comparison is implemented through excitatory and inhibitory connections among retinal cells. A ganglion cell can increase firing for one cone-weighted signal and decrease firing for the opposing signal. Its response therefore describes a difference between inputs, including differences across the center and surround of its receptive field.

Trichromatic and opponent accounts describe different stages of the same system. Three cone classes provide the initial measurements, and retinal circuits recombine them into comparison signals. A color's appearance then depends on those signals, adaptation, and cortical context rather than on three cones independently announcing red, green, and blue percepts.



Carlson and Birkett (2022), Fig. 6.18, p. 170.



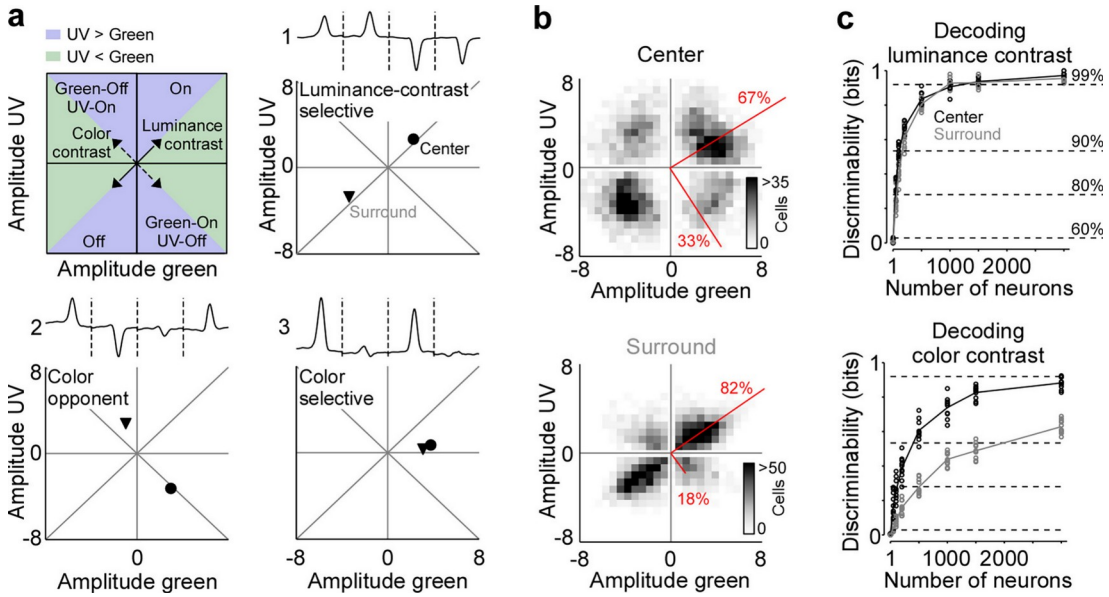
Zhou et al. (2020), Video S4.

Mouse cortex compares ultraviolet and green contrast

Mouse cones provide strong ultraviolet and green-sensitive signals, with their contribution varying across the retina. Franke and colleagues independently changed these stimulus components while measuring calcium activity in V1. They could then distinguish neurons responding mainly to overall brightness from neurons responding to opposing changes in the two spectral channels.

Many cortical neurons showed color opponency in the center of their receptive fields. Their surrounds more often emphasized light-dark contrast. A single cell could therefore combine a central wavelength comparison with surrounding information about general brightness, producing a response that depended on both local color and the surrounding scene.

Color-opponent responses were especially common in posterior V1, which represents the upper visual field. This spatial bias corresponded to differences in the visual information encountered above and below the animal. Modeling suggested that particular opponent responses could improve detection of dark ultraviolet objects against noisy daylight backgrounds, a proposed function tested computationally.



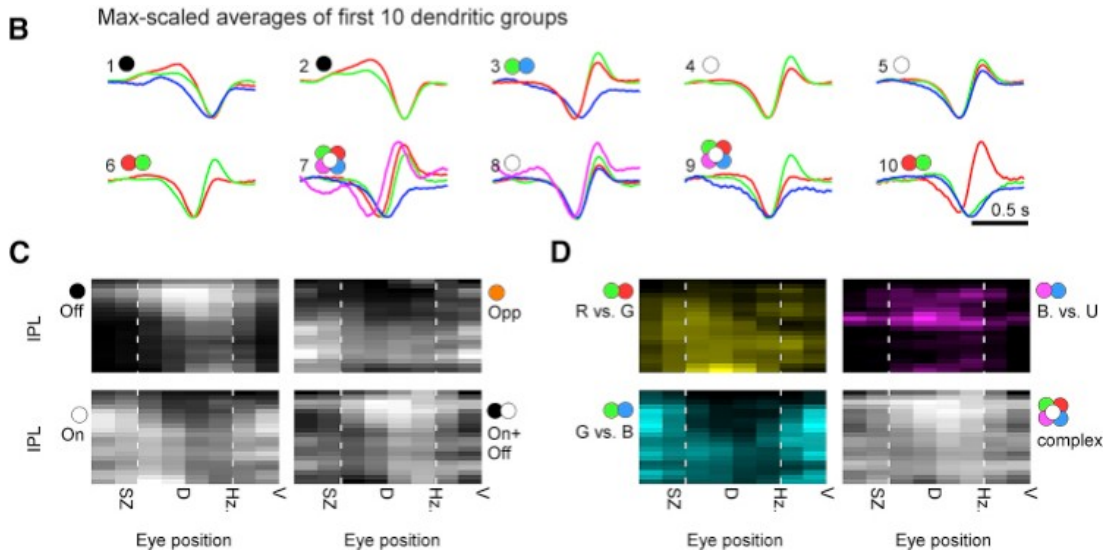
Franke et al. (2024), Fig. 2.

Color and timing are combined in retinal output

Zebrafish ganglion cells respond to combinations of wavelength and time. Some increase activity after one spectral component brightens and another darkens, while others respond with slower or more sustained changes. Zhou, Bear, and colleagues measured these properties by varying four wavelength channels independently and comparing the resulting cellular response patterns.

A prominent set of responses emphasized decreases in blue light and developed relatively slowly. Other populations emphasized ultraviolet increases or different spectral contrasts. Classifying the cells by color preference alone would therefore omit information about when their responses began, how long they lasted, and which changes caused them to recover.

The output reaching the brain contains these combined descriptions. A short ultraviolet-bright event and a prolonged blue-dark event can recruit different ganglion-cell populations even when they occur at the same retinal location. Wavelength and response duration are already represented together before higher visual areas combine the signal with movement and behavioral context.



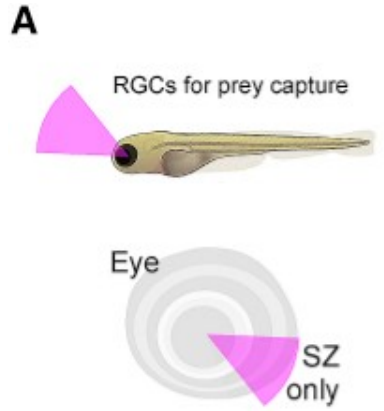
Zhou et al. (2020), Fig. 3, selected panels.

Ultraviolet-sensitive outputs vary across the retina

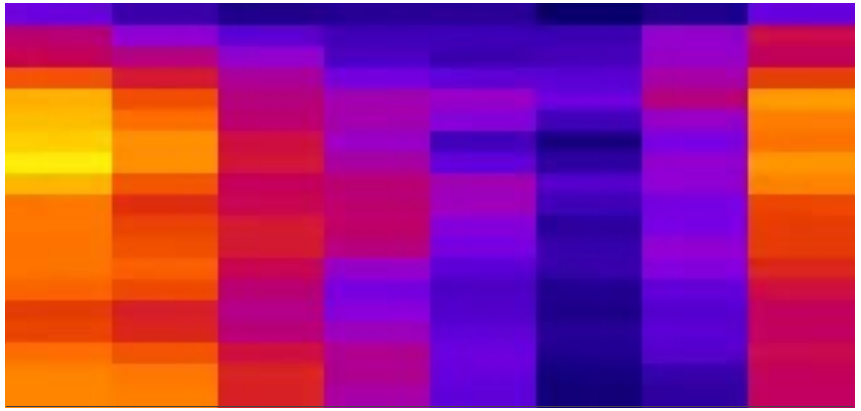
The zebrafish acute zone contains specialized ultraviolet cones, but the regional difference continues into ganglion-cell output. The investigators found a population of sustained ultraviolet-ON responses concentrated in this region. These cells remained active during ultraviolet brightening rather than responding only to the instant when the light changed.

A sustained ultraviolet signal can preserve information about a small bright target while it remains visible. The cells' dendritic organization also differed across retinal regions, changing which layers and inputs were sampled. Structural and functional differences therefore occurred together within the output population associated with the prey-capture region.

Elsewhere in the retina, different mixtures of spectral and temporal responses were present. The same physical light change could consequently produce different population activity depending on where it entered the eye. Retinal position carries information about external location while also selecting circuits adapted to the visual demands of that part of space.



Zhou et al. (2020), Fig. 5, selected panels.



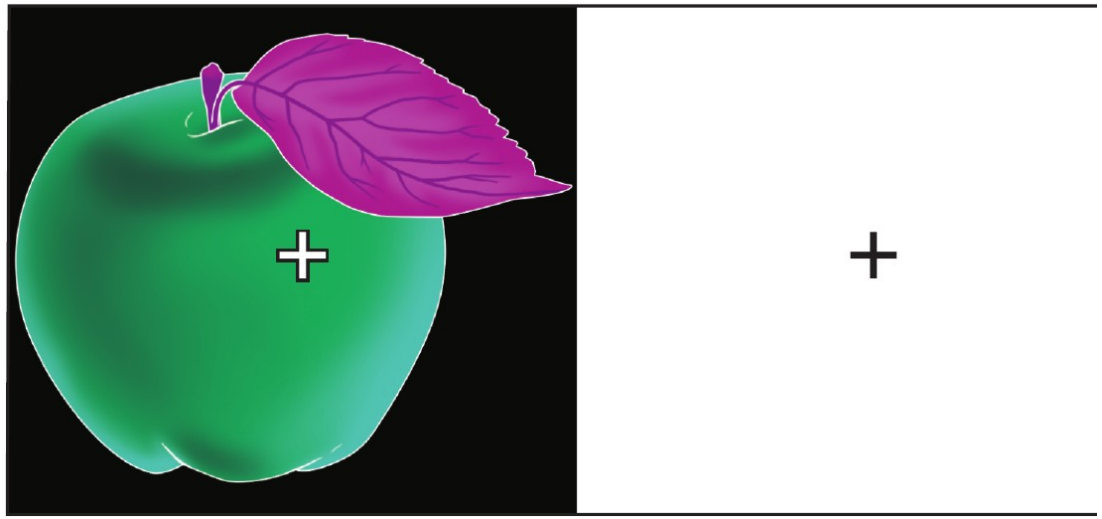
Zhou et al. (2020), Video S3.

Adaptation changes the color balance after viewing

A **negative afterimage** appears in colors complementary to those of a recently viewed image. Fixating a colored region for several seconds changes the responses of the receptors and downstream opponent pathways sampling it. When gaze then shifts to a neutral surface, the adapted pathways no longer respond in their previous balance.

Strong stimulation can reduce a channel's continuing response, while release from prolonged suppression can create a temporary rebound. Neutral light activates the available cone classes, but their adapted contributions are unequal. The resulting opponent signal can therefore favor the color opposite to the one that occupied the same retinal location during adaptation.

The afterimage follows the adapted retinal region rather than remaining attached to the original object. Moving the eyes changes where that region is projected into the surrounding scene. Its color and position arise from the visual system's recent activity and current gaze, even though the neutral surface contains no corresponding colored object.



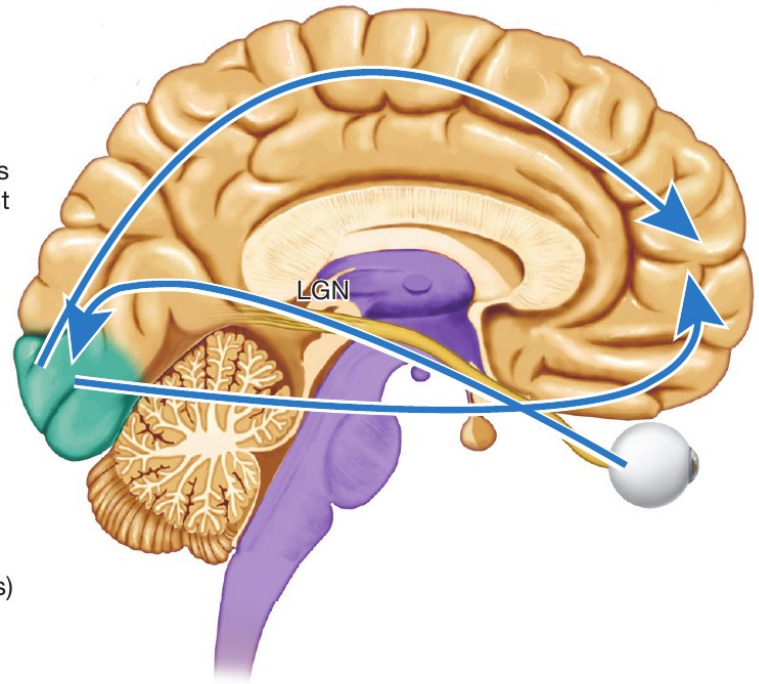
Carlson and Birkett (2022), Fig. 6.20, p. 172.

Cortical context helps stabilize object color

The spectrum reaching the eye depends on both a surface and the light illuminating it. A white wall reflects different wavelength mixtures under daylight and warm indoor illumination. **Color constancy** is the tendency to perceive a surface's color as relatively stable despite these changes in the physical light reaching the retina.

Cortical neurons compare information over regions larger than a single local retinal measurement. Some color-sensitive neurons change their response when the surrounding scene changes, even if the central stimulus remains the same. These interactions provide information about the broader illumination and about which wavelength differences belong to a particular surface.

Damage to ventral visual cortex can impair color perception even when cones remain functional. Other damage can leave color judgments relatively preserved while disrupting recognition of shape. The receptor measurements are therefore necessary inputs, but the conscious appearance of a colored object also depends on how cortical networks combine those measurements with spatial context.



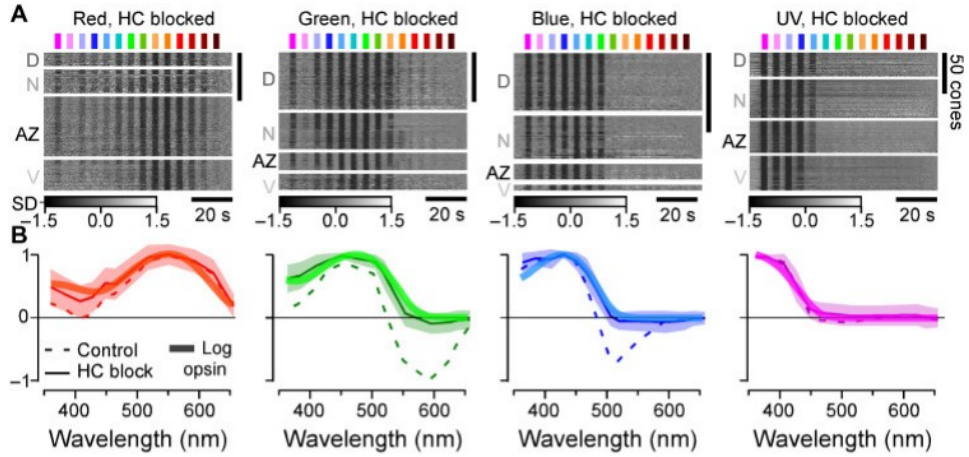
Carlson and Birkett (2022), Fig. 6.21, p. 173.

Horizontal cells reshape cone color responses

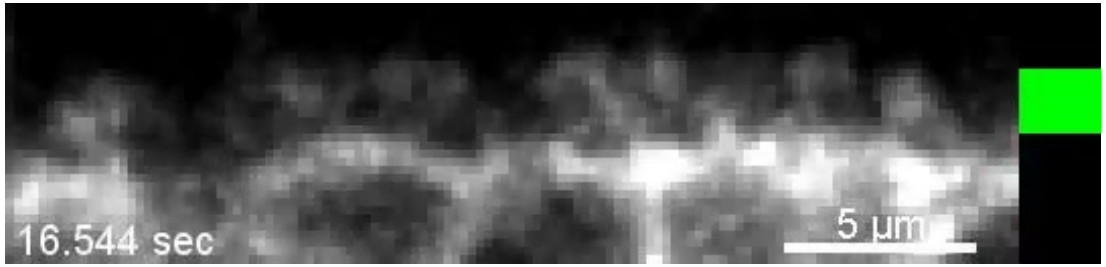
Horizontal cells receive signals from multiple photoreceptors and send feedback that changes cone output. In the zebrafish study, cone-terminal calcium responses included wavelength preferences that were not predicted by the cone's own opsin. Input from surrounding retinal circuits therefore altered the spectral message leaving the photoreceptor terminal.

When the researchers blocked horizontal-cell signaling, cone responses became more similar to the sensitivity expected from their individual opsins. Opponent components were reduced or changed. The experiment separated the cone's direct response to absorbed light from additional wavelength comparisons introduced by interactions among neighboring retinal cells.

The investigators also selectively removed ultraviolet cones and measured responses in other cone classes. Those changes further tested which receptor populations contributed to the network's spectral comparisons. Cone output therefore depended both on the pigment inside the recorded cell and on signals routed through other cells around its synaptic terminal.



Yoshimatsu et al. (2021), Fig. 2, selected panels.



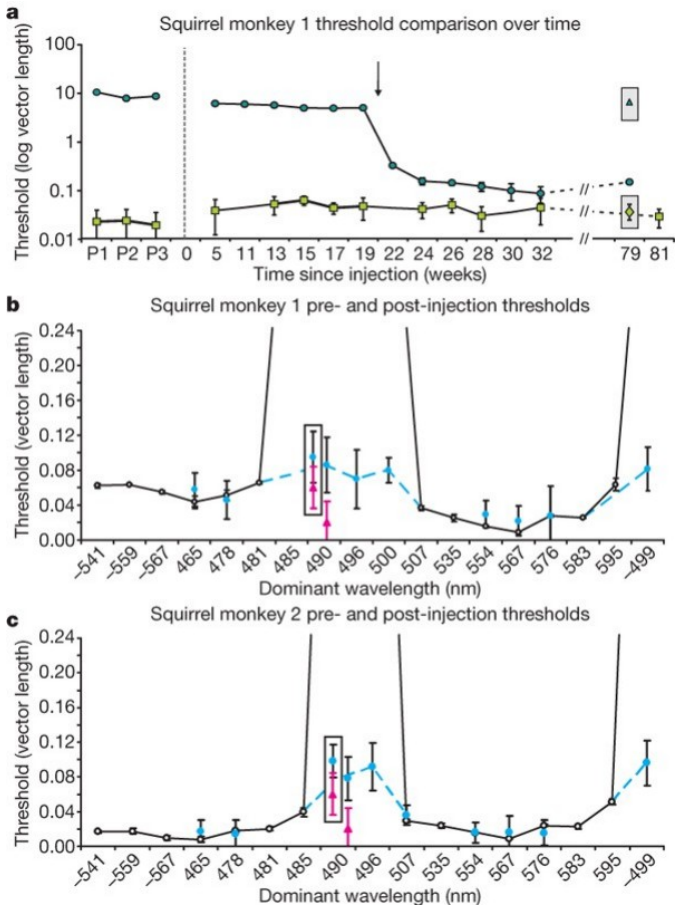
Yoshimatsu et al. (2020), Video S6.

Adding an opsin expanded color discrimination

Some male squirrel monkeys normally have only two cone-pigment classes and cannot distinguish particular red-green combinations that trichromatic individuals separate. Mancuso and colleagues delivered a human L-opsin gene into the retinas of adult dichromatic monkeys. Some existing cones then expressed an additional light-sensitive pigment alongside their original pigment.

Before and after treatment, the monkeys searched for a colored target embedded in a background designed to limit simple brightness cues. After expression developed, they could distinguish additional color combinations that had previously been difficult. The change followed an altered receptor signal rather than surgical replacement of the entire retina or cortex.

The treatment was performed in adult animals, after their visual pathways had already developed with dichromatic input. Existing downstream circuitry could use the newly differentiated cone signals to support additional discriminations. This adult result differs from neonatal pathway rerouting: the principal eye-to-brain route remained present while the spectral information entering it changed.



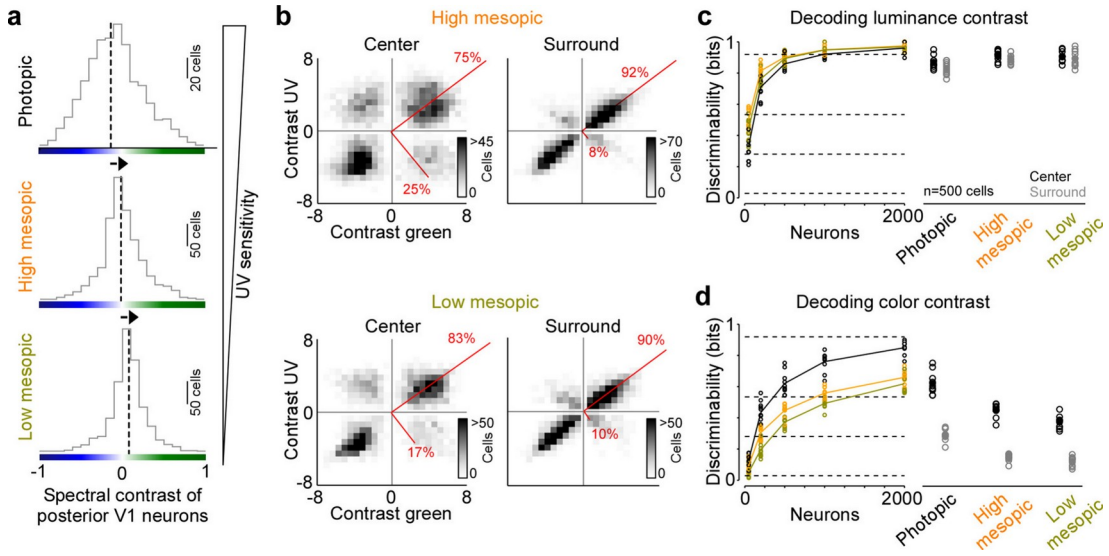
Mancuso et al. (2009), Fig. 3.

Dim light weakens cortical color separation

At lower illumination, rod contributions increase while cone signals become less effective. The mouse V1 study compared responses across different ambient light levels while presenting ultraviolet and green changes. Responses became less ultraviolet-sensitive as illumination declined, changing the spectral information represented by the same broad cortical population.

The ability to separate color contrasts decreased more strongly than the ability to distinguish light increases from decreases. Brightness-related signals therefore remained useful under conditions that weakened the spectral comparisons. A dim scene could still produce detectable borders and changes without preserving the same color distinctions available during brighter viewing.

The shift depended on the receptor inputs and retinal processing reaching cortex. It was not equivalent to simply reducing the brightness of a fully preserved color image. As illumination changed, the relative contributions of different sensory channels changed, altering what distinctions cortical neurons could carry about the scene.



Franke et al. (2024), Fig. 3.

Visual agnosia separates recognition from seeing

Visual agnosia is an impairment in recognizing objects by sight that cannot be explained simply by poor visual acuity. A person may see boundaries and individual features yet fail to identify the object formed by them. Recognizing the same object through touch can remain possible because other sensory routes still reach stored knowledge.

Damage to the ventral visual pathway can interrupt the combination of shape, color, and familiar structure needed for recognition. The deficit differs from forgetting the object's name or purpose. Comparing visual naming with identification through touch or sound helps determine which part of the recognition process has been disrupted.

Different categories can also be affected unequally. A person may recognize a face-like arrangement while failing to identify the individual objects composing it. Such dissociations indicate that seeing an edge, combining edges into an object, and identifying a familiar person depend on overlapping but partly separable neural computations.



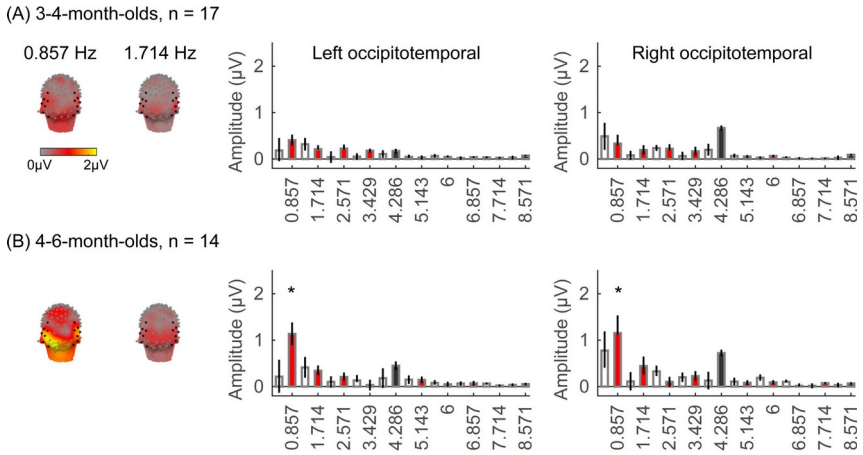
Carlson and Birkett (2022), Fig. 6.26, p. 178.

Infant brain rhythms distinguish face images

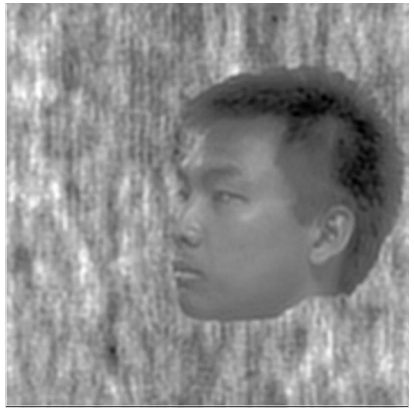
Yan and colleagues measured infants' scalp electrical activity while rapidly presenting images from several visual categories. A particular category, such as faces, appeared at a regular slower rhythm within the sequence. If the brain distinguished that category, its electrical response could contain activity at the category's repetition frequency.

The researchers used controlled grayscale images to reduce differences in average brightness and contrast across categories. Face-related responses emerged reliably in older infant groups before comparable limb and corridor responses. The measured differences concerned the brain's ability to separate image categories rather than an infant verbally naming what appeared on the screen.

The source film contains the actual face-category stimulus sequence, with images changing at a precisely controlled rate. General responses to every image update occur at one frequency, while category-specific responses occur at another. Separating these rhythms allows an experiment to measure visual category processing in observers who cannot follow adult task instructions.



Yan et al. (2024), Fig. 3, selected panels.



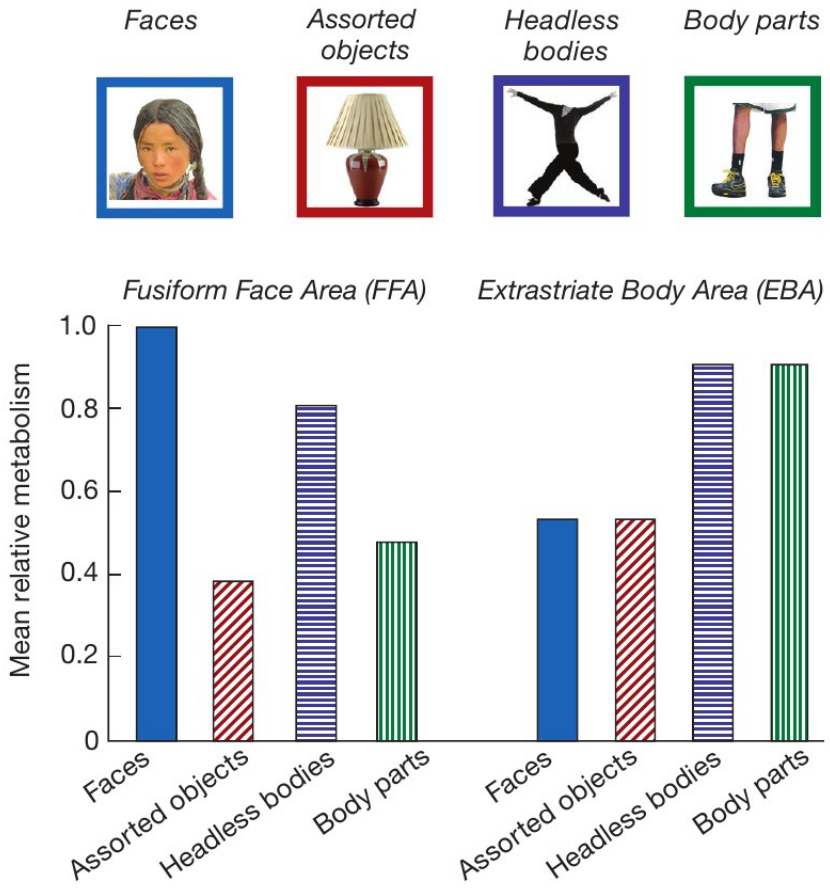
Yan et al. (2024), face-category sequence.

Face and body processing recruit neighboring regions

The **fusiform face area**, or FFA, is a ventral temporal region that responds strongly during face viewing. Nearby, the **extrastriate body area**, or EBA, responds strongly to bodies and body parts. Their different response preferences can be measured by comparing brain activity across categories rather than by observing a single stimulus alone.

Prosopagnosia is difficulty recognizing familiar identities from faces. An affected person may still know who a relative is and recognize their voice, yet fail to identify the relative by appearance. The difficulty can concern the configuration of facial features rather than an inability to detect eyes, a nose, or a mouth.

Face recognition also changes with experience and development. Early preferences for face-like arrangements do not supply a complete adult recognition system at birth. Circuits must distinguish many similar individuals across changes in lighting, expression, and viewpoint, while retaining enough stable information to recognize the same person on another occasion.



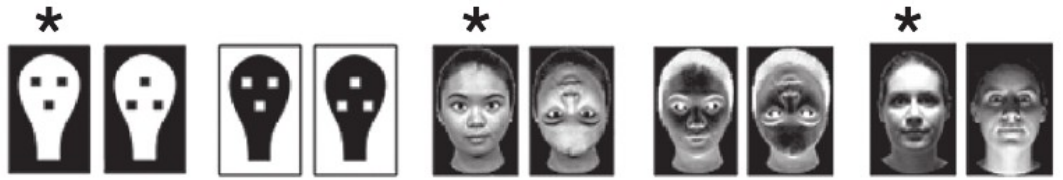
Carlson and Birkett (2022), Fig. 6.27, p. 178.

Turning a face changes more than its visible parts

Face recognition depends partly on the arrangement of features, including the spacing of the eyes and their relation to the nose and mouth. Turning a face upside down preserves the individual parts but changes their familiar configuration relative to the observer. Detecting each part can remain easier than rapidly identifying the person from that unusual arrangement.

Newborn infants prefer some upright face-like patterns over corresponding inverted patterns. The original stimuli also changed contrast polarity, altering whether the eye and mouth regions had the light-dark arrangement typical of a naturally illuminated face. Orientation and contrast arrangement therefore affect viewing preferences before adult face-recognition skill has developed.

An early viewing preference and mature identity recognition are different abilities. A bias toward face-like patterns can increase exposure to faces, while later learning differentiates familiar individuals across expressions and viewpoints. The visual system must preserve stable relationships among features while tolerating the many changes that occur when a real person moves.



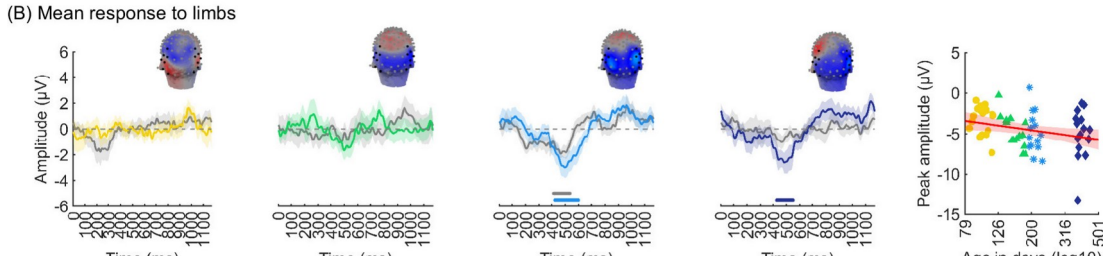
Carlson and Birkett (2022), Fig. 6.30, p. 180.

Limb-category responses emerge later in infancy

The infant study also presented images of limbs at a regular category rhythm among other pictures. A distinct electrical response to this rhythm indicated that the brain was processing limb images differently from the intervening categories. This measurement separated category sensitivity from the general response produced whenever any image appeared.

Reliable limb-related responses emerged later than the face-related response in the tested age groups. The difference occurred even though the images were controlled for several low-level properties. Different visual categories therefore did not acquire equally distinct population responses at the same point in infant development.

The developing response patterns also became more distinguishable across categories with age. A classifier trained on the electrical signals could increasingly identify which image category an infant was viewing. That readout measured information available in the recorded brain activity, rather than assuming that a larger response alone meant better recognition.



Yan et al. (2024), Fig. 4, selected panels.



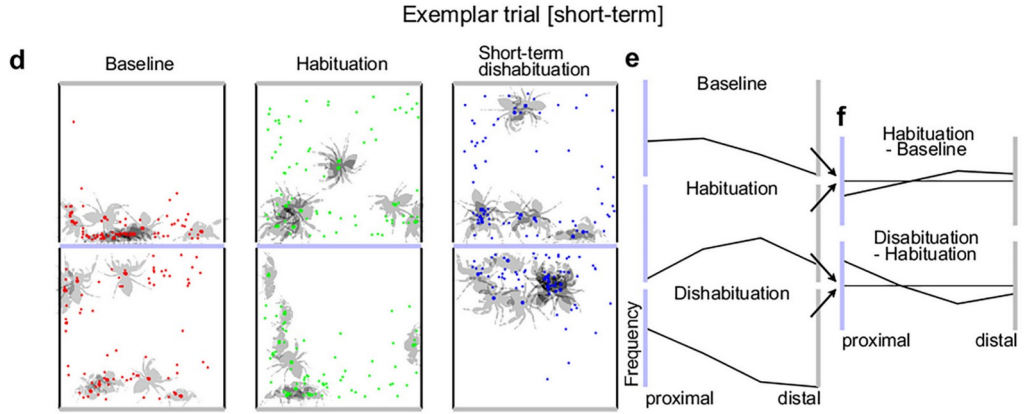
Yan et al. (2024), limb-category sequence.

Spider familiarity changes visual approach

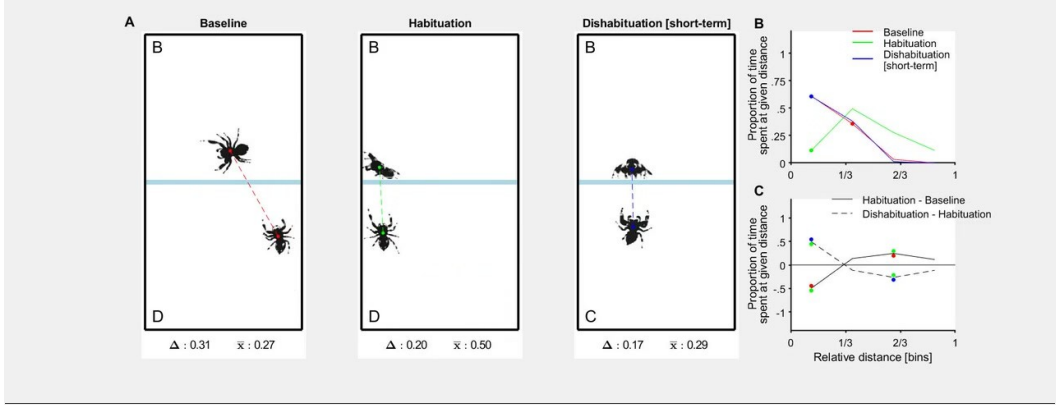
Regal jumping spiders have visually guided hunting behavior and can inspect other spiders across a transparent barrier. Dahl and Cheng repeatedly presented particular individuals while tracking the distance between them. The animals remained physically separated, so visual information about the partner could be tested without allowing direct contact during the encounter.

During repeated encounters with the same partner, the spiders tended to remain farther apart than during the initial meeting. This reduction in approach is **habituation**, a decreased response after repeated exposure. Changing to a different partner could restore approach, a response called **dishabituation** when the change reverses the earlier reduction.

The comparison asks whether the animal retains information specific to the previously seen individual. A general decline in movement would predict reduced approach to both familiar and new partners. Renewed approach to a new partner instead provides behavioral evidence that the visual input was distinguished from the earlier individual's appearance.



Dahl et al. (2025), Fig. 1, selected panels.



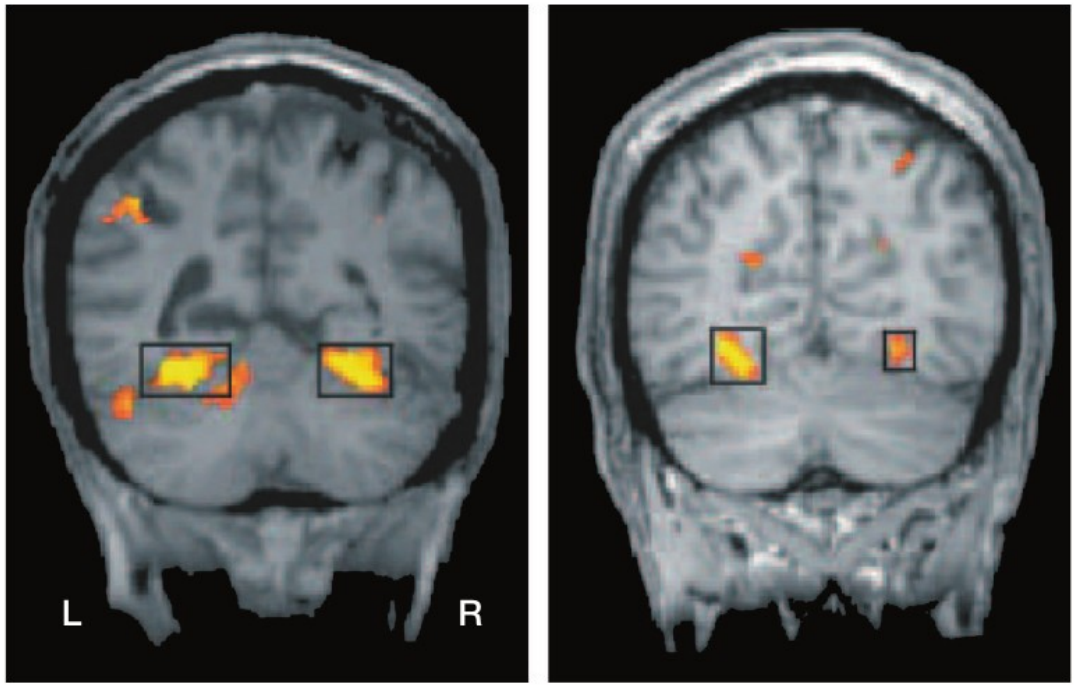
Dahl et al. (2025), 97146 Fig. 2 Video 1.

Scene recognition can survive object-recognition loss

The **parahippocampal place area**, or PPA, responds strongly to scenes and spatial layouts. A scene contains information extending across many objects, including boundaries, textures, and the arrangement of surfaces. Recognizing a room or a beach can therefore draw on broader structure than identifying each object located within it.

A patient with severe visual object agnosia could still classify familiar kinds of scenes. Functional imaging detected responses in preserved scene-sensitive cortex while object-recognition regions were damaged. The preserved ability allowed scene processing to be studied without assuming that successful recognition required first naming all of the objects within the view.

An object and its setting supply related but different visual information. A chair can be recognized across several rooms, and a room can be recognized after its furniture changes. Separate response preferences within ventral visual cortex help preserve both the identity of particular objects and information about the surrounding spatial environment.



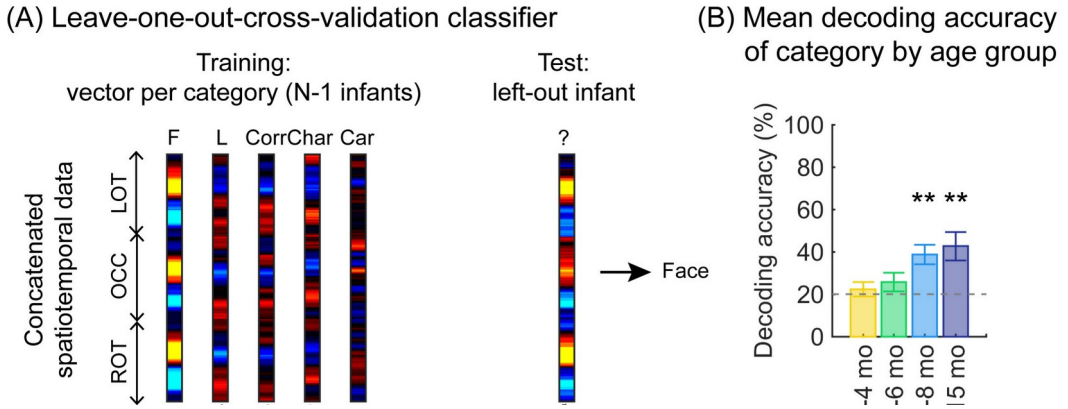
Carlson and Birkett (2022), Fig. 6.28, p. 179.

Infant scene responses become more distinct with age

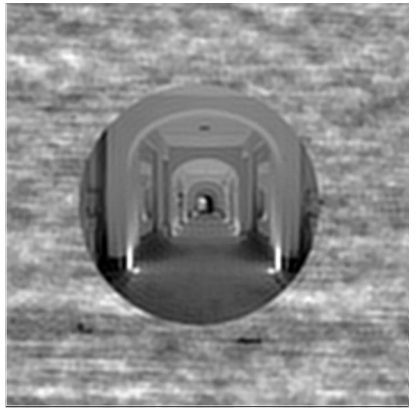
The corridor category in the infant experiment supplied scene-like images with extended surfaces and spatial layout. The source sequence alternated these images with other categories at controlled intervals. A category-frequency electrical response indicated that the brain was distinguishing corridor images from faces, limbs, characters, and cars in the same sequence.

Reliable corridor-related responses emerged after the face-related response in the sampled age groups. The result identified different developmental timing for these category representations. It did not require the infants to recognize a particular building or navigate through the pictured environment, because the task measured responses while they viewed the images.

Across later infancy, the overall pattern of category responses became more separable. Distinct responses to faces, limbs, and scenes could coexist within the same developing visual system. Their different timing suggests that visual category organization is refined over experience rather than appearing as one fully differentiated set of adult response patterns at birth.



Yan et al. (2024), Fig. 5, selected panels.



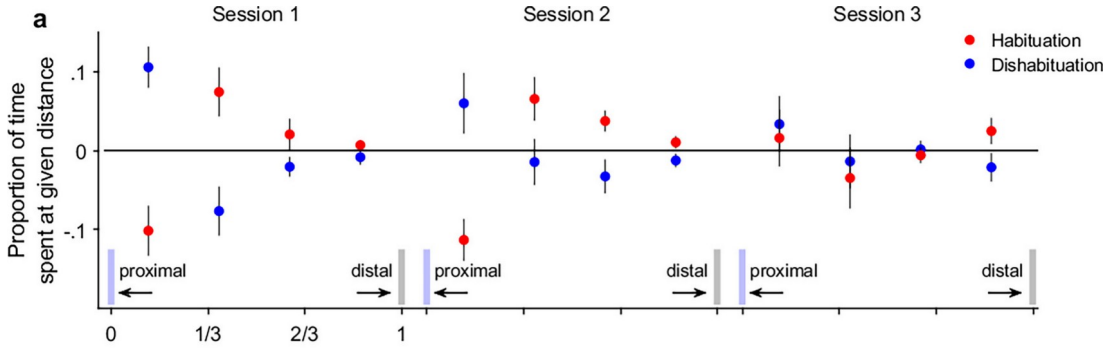
Yan et al. (2024), corridor-category sequence.

A new individual restores visual investigation

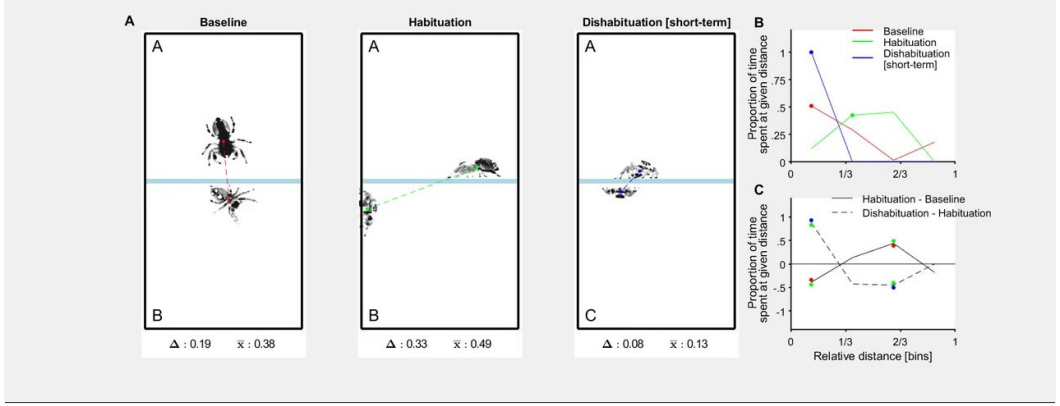
In the jumping-spider experiments, the familiar partner could be replaced after repeated encounters. The spiders then approached the new individual more than they had approached the repeatedly viewed partner. Because the new partner remained behind a transparent barrier, the changed behavior could be compared while maintaining physical separation between the animals.

The researchers also examined encounters across longer intervals and successive sessions. A particularly novel individual introduced after extended testing could still produce renewed approach. This pattern distinguished sensitivity to the encountered individual's novelty from a simple progressive loss of willingness to move during the experimental session.

The measured behavior was the distribution of distances between the spiders through time. It supplied evidence about visual discrimination and retained familiarity, without requiring a human-style face judgment. The animal had to compare the current partner's appearance with information retained from earlier viewing and alter its approach according to that comparison.



Dahl et al. (2025), Fig. 2, selected panels.



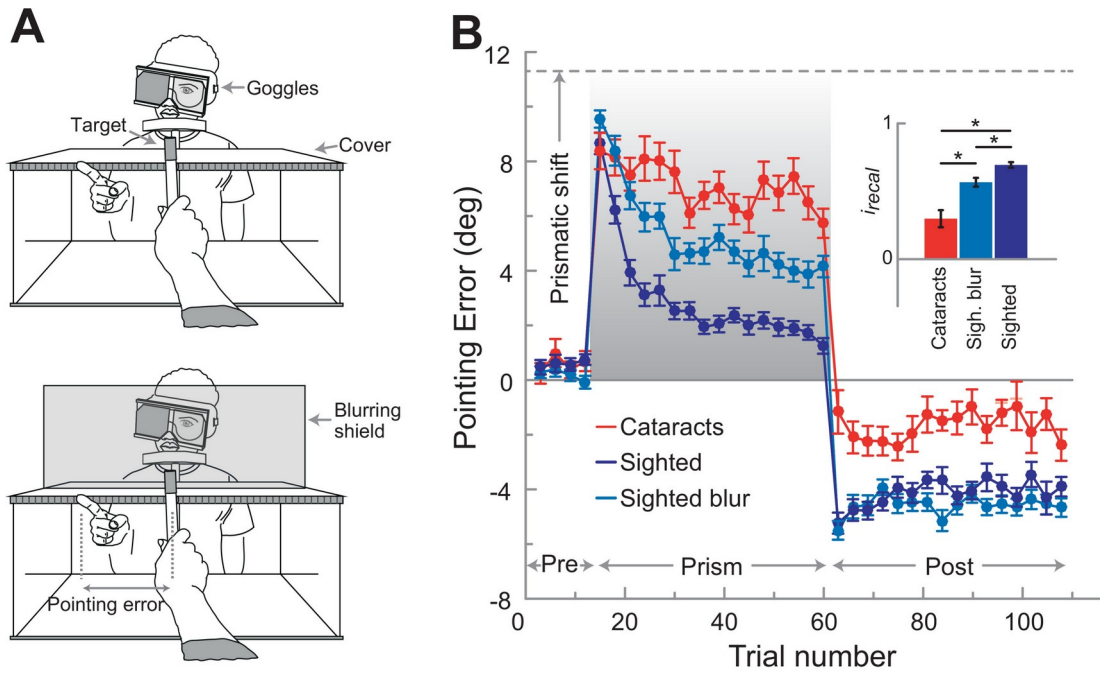
Dahl et al. (2025), 97146 Fig. 2 Video 2.

Restored sight requires visuomotor recalibration

A **cataract** is an opacity in the lens that reduces the pattern of light reaching the retina. Children born with dense cataracts in both eyes can receive patterned vision only after surgery. Senna and colleagues tested how these children and adolescents adjusted visually guided pointing after gaining access to clearer visual input.

Prism goggles shifted the apparent target position sideways. Normally sighted participants used feedback about their pointing errors to adjust subsequent reaches and showed an opposite error when the prisms were removed. Recently treated participants recalibrated less, and blurring vision in comparison participants did not reproduce the full reduction.

Recalibration improved with additional months and years of visual experience after surgery. Correcting the optical obstruction did not immediately supply a fully calibrated mapping from visual position to arm movement. The developing system continued learning from the relation between the seen target, the motor command, and the resulting hand position.



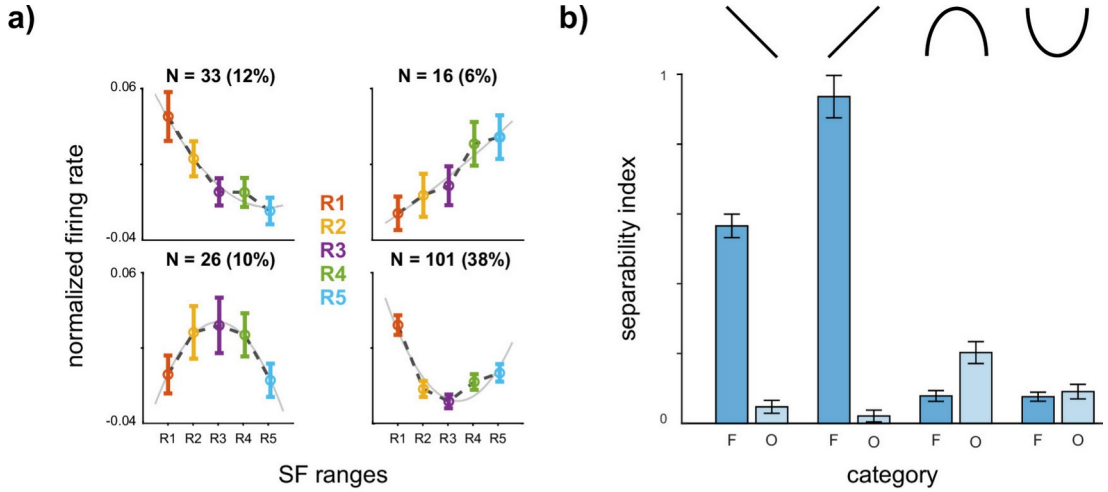
Senna et al. (2022), Fig. 1, selected panels.

Fine spatial information helps separate face images

The macaque IT study compared neurons preferring different spatial-frequency ranges with their ability to distinguish image categories. Neurons preferring high spatial frequencies carried particularly useful information for separating faces from other images. Fine spatial structure therefore remained relevant within a population involved in recognizing complex visual content.

Different frequency preferences were associated with different category-response patterns. The researchers separated this population association from the information carried by individual cells. A neuron could contribute strongly to identifying a frequency band without making an equally strong contribution to distinguishing a face from another category of object.

Spatial scale and object category are therefore related but separable properties of the recorded activity. A face contains fine edges and broad shape, while another object can share either kind of structure. Comparing filtered and scrambled images allowed the investigators to examine these contributions without treating a preference for fine detail as equivalent to a face identity code.



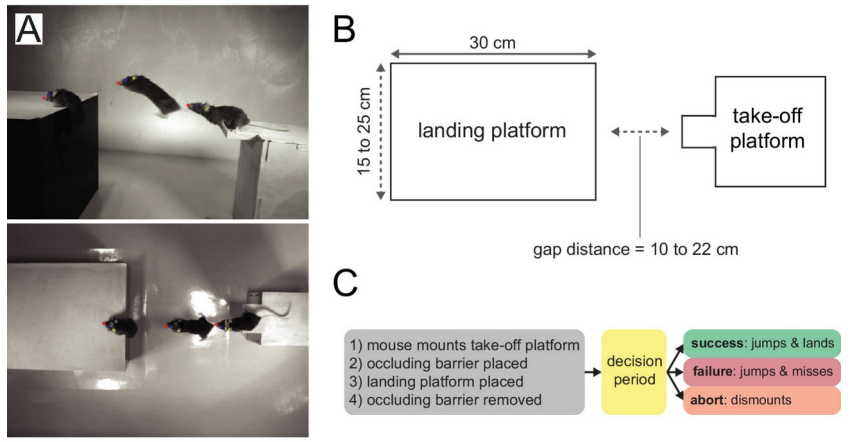
Toosi et al. (2025), Fig. 3, selected panels.

Binocular disparity provides a cue to depth

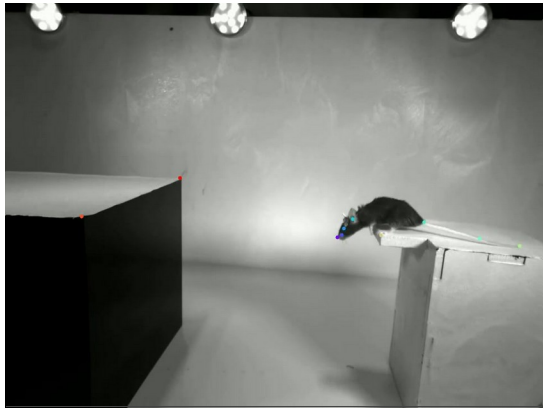
The eyes view a nearby object from slightly different positions. **Binocular disparity** is the difference between where corresponding features fall in their two images. When the eyes fixate one distance, objects nearer or farther away produce different disparities that binocular cortical neurons can use as evidence about depth.

The brain must match features belonging to the same object before comparing their positions. A neuron may respond most strongly when a feature occupies one location in the left-eye image and a slightly shifted location in the right-eye image. Different preferred shifts provide a population description of relative distance.

Other depth cues remain available to one eye alone. Overlap, perspective, relative size, and movement-generated image shifts can all help estimate spatial layout. An animal judging a gap therefore need not rely entirely on stereoscopic depth, especially if its movements produce additional information about the landing surface and surrounding objects.



Parker et al. (2022), Fig. 1.



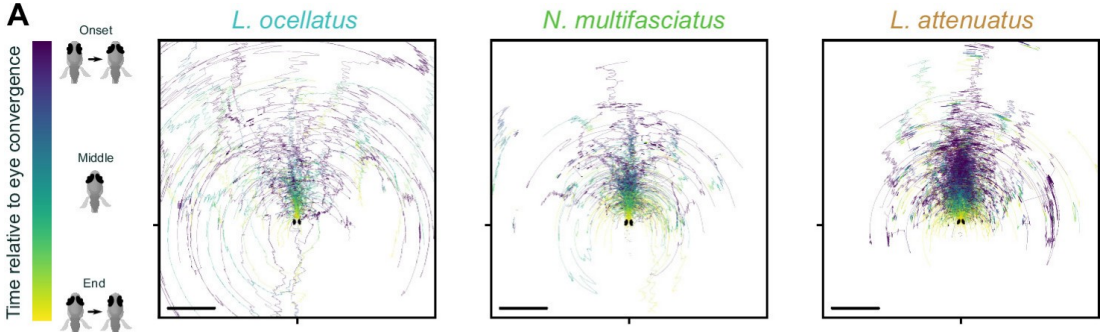
Parker et al. (2022), 74708 Video 1.

Cichlids bring prey toward the frontal field

Mearns and colleagues compared eye movements across several cichlid species during capture of small prey. The animals often rotated their eyes inward and positioned prey in front of the head. The shared frontal view changed as the animal approached, bringing information from both eyes into a region useful for the final capture movement.

The investigators measured the angle of each eye and the prey's position relative to the head. A frontal target and an inward eye rotation could therefore be identified separately from a turn of the whole animal. These measurements described the visual geometry accompanying the behavior rather than assuming that every body turn represented a change in gaze.

The fish also used discrete movement bouts with different tail patterns and capture strikes. Positioning the prey centrally was one component of a coordinated sequence, not a complete account of distance estimation. Eye orientation, approach direction, and the remaining target distance jointly constrained when and where the final strike occurred.



Mearns et al. (2025), Fig. 4, selected panels.



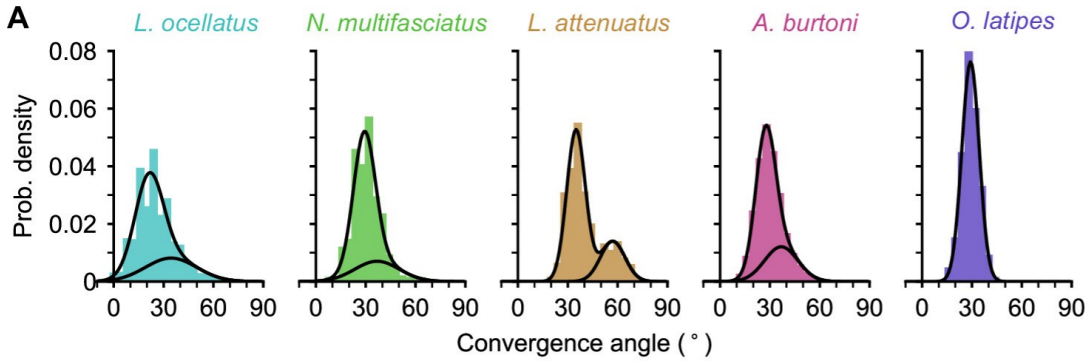
Mearns et al. (2025), 98347 Fig. 1 Video 3.

Medaka capture prey without eye convergence

Medaka larvae used a hunting pattern different from that of the tested cichlids. They often swam continuously, kept prey toward one side, and captured it with a lateral head swing. Their eyes did not show the same inward convergence that brought prey into the frontal binocular field in the cichlid species.

A side-positioned target can be followed primarily through one eye. Movement of the head and body then changes the image position of nearby prey relative to more distant surroundings. Such **motion parallax** is a possible source of depth information that does not require corresponding features to be compared between two eyes.

The observed eye and body movements are consistent with different visual strategies across species. They establish how the target was positioned and approached; they do not directly measure a medaka neuron's depth computation. The behavior supplies specific geometric conditions under which monocular movement cues could contribute to the capture response.



Mearns et al. (2025), Fig. 3, selected panels.



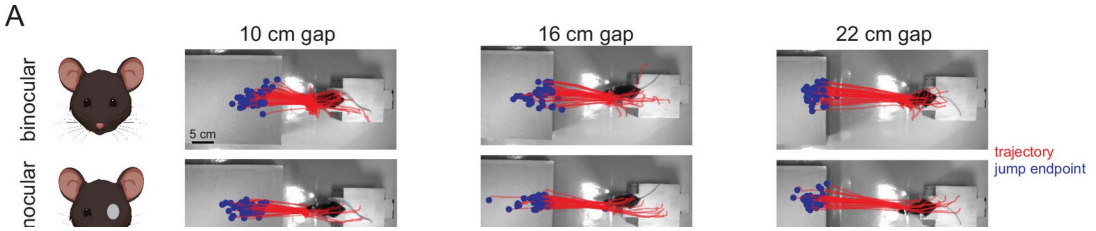
Mearns et al. (2025), 98347 Fig. 1 Video 6.

Mice can judge a gap using one eye

Parker and colleagues trained freely moving mice to jump from a starting platform to a landing platform. Changing gap distance and landing width required the animal to adjust the jump to the visible target. The measured landing position supplied a behavioral readout of how visual distance information was converted into movement.

The mice could still perform after one eye was closed. Accurate jumping therefore did not require binocular disparity under these conditions. The animal could use information available to a single eye, including the changing view produced by its own head movements before it committed to the jump.

Under monocular viewing, the mice changed their head position and made more vertical movements during the decision period. Moving the eye through space changes the relative image positions of surfaces at different distances. These behavioral adjustments provided additional opportunities to obtain depth information from the remaining visual input.



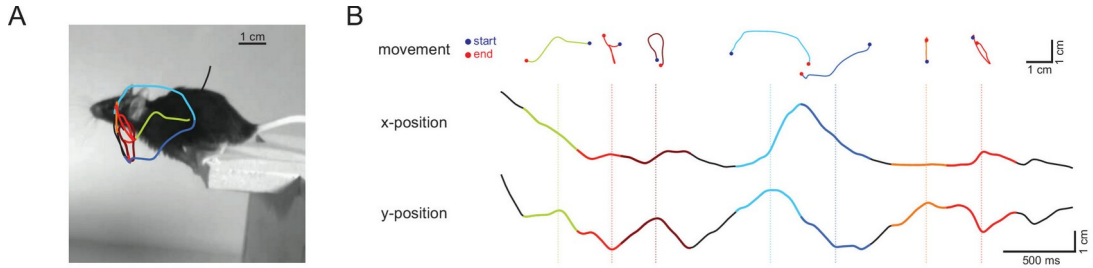
Parker et al. (2022), Fig. 2, selected panels.

Head movement supplies depth through parallax

Motion parallax occurs because objects at different distances move across the retinal image by different amounts when the observer translates through space. Nearby objects usually shift more than distant objects for the same movement. Comparing those image changes provides information about depth even when only one eye is available.

Moving the head vertically before a jump changes the viewpoint of the landing surface and the gap. The brain can combine that image displacement with information about the head's movement. An image shift alone is ambiguous because it could result from the object moving, the observer moving, or both movements occurring together.

A self-generated movement can therefore become part of the sensory measurement. The animal does not merely wait for a better retinal picture; it changes the viewpoint that produces the picture. Head position, eye position, and the location of the landing edge determine which depth cues are available at the moment the jump is planned.



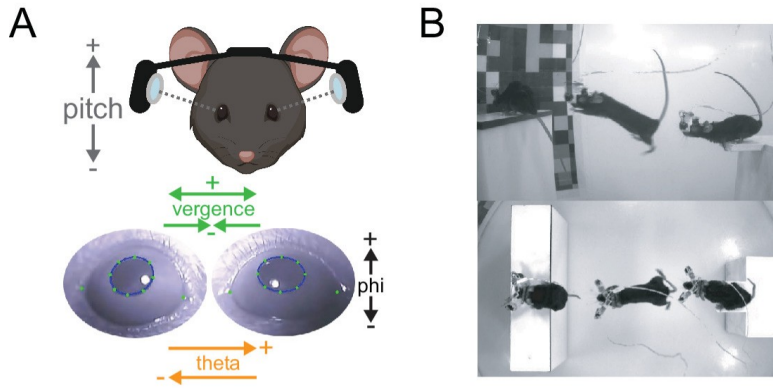
Parker et al. (2022), Fig. 3, selected panels.

Eye movements stabilize vision before a jump

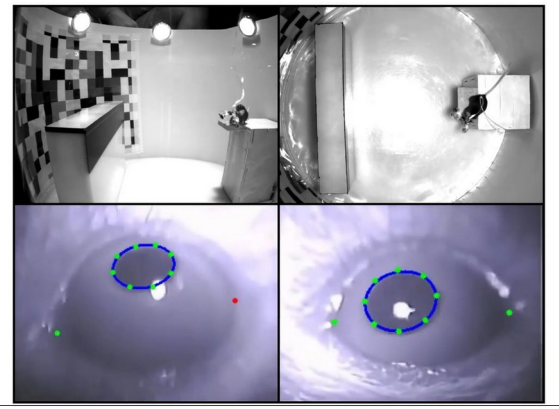
The gap-jumping study used small head-mounted cameras to record both eyes while external cameras tracked the head and body. Eye rotations compensated for parts of the head movement, keeping gaze more stable than it would have been if the eyes had remained fixed within the skull throughout the decision period.

This stabilization did not eliminate the change in viewing position created by moving the head through space. The animal could therefore obtain movement-related depth information while reducing disruptive image rotation. Translating the viewpoint and rotating the eye have different effects on the retinal image, even when both occur during the same head movement.

Eye and head coordination also changed near the jump. Measuring both was necessary to identify the visual input available before takeoff. Tracking only the animal's body or only the pupil would omit part of the transformation between the physical landing surface, the changing retinal image, and the eventual movement across the gap.



Parker et al. (2022), Fig. 4, selected panels.



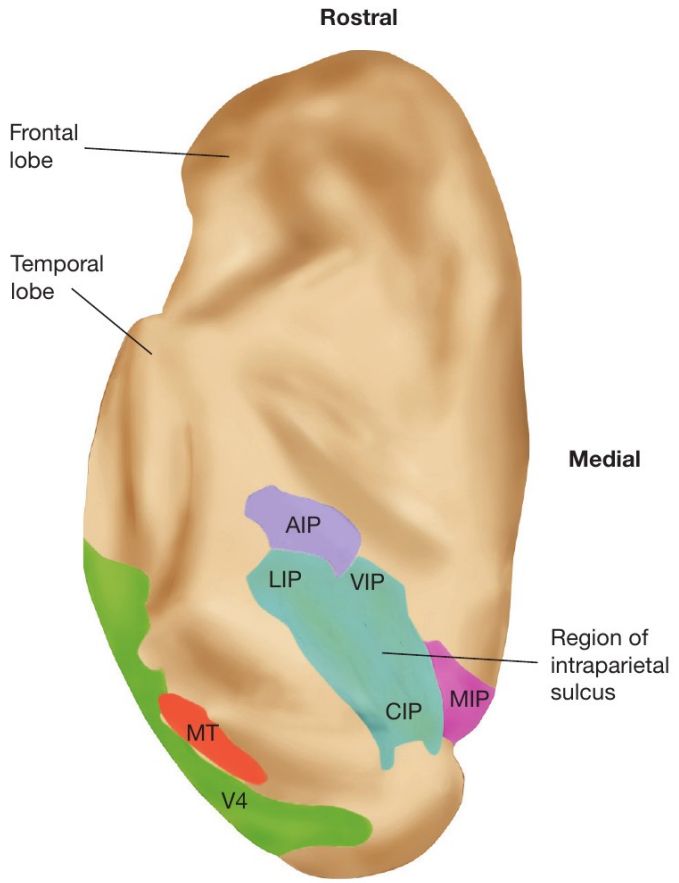
Parker et al. (2022), 74708 Video 3.

Parietal cortex turns visual estimates into actions

The **posterior parietal cortex** receives visual information through the dorsal stream and combines it with signals about body position and movement. Different regions contribute to selecting gaze targets, reaching toward objects, and shaping the hand for grasping. Their outputs communicate with frontal regions involved in preparing and controlling actions.

Reaching requires the target's location to be expressed relative to the moving limb. Grasping additionally requires the fingers to open by an appropriate amount and approach from a useful direction. The visual description needed for these actions changes continuously as the eyes, head, arm, and object move relative to one another.

Damage can impair these transformations while leaving object recognition relatively intact. A person may correctly judge that two objects differ in size yet fail to scale the opening of the hand before reaching. The impaired behavior concerns using visual dimensions to control an action, rather than failing to see that those dimensions differ.



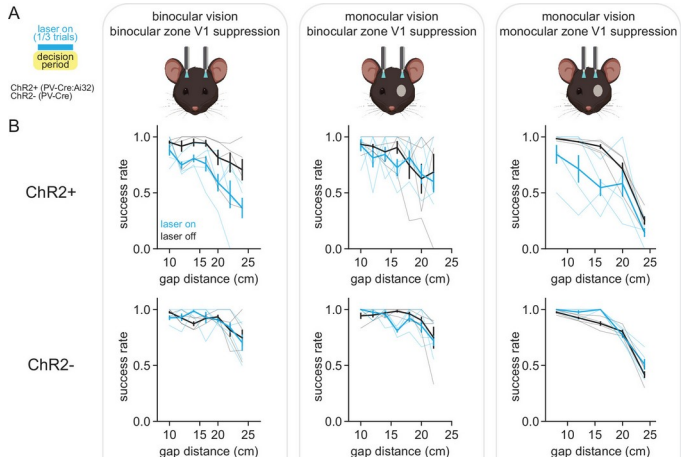
Carlson and Birkett (2022), Fig. 6.35, p. 183.

Suppressing V1 disrupts visually guided jumping

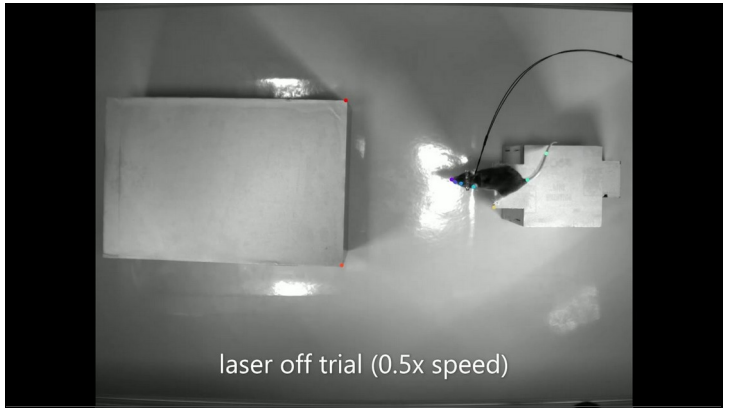
The researchers used **optogenetic suppression** to test whether visual cortex contributed to gap jumping. They expressed a light-sensitive channel in inhibitory interneurons, cells that reduce the activity of neighboring neurons. Illuminating those interneurons increased inhibition and temporarily reduced visual-cortical activity during the period when the mouse judged the gap.

Suppressing the relevant part of V1 impaired performance under binocular or monocular viewing conditions. The affected cortical region depended on the visual input available to the animal. This intervention tested a contribution of the cortical activity itself, beyond observing that neural responses might accompany successful jumps.

The source film compares trials with and without the suppression light. The animal still had to initiate a movement, cross the gap, and land on the platform. Recording where the jump ended separated a disruption of visually guided targeting from a description limited to whether the animal moved at all.



Parker et al. (2022), Fig. 5, selected panels.



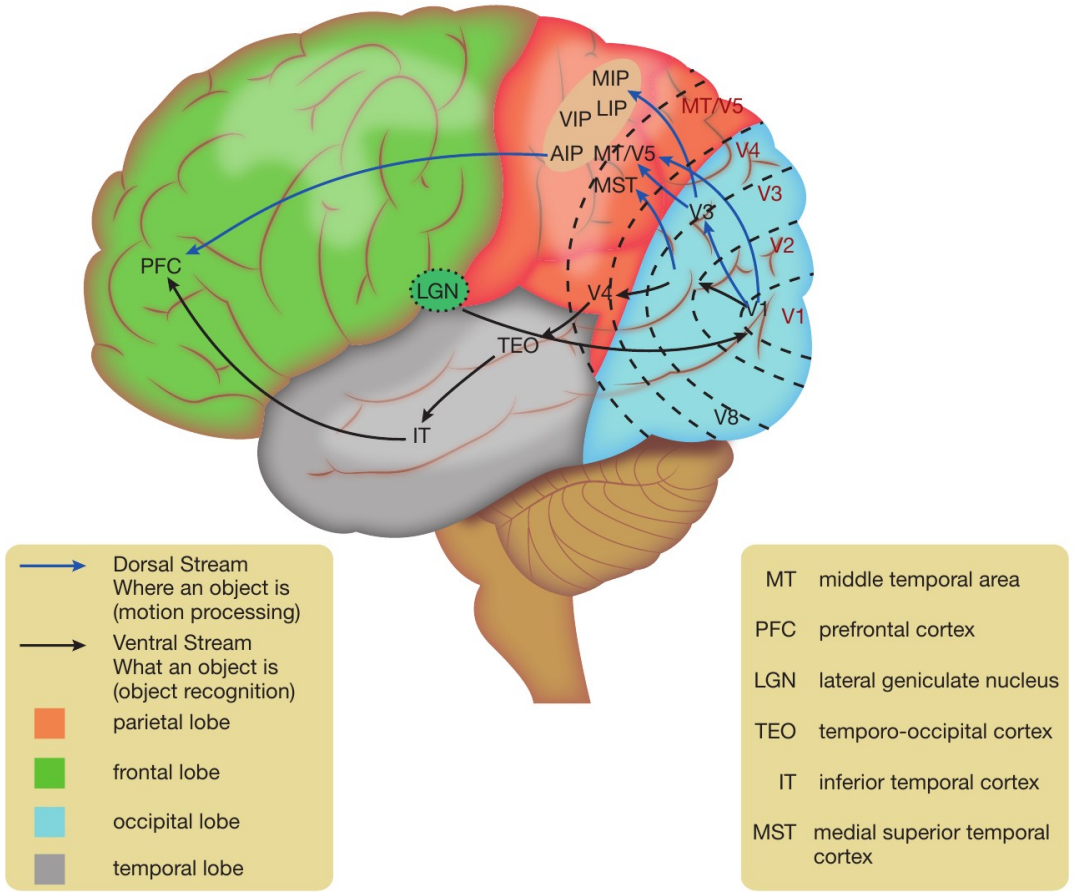
Parker et al. (2022), 74708 Video 4.

Motion-sensitive cortex estimates trajectories

In primates, area **MT**, also called V5, contains many neurons sensitive to visual motion. Individual cells can prefer particular directions and speeds. Population activity therefore provides information about how image features move over time, rather than describing only which positions or orientations are occupied in a single moment.

Area **MST** receives motion-related input and responds to more complex movement patterns, including expansion and rotation. Such patterns occur when an observer moves through the environment. They can help distinguish a target's local motion from the broad displacement of image features caused by movement of the eyes, head, or body.

Damage involving motion-sensitive cortex can produce **akinetopsia**, difficulty experiencing continuous movement. An affected observer may identify an object's successive positions without perceiving the usual smooth motion between them. Detecting that an object has changed location and representing its trajectory are therefore related but separable functions within the visual system.



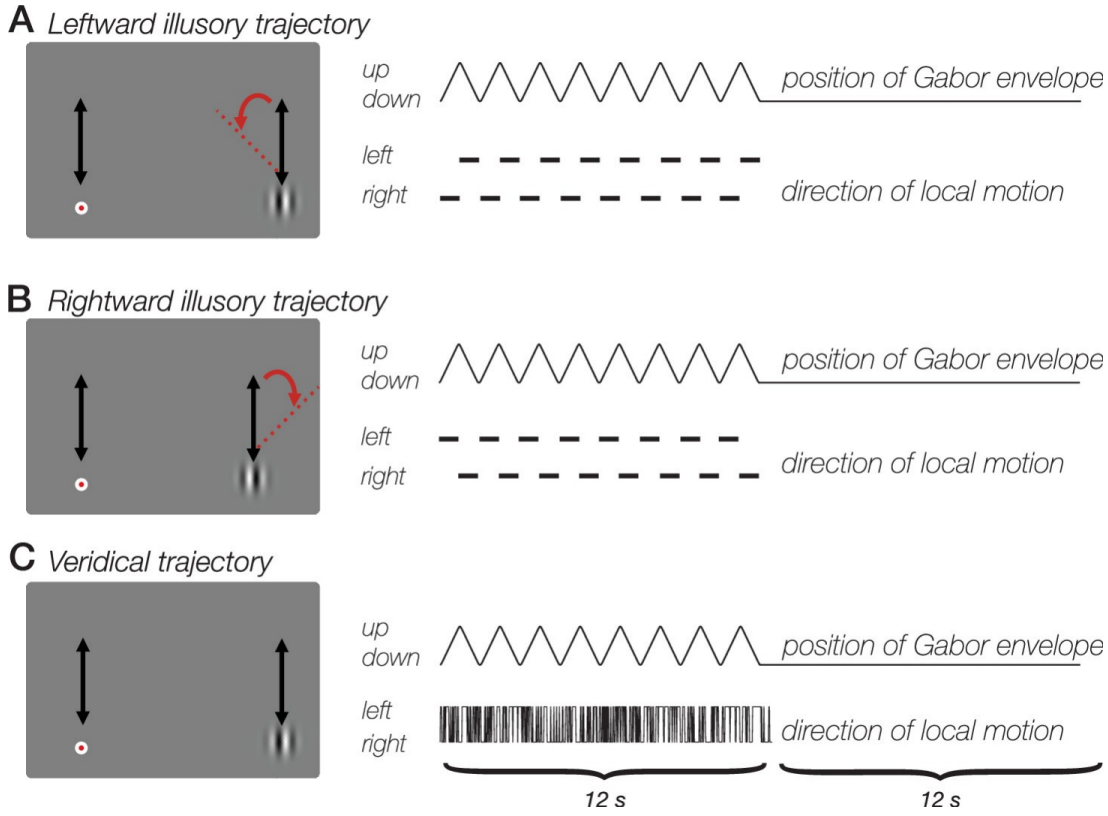
Carlson and Birkett (2022), Fig. 6.14, p. 166.

Internal motion can bend an object's perceived path

The **double-drift illusion** combines two motions: a visible window moves through space while the texture inside it drifts in another direction. Observers can perceive the window traveling along a path that differs from its physical trajectory. Motion within the object changes the estimated position of the object containing it.

Steinberg and colleagues examined the illusion while observers tracked a moving fixation target. The target's movement kept the drifting stimulus relatively stable on the retina, yet the perceived trajectory could still deviate. The response therefore required considering information about eye movement together with the motion signals within the retinal image.

The illusion separates the physical path, retinal stimulation, and experienced path. These normally agree closely enough that their distinct contributions are difficult to notice. Here, combining the texture's drift with the window's motion shifted the position estimate over time, producing a systematic trajectory error rather than merely making the object difficult to see.



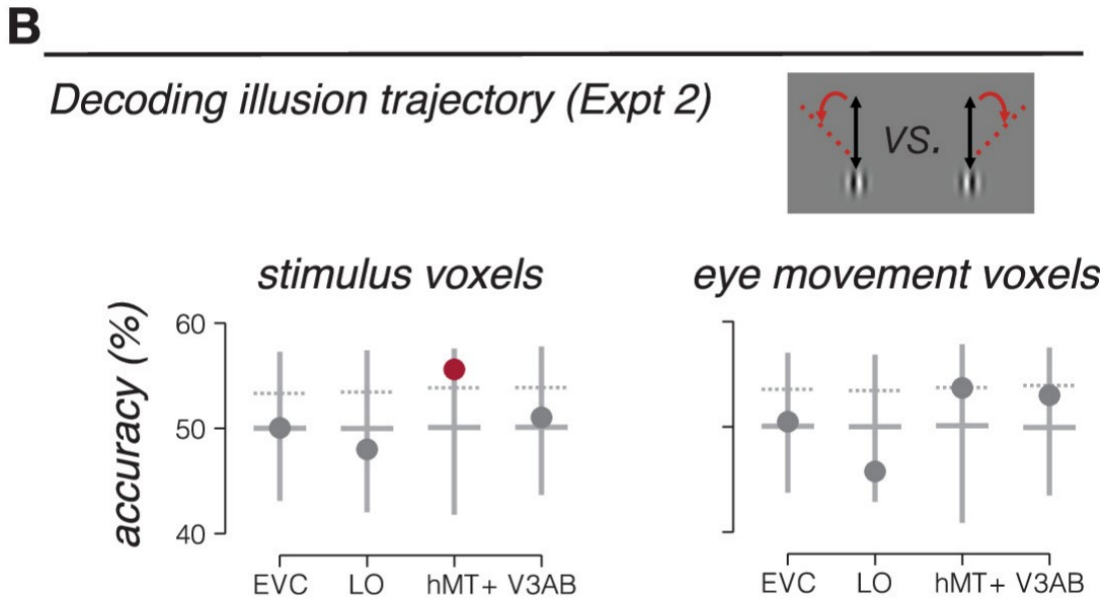
Steinberg et al. (2024), Fig. 1.

Motion cortex carried the illusory trajectory signal

The double-drift study measured **BOLD fMRI**, a blood-oxygenation response related to local neural activity, while observers experienced different illusory paths. Several visual areas responded more strongly during the illusion than during a comparison condition. The researchers then asked whether patterns of activity contained information about which illusory trajectory was perceived.

Activity patterns in **hMT+**, the human motion-sensitive cortical complex, distinguished the illusory trajectories. Other measured visual regions did not provide the same successful readout. The result identified information within a regional population response rather than assuming that any increase in activity necessarily represented the object's perceived path.

Eye movements were recorded during scanning, and the trajectories could not be recovered from those eye-position measurements. The cortical result therefore was not explained by observers simply following different paths with their eyes. Retinal motion and information about gaze were combined within a representation carrying the systematically shifted trajectory.



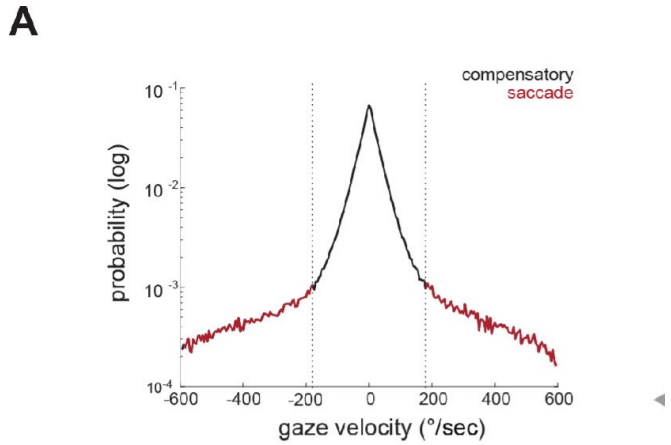
Steinberg et al. (2024), Fig. 3, selected panels.

Gaze shifts alternate with stabilized viewing

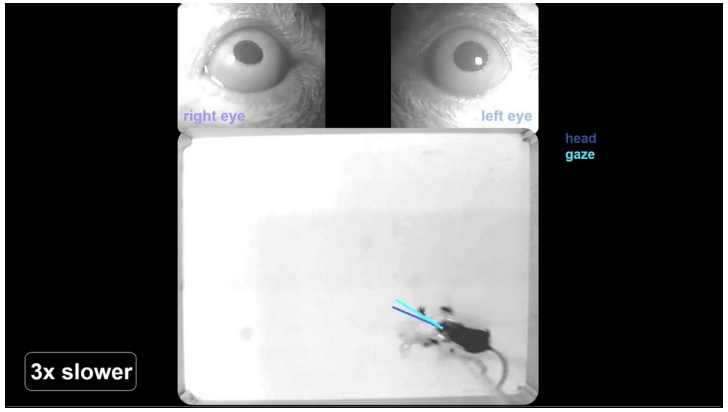
Michaiel and colleagues recorded the eyes and head of mice pursuing crickets. When the head turned, the eyes often moved in the opposite direction, temporarily keeping gaze stable. The visual scene therefore moved less across the retina than would be predicted from the head rotation alone.

These stabilized periods were interrupted by rapid **saccades**, gaze shifts that moved the viewing direction toward a new location. During pursuit, noncompensatory eye movements helped bring gaze into alignment with the head as the animal targeted the cricket. The gaze sequence contained both compensation for movement and deliberate relocation of the visual sample.

A camera attached only to the body would combine these different phases into one apparent turn. Recording eye position revealed when retinal stimulation remained relatively stable and when it shifted rapidly. The changing pattern determined which part of the scene was sampled during each stage of the approach toward moving prey.



Michaiel et al. (2020), Fig. 4, selected panels.



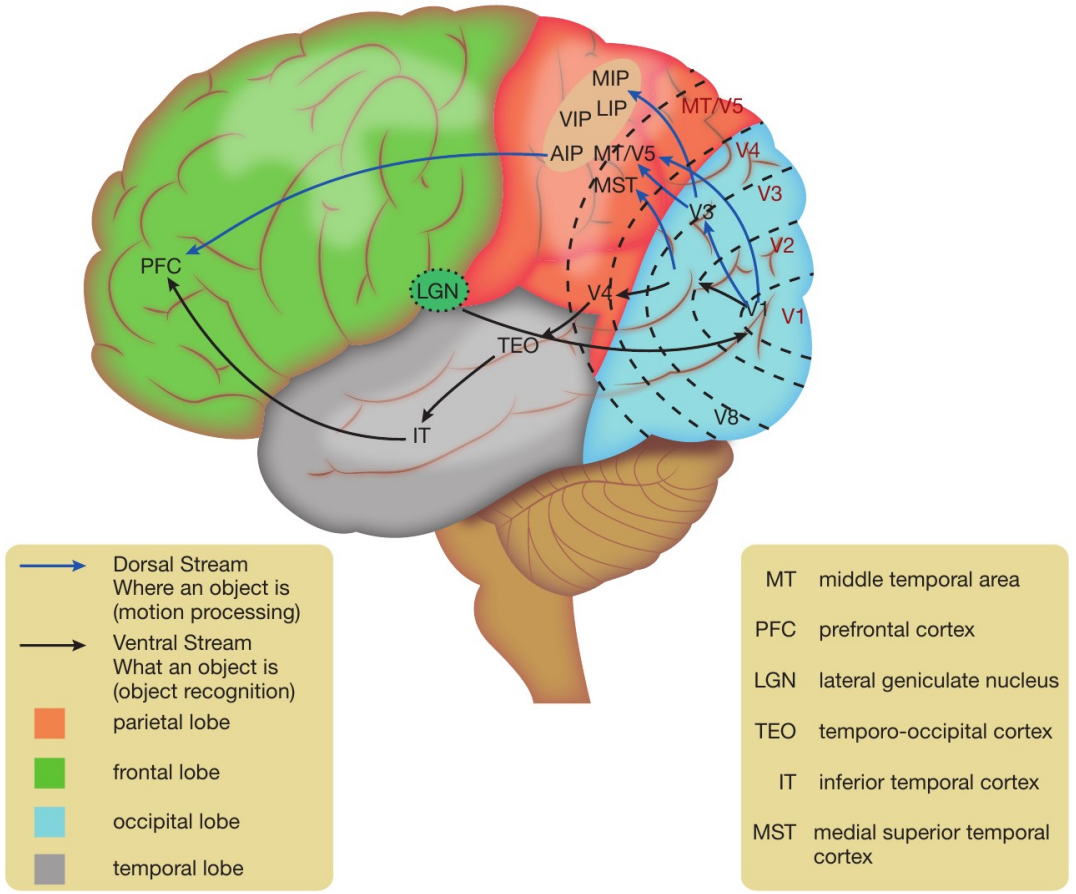
Michaiel et al. (2020), 57458 Video 2.

Retinal motion and perceived motion can differ

An image can move across the retina because an object moved or because the eyes moved across a stationary scene. During a saccade across a page, the retinal image of the text shifts rapidly even though the words appear stationary. Interpreting motion therefore requires information about the cause of the image displacement.

A copy of an eye-movement command can inform sensory circuits that the retinal shift was self-generated. Signals from head movement and other senses contribute additional information about the observer's motion. Visual networks use these signals together with the pattern of image change to estimate which objects moved relative to the surrounding environment.

The converse also occurs during tracking: a pursued object can remain near the same retinal location while being perceived as moving through the world. Stable retinal position is therefore compatible with perceived motion when gaze changes appropriately. The estimated trajectory depends on combining retinal location over time with the changing direction of the eyes.



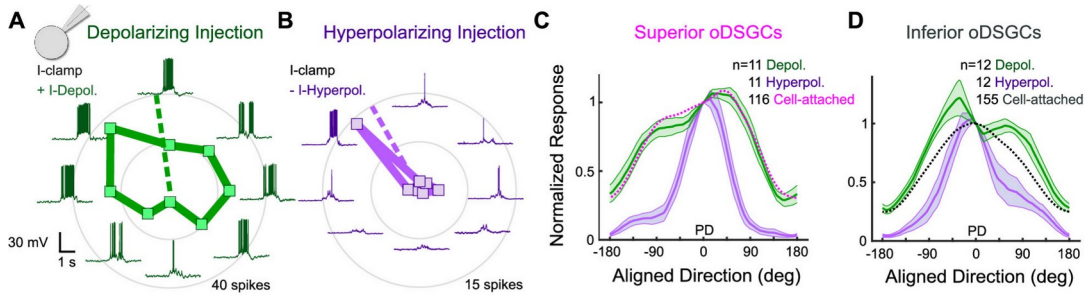
Carlson and Birkett (2022), Fig. 6.14, p. 166.

Retinal motion channels predict eye reflexes

The **optokinetic reflex** stabilizes vision when a broad visual pattern moves across the retina. Direction-selective ganglion cells respond preferentially to particular directions of motion and supply information used to drive compensatory eye movements. The retinal input therefore carries an estimate of image motion before the stabilizing response is generated.

The mouse study compared ON direction-selective cells preferring opposite vertical directions. Their tuning was asymmetric and changed with contrast. Differences in excitatory and inhibitory synaptic inputs, together with the firing threshold, contributed to these unequal outputs rather than producing two perfectly matched channels with reversed preferences.

Combining the retinal measurements with a central subtraction of competing directional outputs predicted vertical eye-movement trajectories across stimulus conditions. A change in the balance of the retinal channels could therefore alter the reflex's direction or time course. The behavioral response depended on how the brain compared their signals, not just whether any motion-sensitive cell fired.



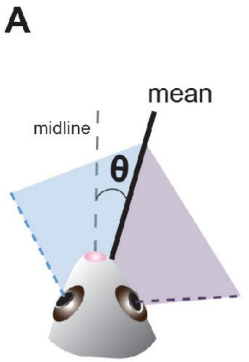
Harris et al. (2023), Fig. 5, selected panels.

A pursuing mouse coordinates its eyes and head

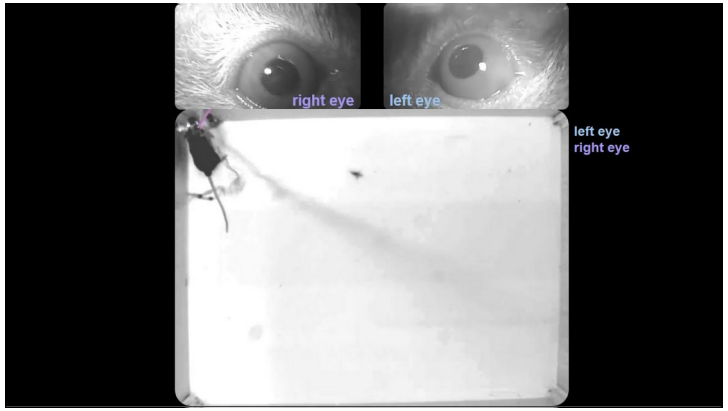
During cricket pursuit, a mouse must keep the target available while moving its own head and body. Simultaneous eye-camera and overhead recordings located the eyes relative to the head and the head relative to the arena. These measurements distinguished a change in eye rotation from an actual change in the direction of gaze.

The direction of each eye was estimated from pupil position and combined with head orientation. This produced a time-varying description of the animal's visual sampling during turns and approaches. The cricket's location could then be compared with the viewing direction as the distance between predator and prey decreased.

The eyes were not fixed cameras mounted on a moving body. Their compensatory rotations and rapid gaze shifts changed the image reaching the retina throughout the pursuit. A successful approach required the animal to obtain useful target information while the sensory surface collecting that information was itself continually moving.



Michaiel et al. (2020), Fig. 3, selected panels.



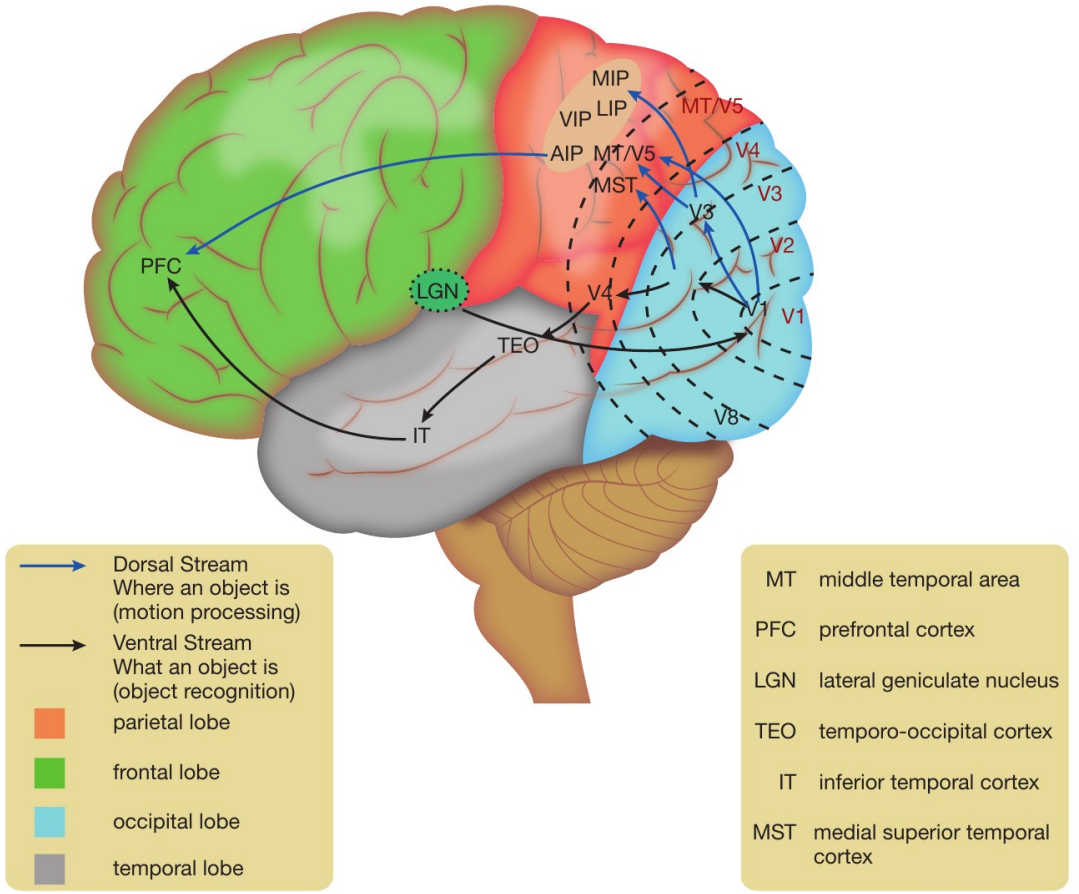
Michaiel et al. (2020), 57458 Video 1.

Optic flow changes as the observer moves

Optic flow is the pattern of image motion produced across the visual field as the observer moves relative to the environment. During forward travel, many visible features expand outward from a region near the direction of heading. Nearby surfaces usually produce faster image changes than surfaces farther away.

The flow also changes when the eyes rotate or the head turns. A broad rotation of the image can occur without the observer translating toward a new location. Estimating heading therefore requires distinguishing image changes caused by translation from those caused by rotation, using visual patterns and signals about the observer's own movements.

Motion-sensitive cortical areas analyze these larger patterns rather than only following one isolated feature. Expansion, contraction, and curved flow can provide information about approach and turning. The contribution of each pattern depends on where the observer looks and which surfaces occupy the visible scene at that moment.



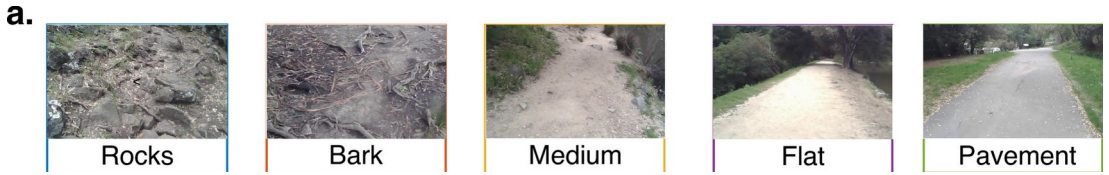
Carlson and Birkett (2022), Fig. 6.14, p. 166.

Human walkers stabilize gaze on chosen locations

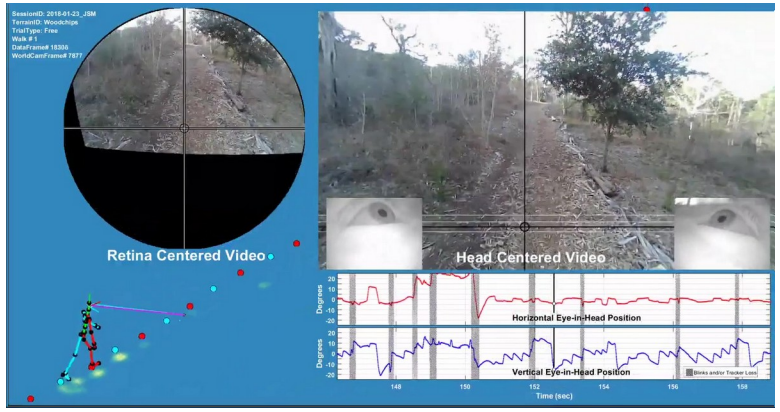
Muller and colleagues measured eye position, body movement, and the three-dimensional environment while people walked naturally. These measurements allowed them to estimate the retinal motion created by real locomotion. The source recording combines the observer's changing view with information about gaze and body position during the walk.

When gaze remained on a chosen location, the eyes compensated for the body's forward movement. That compensation shaped the retinal flow across the surrounding scene. Motion near the fixated region differed from motion farther away, so the image pattern depended on both the environment's geometry and the observer's gaze behavior.

Different gaze directions changed the distribution of motion speeds and directions across the retina. Looking down toward the walking surface produced a different input from looking farther ahead or to one side. The visual system therefore receives a behavior-dependent distribution of motion signals rather than one fixed flow pattern determined by walking speed alone.



Muller et al. (2023), Fig. 2, selected panels.



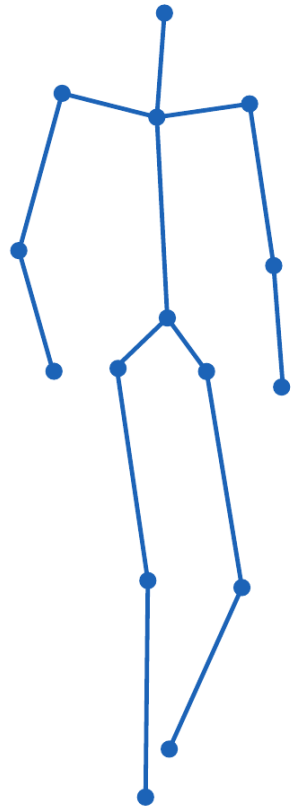
Muller et al. (2023), 82410 Video 1.

Sparse motion can reveal a biological action

A moving person can be recognized from a small set of lights attached near major joints. Each individual point provides little information about a body, but their coordinated trajectories preserve the relationships among limbs. This is **biological motion**, movement whose organization can reveal the action of a living body.

Changing joint positions through time constrains which points belong together. Opposing arm and leg movements, bending knees, and the relative motion of shoulders and hips can specify walking without a visible outline or textured surface. The nervous system can recover structured form from the way sparse elements move together.

Regions around the posterior superior temporal sulcus contribute to processing such organized actions. Their responses differ from a simple detector for any moving dot. Recognizing the action requires integrating movements across body locations and time, while distinguishing their biological arrangement from an equally active but spatially scrambled set of trajectories.



Carlson and Birkett (2022), Fig. 6.37, p. 187.

A moving scene evokes compensatory fly gaze

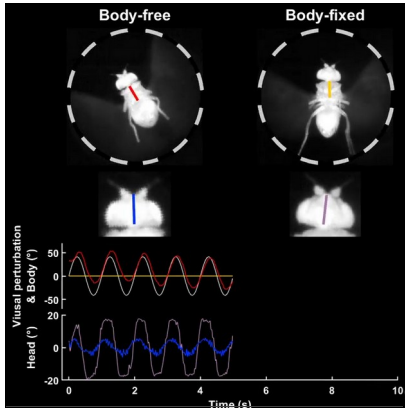
A fruit fly uses visual feedback to stabilize its head during movement. Cellini and colleagues presented moving patterns while comparing flies whose bodies could rotate with flies whose bodies were fixed. Cameras measured the head's response, separating movement of the head relative to the body from rotation of the entire animal.

The body-fixed flies made larger head movements in response to the same visual perturbation. Allowing the body to move changed the mechanical and sensory conditions under which the head stabilized gaze. A visual pattern therefore did not produce one invariant head response independently of the animal's other available movements.

The recordings preserve the original time course of the visual perturbation and resulting motion. Comparing the head relative to the room with the head relative to the body distinguishes two stabilization demands. The nervous system must coordinate them because the eyes are carried on a head that can move on an independently moving body.



Cellini et al. (2022), Fig. 3, selected panels.



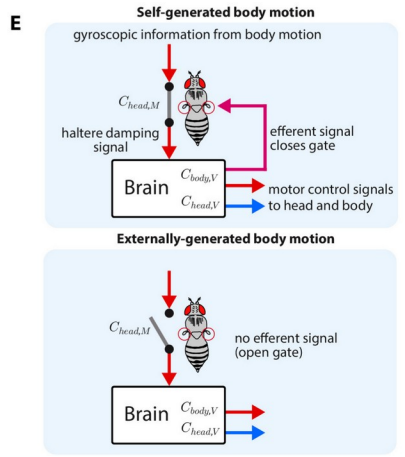
Cellini et al. (2022), 80880 Fig. 3 Video 1.

Body feedback damps visually driven head movement

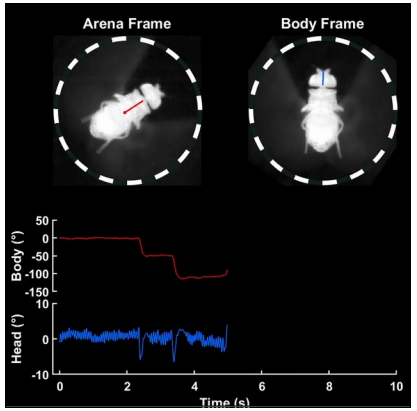
Mechanosensory feedback reports physical movement or deformation through sensory structures in the body. In the fly experiment, signals generated by body motion reduced the size of the visually driven head response. This **damping** limited excessive movement that could otherwise destabilize gaze while the animal corrected a visual disturbance.

The researchers compared self-generated body movements with rotations imposed by a motor. The resulting head responses differed, indicating that feedback depended on the movement's motor context. A signal accompanying the animal's own turn was therefore not processed identically to a similar physical turn imposed from outside.

The source recording captures simultaneous body and head saccades against a stationary visual background. Coordinated movements relocate gaze without requiring the body and head to remain locked together. The visual and mechanosensory loops adjust different components of the movement while interacting through the head's changing position on the body.



Cellini et al. (2022), Fig. 7, selected panels.



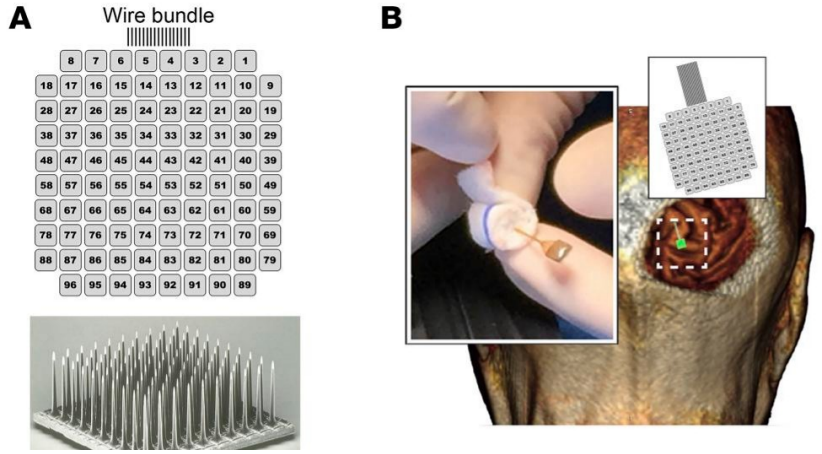
Cellini et al. (2022), 80880 Fig. 7 Video 1.

Cortical electrodes evoke localized light percepts

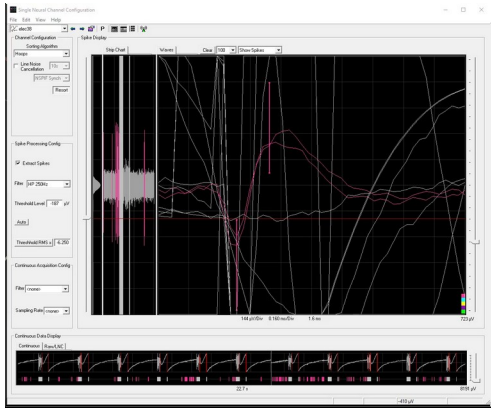
A **phosphene** is a perceived spot or region of light produced without light entering the eye at that location. Fernández and colleagues implanted a small electrode array in the occipital cortex of a blind participant. Electrical pulses could activate nearby neural tissue and produce visual percepts associated with particular stimulation sites.

The participant described the location and appearance of the evoked percepts. Different electrodes could produce percepts in different parts of visual space, allowing the investigators to map the available signals. Changing stimulation strength also altered properties such as detectability, making the stimulation parameters part of the information delivered to the visual system.

Some experiments recorded neural firing around the electrodes while stimulation was delivered. Neural activation and a reported phosphene were measured separately, because increased local firing did not guarantee a conscious light percept on every trial. A cortical interface therefore required both physiological measurements and the participant's report of the resulting visual experience.



Fernández et al. (2021), Fig. 1, selected panels.



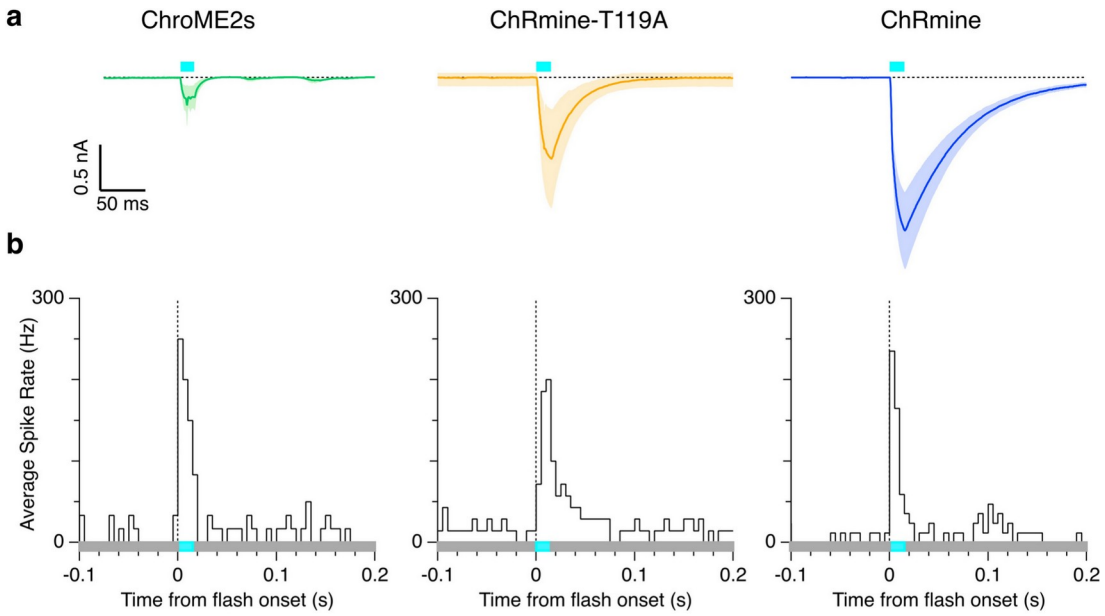
Fernández et al. (2021), Supplemental Video 1.

Light-sensitive channels activate surviving retinal cells

In retinal degeneration, photoreceptors can die while some inner-retinal neurons remain. **Optogenetic restoration** introduces a light-sensitive protein into those surviving cells. Illumination can then change their membrane current directly, bypassing part of the lost photoreceptor pathway while retaining a route for signals to leave the retina and reach the brain.

The 2025 mouse study expressed channelrhodopsin variants in retinal ganglion cells of animals with severe photoreceptor degeneration. Light evoked action potentials in treated cells. A faster-closing ChRmine variant also improved performance in a light-avoidance test at a lower tested illumination than the other variants, linking channel properties to a measured behavior.

Closing speed and light sensitivity describe different aspects of the introduced channel. A channel that closes slowly can blur rapid changes, while one requiring intense light may fail under dim conditions. In the behavioral test, a change in preference between bright and dark compartments measured how the restored light signal influenced the animal's location.



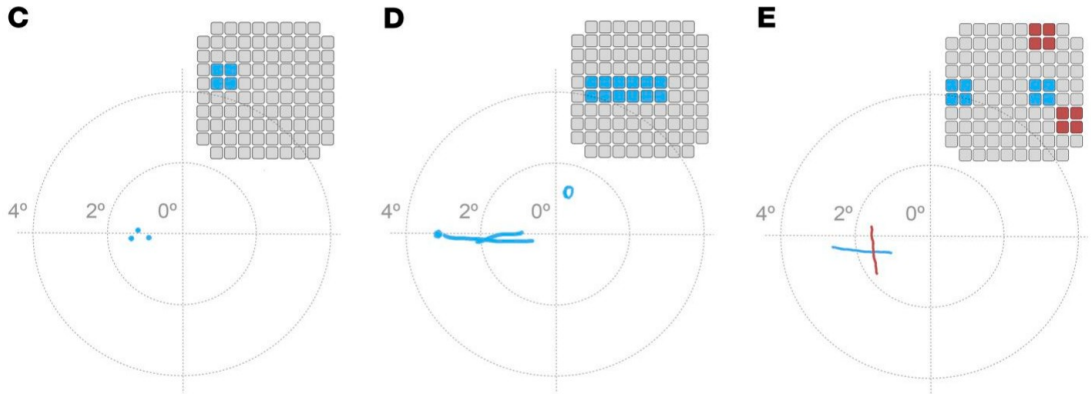
Fong et al. (2025), Fig. 2, selected panels.

Electrical patterns can convey a visual choice

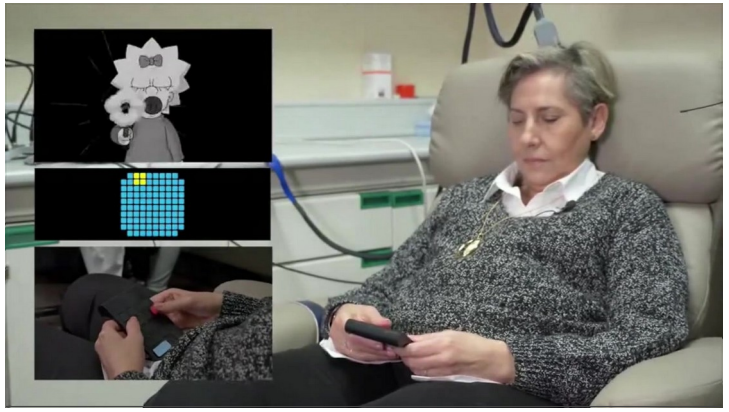
A cortical prosthesis must deliver distinguishable patterns rather than simply produce any detectable sensation. In the intracortical study, stimulating different groups of electrodes produced patterns the participant learned to discriminate. A two-choice task associated particular stimulation patterns with responses, allowing the investigators to measure whether the delivered alternatives could be distinguished.

One task used a game display in which each direction corresponded to a particular electrode group. The participant pressed the matching response button after stimulation. The displayed game organized the trials; the experimentally conveyed information was the difference between the electrical patterns delivered to visual cortex.

Electrode positions did not translate one for one into visual pixels. Four neighboring electrodes could produce three perceived dots, while a larger group could produce a horizontal line. Mapping these actual percepts allowed researchers to select distinguishable patterns and teach the participant to use their differences as cues for the appropriate response.



Fernández et al. (2021), Fig. 7, selected panels.



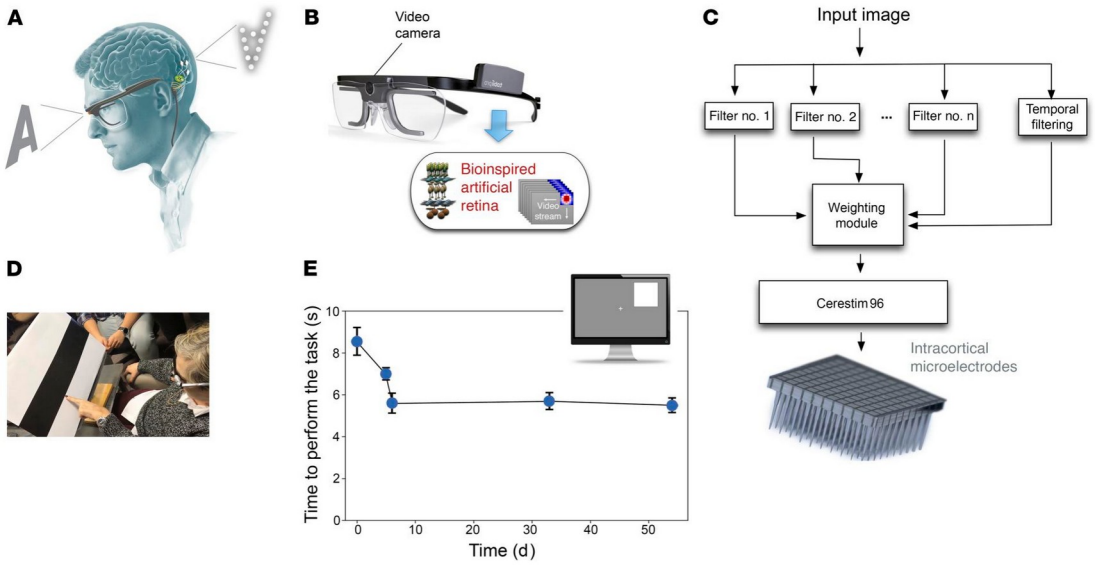
Fernández et al. (2021), Supplemental Video 2.

A camera can supply input to cortical stimulation

The study also used a camera mounted on eyeglasses to capture the scene. A processor transformed the camera image into a pattern of electrode commands, emphasizing selected spatial and temporal features. The stimulator then converted those commands into electrical pulses delivered through the implanted cortical array.

This pathway bypassed the usual optical-to-retinal conversion for the encoded information. The camera measured light, software selected and transformed features, and cortical electrodes activated neural tissue. The participant's remaining visual pathways and experience then determined how the resulting percepts could be used to perform the experimental task.

The transformation required a mapping between image features and electrode responses. An electrode did not function as an ordinary retinal photoreceptor with its natural connections preserved. The investigators first measured the available percepts and trained the participant to interpret selected patterns before testing camera-guided localization of simple high-contrast targets.



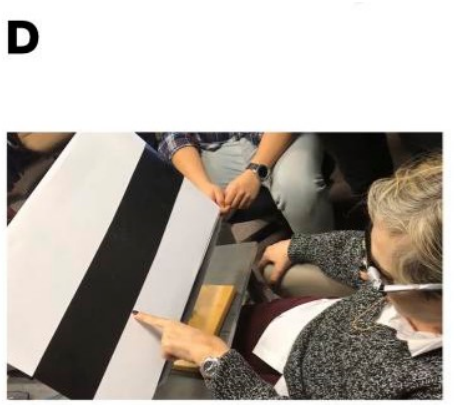
Fernández et al. (2021), Fig. 9.

Head scanning helps locate camera-encoded edges

Moving a camera across an edge changes which parts of its image are bright and dark. In the cortical-interface experiment, the participant used head scanning while the eyeglass camera supplied stimulation commands. Changes in the evoked percepts could then be compared with the direction of the self-generated head movement.

The participant localized the border between dark and light bars and pointed toward simple bright targets at different screen locations. These tasks tested whether the camera-to-cortex signal could support a spatial action. The participant's own movement helped convert a limited set of evoked signals into information about where a target lay.

Localization became faster with practice over the tested sessions. The improvement concerned using the interface's particular signals to guide a response. As in natural active vision, useful spatial information emerged from a sequence of samples combined with movement, although the camera, encoding algorithm, and cortical stimulation replaced several stages of the normal retinal pathway.



Fernández et al. (2021), Fig. 9, selected panels.



Fernández et al. (2021), Supplemental Video 3.

Key takeaways

Light reduces photoreceptor output. Activated pigment closes cyclic-GMP-dependent channels; hyperpolarization reduces calcium entry and glutamate release.

Retina computes before cortex. ON/OFF pathways, center-surround interactions, spectral comparisons, and synaptic timing create parallel visual signals.

Seeing depends on context. Adaptation, surrounding contrast, contour completion, and motion integration can change appearance without matching changes in the stimulus.

Cortical pathways support different functions. Ventral networks contribute to recognition; dorsal networks transform visual estimates into gaze, reaching, grasping, and navigation.

Plasticity depends on developmental conditions. Early retinal rerouting changes auditory-cortical visual responses; later visual experience can recalibrate existing pathways and actions.

Vision is an active process. Eye and head movements change retinal sampling and depth cues; neural interfaces can supply limited visual information through alternative input routes.

Carlson & Birkett (2022); primary sources in slide notes